$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \qquad 13.5$$

The formula may be applied to both types of lenses. As with the mirror, two sign conventions are applicable.

Real-is-positive	New Cartesian
$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$
f is +ve for a converging lens	f is +ve for a converging lens
f is –ve for diverging lens	f is –ve for diverging lens
Distances of real objects and real images are positive. Distances of virtual objects and images are negative	All distances to the left of the lens are negative and all those to the right are positive

Magnification (M) is defined as

$$M=\frac{\text{Image height}}{\text{Object height}}=\frac{\text{distance of image from lens}}{\text{distance of object from lens}}$$

$$M=\frac{v}{u} \qquad 13.6$$

Example 13.5

An object 5 cm high is placed at a distance 12 cm from a convex lens of focal length 8 cm. Calculate the position, size and nature of the image.

Solution

Real-is-positive	New Cartesian
$f=+8$ cm	$f=+8$ cm
$u=+12$ cm	$u=-12$ cm (object left of lens)
Using $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	Using $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$
$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{8}-\frac{1}{12}$	$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}$
$\frac{1}{v}=\frac{1}{24}$	$\frac{1}{v}=\frac{1}{8}+\frac{1}{-12}=\frac{1}{8}-\frac{1}{12}$
$v=24$ cm (real image)	$\frac{1}{v}=\frac{1}{24}$
$M=\frac{v}{u}=\frac{24}{12}=2$	$v=24$ cm (on the right real image)
$M=\frac{\text{image height}}{\text{object height}}=\frac{h}{5}$	
$2=\frac{h}{5}\ \therefore\ h=\underline{10\text{ cm}}$	

The image is real, inverted, magnified and at a distance of 24 cm from the lens on the side opposite to object. The height of the image is 10 cm.

The above example shows that we obtain the same result whatever the sign convention we use.

Example 13.6

An object placed 5 cm in front of a converging lens of focal length 10 cm produces a magnified image 8 cm high. Find the size and the position of the image.

Solution

Using the real-is-positive convention, we have

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{5}=\frac{1}{10}$$

$$\frac{1}{v}=\frac{1}{10}-\frac{1}{5}=\frac{-1}{10}$$

$$v=\frac{-10}{1}=\underline{-10\text{ cm}}$$

Therefore the image is virtual,

$$M=\frac{v}{u}=\frac{\text{height of image}}{\text{height of object}}$$

$$\frac{10}{5}=\frac{\text{height of image}}{\text{height of object}}$$

$$2=\frac{8}{h};\ \text{where } h=\text{height of object}$$

$$h=\frac{8}{2}=\underline{4\text{ cm}}$$

Example 13.7

Find the nature and position of the image of an object placed 6 cm from a diverging lens of focal length 15 cm.

Solution

$u=6$ cm

$f=-15$ cm

Using $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

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$$\frac{1}{v} = -\frac{1}{6} - \frac{1}{15} = -\frac{7}{30}$$

$$v = -\frac{30}{7} = \underline{-4.28 \text{ cm}}$$

The image is virtual and 4.28 cm from the lens on the same side as the object.

Example 13.8

A converging lens produces a four times magnified and upright image of an object placed in front of it. If the focal length of the lens is 11 cm, calculate the object distance.

Solution

$$M = \frac{v}{u} = 4$$

$$\therefore \quad v = 4u$$

An upright image is virtual. Hence image distance, v, is –ve.

Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{-v} + \frac{1}{u} = \frac{1}{10}$$

$$\frac{1}{-4u} + \frac{1}{u} = \frac{1}{10}$$

$$\frac{3}{4u} = \frac{1}{10}$$

$$u = \frac{30}{4}$$

$$= \underline{7.5 \text{ cm}}$$

Experimental determination of the focal length of a converging lens

1 Quick but approximate method

A quick method of finding the focal length of a converging lens is to focus the image of a distant object on a screen using the lens. The lens is placed in front of a screen and its position is adjusted until a sharp image of a distant object (e.g. window panes) is obtained on the screen. The distance between the lens and the screen gives an approximate value of the focal length. Rays coming from the distant object are parallel rays. They therefore converge to the principal focus of the lens after refraction.

This method is used to obtain an approximate value of the focal length very quickly.

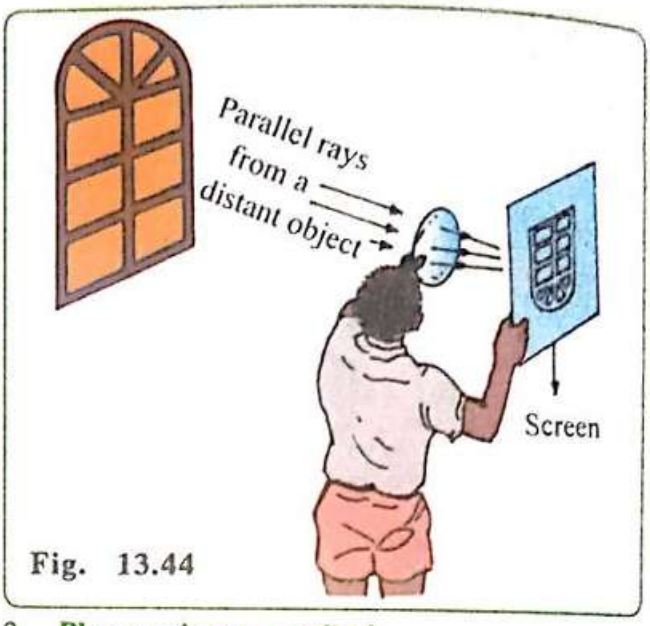
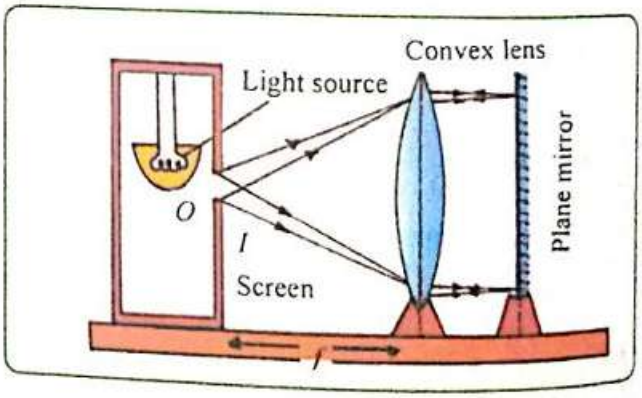


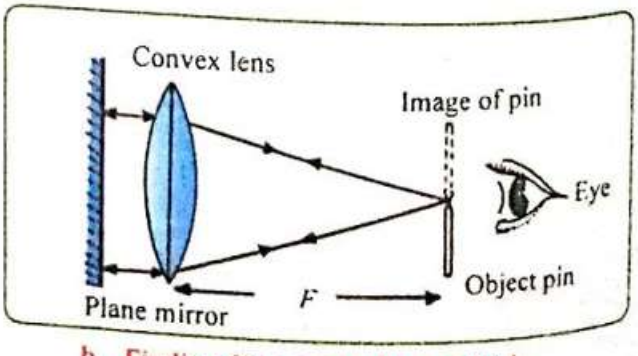
Fig. 13.44

2 Plane mirror method

The lens is mounted on a lens holder and placed in front of a plane mirror such that the planes of both lens and mirror are vertical. An illuminated object (e.g. the crosswire of a ray box) is placed in front of the lens as shown in fig. 13.45a.

a. Finding f by plane mirror and illuminated object method





b. Finding f by no-parallax method

Fig. 13.45

The object is moved to and fro until a sharp image of the object is obtained on the surface of the ray box besides the object. The distance of the object from the lens gives the focal length of the lens.

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Alternatively, we can use an optical pin stuck in a cork as the object. The pin is adjusted so that its tip is at the same horizontal level as the centre of the lens. The pin's position in front of the lens is adjusted until there is no parallax between it and its image formed by the combination of lens and mirror. The distance between the pin and the lens gives the focal length of the lens. (fig. 13.45b).

For both methods it is necessary to repeat the experiment about three more times to obtain an average value of focal length.

Explanation of the experiment

When the position of the image and the object coincide, it means that the rays of light incident on the mirror are returning from it along the original path as shown in fig. 13.45. This can only occur when the beam is incident normally on the mirror. The beam incident on the lens from the mirror is therefore a parallel beam, which after refraction is brought to a focus at the object position.

3 Using the Lens Formula

We can use the lens formula to obtain the focal length of a converging lens, by measuring the object and image distances. An approximate value of f is first obtained by the method already described above. We then place an illuminated object at a known distance, u, greater than f, in front of the lens.

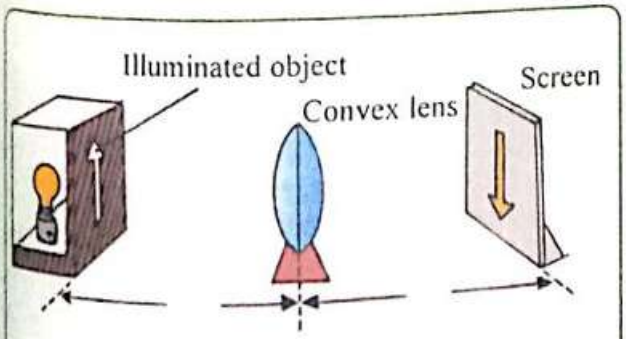


Fig. 13.46 To verify the lens formula for a convex lens

The position of the screen is adjusted on the other side of the lens until a sharp image of the object is obtained on the screen. The distance (v) from the screen to the lens is measured. The experiment is repeated for five other values of u. The results are tabulated as in table 13.3.

Table 13.3

u	v	$\frac{1}{u}$	$\frac{1}{v}$

A graph of $\frac{1}{v}$ against $\frac{1}{u}$ is plotted as shown in fig. 13.47.

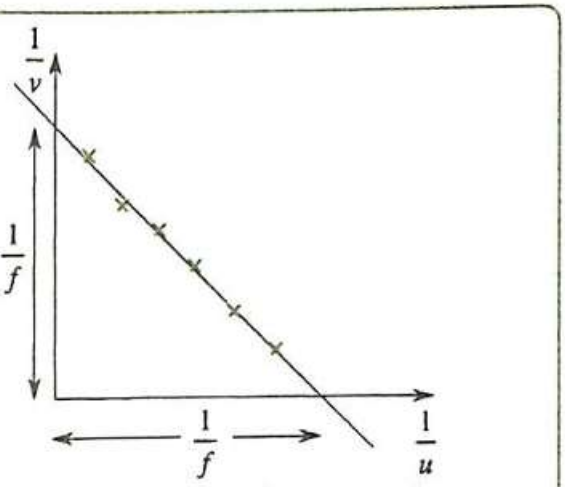


Fig. 13.47 Graph of $\frac{1}{v}$ against $\frac{1}{u}$

The intercept on either axis will be equal to $\frac{1}{f}$. Hence the reciprocal of the intercept gives the value of f. The use of the intercept to find f follows from the lens formula.

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

When $\frac{1}{u}=0$, then $\frac{1}{v}=\frac{1}{f}$. But $\frac{1}{u}$ equals zero on the $\frac{1}{v}$ axis. Similarly $\frac{1}{v}$ equals zero on the $\frac{1}{u}$ axis. So we can use the intercept on either axis to obtain f.

In the absence of a ray box, we can use two pins to obtain the focal length of the lens by the no-parallax method. The object pin is placed before the lens and its image is located by the no-parallax method using a search pin. (fig. 13.48) A set of values of u and v are obtained and their reciprocals plotted as shown in fig. 13.47.

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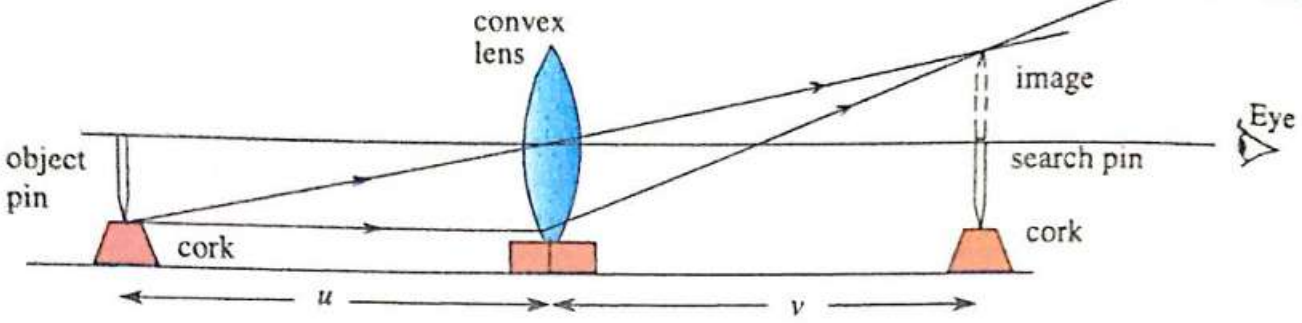


Fig. 13.48 Determination of focal length from u and v by the no-parallax method

In optical experiments, we should take such precautions as

(i) avoiding parallax in measuring with the metre rule,

(ii) ensuring a proper alignment of object, lens or mirror, and screen

(iii) ensuring a sharp image is properly focused on the screen

(iv) optical pins should be vertical and well spaced.

Activity 13.2

1 Obtain a converging lens from your school laboratory. Find the approximate focal length by focusing a distant object on a screen.

2 Stand an optical pin in front of the lens at a distance greater than the focal length. Use the method of non-parallax to locate the image position. Measure the object distance u, and the image distance v.

3 Repeat the experiment for various values of u and obtain the corresponding values of v.

4 Tabulate your readings as shown on the table:

u	v	$M = \frac{v}{u}$	$\frac{1}{u}$	$\frac{1}{v}$

5 Plot a graph of $\frac{1}{v}$ against $\frac{1}{u}$ and measure the intercept on both axis. Find the reciprocal of the intercept.

6 Plot values of M on the y-axis and values of v on the x-axis. Find the slope of the graph, and the intercept on the v-axis.

7 Compare the intercept on both axis of the first graph with the slope and the intercept on the v-axis of the second graph.

8 Interprete your results.

Activity 13.3

Place a rectangular glass block on a drawing paper and draw the outline $WXYZ$. Remove the block. Mark a point O along WX as shown and draw NO perpendicular to WX.

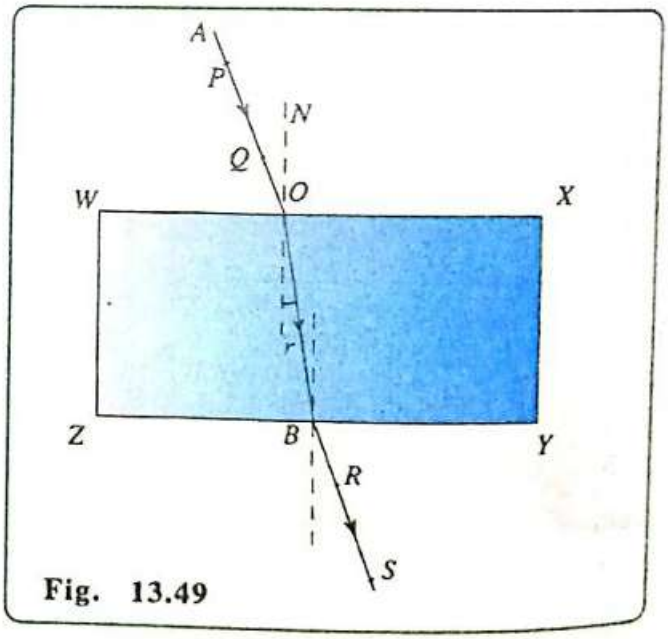


Fig. 13.49

Draw AO such that the angle of incidence i is 20°. Fix one pin at P and another at Q along AO. Replace the block. Fix one pin at R and another at S so that they are in a straight line with the pins at P and Q when viewed through the block. Remove the block and the pins. Join SR and produce it to meet ZY at B. Also join OB and measure the angle r. Repeat the experiment four more times with i taking values of 30°, 40°, 45°, and 50°, and find the corresponding values of r. Tabulate the readings.

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Plot a graph with ($\sin i - \sin r$) on the vertical axis and $\sin r$ on the horizontal axis. Calculate the slope of the graph.

State two precautions you took when performing this experiment.

Summary

1 When a ray of light travels from one medium to a denser medium, the ray will be refracted towards the normal. When it travels from a dense medium to a less dense medium, it will be refracted away from the normal.

2 Laws of refraction. (i) the incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane (ii) The ratio of $\frac{\sin i}{\sin r}$ is constant for a given pair of media (Snell's law).

3 The ratio $\frac{\sin i}{\sin r}$ is called the refractive index, n.

4 The refractive index of a medium is defined as the ratio

$$n = \frac{\sin i \text{ (in air or vacuum)}}{\sin r \text{ (in the medium)}}$$

5 $\frac{\text{Real depth}}{\text{apparent depth}} = n$ (refractive index) at near normal incidence.

6 The critical angle C is the angle of incidence in the denser medium when the angle of refraction in the less dense medium is 90°. If the less dense medium is air, the refractive index of the denser medium is $n = \frac{1}{\sin C}$.

7 Total internal reflection is the reflection of an incident ray of light at the interface between the medium of incidence and another medium of lower refractive index when the angle of incidence exceeds the critical angle. For total internal reflection to occur (i) light must be travelling from an optically more dense to an optically less dense medium. (ii) the angle of incidence in the denser medium must be greater than the critical angle.

8 The angle of deviation, d, of a light ray is the angle between the incident and emergent rays.

9 The separation of white light into its constituent colours is known as *dispersion*. The seven colours are Red, Orange, Yellow, Green, blue, Indigo, Violet. The colour of the light is related to the wavelength or frequency of the light.

10 A prism produces dispersion and deviation of colours at its first face, and further deviation at its second face. Diamonds achieve their brilliance from a combination of dispersion and total internal reflection.

11 A pure spectrum is one in which the colours are clearly separate or distinct from each other. Such a spectrum can be produced by using a bright source of white light, a narrow slit, two converging lenses, a prism and a screen.

12. The colour of an object depends on the colours in the light which falls on the object and also on the absorption and reflection of this light by the object. A red flower appears red when exposed to white light because it absorbs all the colours except red, but it appears black or dark when exposed to blue light.

13 The three primary colours are red, green and blue. Secondary colours obtained by the additive mixing of any two of the primary colours are yellow, cyan and magenta.

14 The principal focus of a lens is the point on the principal axis to which rays parallel and close to the principal axis converge after refraction through a converging lens or appear to diverge from after refraction through a diverging lens.

15 Focal length of a lens is the distance from the lens to the principal focus. The power of a lens is the reciprocal of the focal length expressed in metres. The unit is the Dioptre when the focal length is in metres. $1\ \text{D} = \frac{1}{f}\ (\text{m})$

16 The images formed by a convex (converging) lens may be either real or virtual depending on the position of the object. The images formed by a concave(diverging) lens are always virtual, erect and diminished for all positions of the object.

17 Lens formula:

Real-is-positive (i) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}; M = \frac{v}{u}$

New Cartesian (ii) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; M = \frac{v}{u}$

18 Focal length of a convex lens is +ve, focal length of a concave lens is –ve.

19 To distinguish between a convex lens and a concave lens without touching the lens, we bring a candle flame from a great distance towards the lens, looking at it through the lens. If the image is always erect or upright and smaller than the candle, the lens is a diverging lens. If the image is inverted and

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gradually increases in size as the candle is brought nearer the lens, then disappears at a stage and reappears as an enlarged upright image, then the lens is a converging lens.

Exercise 13

1 A transparent rectangular block 5.0 cm thick is placed on a black dot. The dot when viewed from above is seen 3.0 cm from the top of the block. Calculate the refractive index of the material of the block

A $\frac{2}{5}$ B $\frac{3}{5}$ C $\frac{3}{2}$ D $\frac{5}{3}$ E $\frac{5}{2}$

2 An object placed 36 cm from a converging lens of focal length 24 cm, forms a real image which is 6 cm high. What is the height of the object?

A 4. 0 cm B 3.0 cm C 2.0 cm
D 12.0 cm E 6.0 cm

3 An object is placed 20.0 cm from a converging lens. If the real image formed is 80.0 cm from the object, the focal length of the lens is:

A 15.0 cm B 30.0 cm
C 10.0 cm D 22.5 cm
E 16.0 cm

4 When a ray of sunlight passes obliquely through a rectangular glass block,

A it emerges without displacement parallel to the incident ray
B It gets dispersed into seven visible colours without any deviation at all.
C It deviates without dispersion.
D It gets laterally displaced, and the emergent ray is parallel to the incident ray.
E both dispersion and deviation take place.

(SC/GCE)

5 The direction of a light ray changes as it passes from one medium to another. The phenomenon is called

A diffraction B reflection
C dispersion D refraction
E deviation

6 The velocities of light in air and glass are 3.0×10^8 ms^{-1} and 1.8×10^8 ms^{-1} respectively. Calculate the sine of the angle of incidence that will produce an angle of refraction of 30° for a ray of light incident on glass.

A 1.2 B 1.0 C 0.8 D 0.6 E 0.3

(SSCE)

7 For the production of a pure spectrum, two converging lenses, a prism, and a light source are used. What is the function of the lens near the light source?

A To separate the light colours
B To produce parallel rays which are incident on the prism
C To diverge the light rays incident on the prism
D To converge light rays on to the prism
E To produce the dispersion necessary for the spectrum.

8 A ray of light travels from glass to water. What other condition is required for this ray to be totally internally reflected? The

A angle of incidence in the denser medium must be equal to 30°
B angle of incidence at the glass-water boundary must be greater than the critical angle for the two media
C incident ray must lie along the boundary of the two media
D ratio of the sine of the angle of incidence to the sine of the angle of refraction must be equal to the sine of the critical angle
E Incident ray must strike the boundary between the two media perpendicularly.

9 The image of an object formed by a lens is observed to be virtual, erect and enlarged. The arrangement of lens and object consists of a

A Concave lens with the object between the lens and its principal focus
B Convex lens with the object at a distance from the lens and its principal focus
C Concave lens with the object at a distance from the lens equal to the focal length of the lens
D Convex lens with the object between the lens and its principal focus
E Convex lens with the object at a distance from the lens equal to twice the focal length of the lens.

(SC/GCE)

10 A ray of light is incident at an angle of 30° on a glass prism of refractive index 1.5. Calculate the angle through which the ray is minimally deviated in the prism.
(The medium surrounding the prism is air).

A 10.5° B 19.5° C 21.1°
D 38.9° E 40.5°

(SSCE)

11 State the laws of refraction of light and describe how you can verify the laws in the laboratory using a rectangular glass slab and optical pins.

Explain with the aid of diagram why
(i) a straight pencil partly immersed in water appears bent.

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(ii) to an observer immediately above a pool of water, a coin at the floor of the pool appears nearer the surface than it really is.

A ray of light passes from air through a rectangular block of glass with parallel faces 4.5 cm apart at an angle of incidence of 52°. Find

a. the lateral displacement of the ray and
b. the angle of refraction.

Refractive index of glass = 1.5.

12 What is meant by *critical angle*, *total internal reflection*? Under what conditions does total internal reflection occur? Describe one phenomenon of total internal reflection in nature. Derive a relationship between the critical angle and the refractive index of a substance.

13 Explain with the aid of diagrams how a converging lens could be used to:

(i) ignite a piece of carbon paper
(ii) produce an enlarged picture on a screen
(iii) correct an eye defect.

(SSCE)

14 a. Define the terms: *principal focus* and *focal length of a converging* lens.

b. Distinguish between a real and a virtual image and draw ray diagrams to illustrate how a converging lens can be used to produce each type of image.

c. A small image is viewed through a converging lens held close to the eye. If the focal length of the lens is 10 cm and a virtual image of height 2 cm is formed 30 cm away from the lens, obtain by calculation or scale diagram:

(i) the distance of the object from the lens
(ii) the size of the object

d. Draw a ray diagram to show how two suitable lenses may be arranged to form a compound microscope.

(SC/GCE)

15 Explain what is meant by the *pure spectrum of light*. Draw a labelled ray diagram showing the arrangement of the apparatus necessary to obtain a pure spectrum of white light. Write down the function of each apparatus. Why does

(i) green grass appear green and
(ii) red glass appear red when looked through in sunlight.

16 a. Describe how you would determine the focal length of a converging lens by means of a plane mirror and a pin.

b. Explain why paper with a dull or mat surface is usually chosen for the pages of books.

c. An object 1 cm high is placed on the axis of a converging lens at a distance of 7.5 cm from the centre of the lens. If the focal length of the lens is 15.0 cm, find by scale drawing the position, height, magnification and the nature of the image to be expected.

(SC/GCE)

17 Distinguish between a real image and a virtual image. Draw diagrams to show how a converging lens may be used to produce each type of image. Describe an accurate method to determine experimentally the focal length of a converging lens.

An object is placed between a converging lens and a plane mirror. Explain how two real images of the object may be produced by the system. If the focal length of the lens is 15 cm and the object is 20 cm from both the lens and the mirror. Calculate the distances of the two images from the lens.

(SC/GCE)

18 (i) Find the position, magnification and nature of the image formed by a lens of power +5 *Dioptres* when the object is 20 m away from it.

(ii) What is the power in dioptres of

a. converging lens of focal lengths 2 m, 0.5 m, 50 mm
b. diverging lens of focal lengths 0.4 m, 0.1 m, 80 mm?

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19 Complete the following table:

	Magnification	Nature of image	Focal length in m	Object distance in m	Image distance in m
1	2	real	0.2		
2	2	virtual	0.2		
3	1.5	real		0.2	
4	3	virtual		0.1	
5	2.5	real			0.5
6	2.5	virtual			0.5

20 a. Define *refractive index*. Describe an experiment to find the refractive index of water.

b. A coin is placed at the bottom of a pond 3.00 m deep. Calculate the apparent displacement of the coin when it is viewed directly from above (Refractive index of water $= \frac{4}{3}$)

c. Distinguish between (i) *dispersion*, *refraction* and *deviation* (ii) *Real* image and *virtual* image. In what two positions will a converging lens of focal length 7.5 cm form images of an object on a screen 40 cm from the object?

21 (i) Explain what is meant by *refraction* as it applies to light.

(ii) A ray of light incident normally on face *AB* of the prism *ABC* strikes face *AC* of the prism. Draw a ray diagram to show the path of the ray as it passes through the prism.

(iii) State the conditions necessary for the incident ray in the experiment to be totally internally reflected. On what face will this occur? [Take the refractive index of glass as 1.5].

(SSCE)

22 (a) (i) State the laws of refraction of light.

(ii) Describe an experiment to determine the refractive index, n, of the material of an equilateral triangular glass prism using the minimum deviation method.

(b) A rectangular glass prism of thickness 12cm is placed on a mark on a piece of paper resting on a horizontal bench.

(i) Draw a ray diagram to show the apparent position of the mark in the glass prism.

(ii) If the refractive index of the material of the prism is 1.5, calculate the apparent displacement of the mark.

(SSCE)

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14 APPLICATION OF LIGHT WAVES

This chapter discusses the application of light waves, and the application of the properties of lenses in such instruments as the human eye, the camera, the microscope, the projector and the telescope.

At the end of the chapter the student should be able to:

1. Construct a model of a box camera.
2. Explain the optical principles involved in the snapshot camera.
3. Set up a single lens project and use it to project a film strip on a screen.
4. Explain the formation of images by the camera and the projector by tracing rays of light through them.
5. Explain the role played by some parts of the eye in the formation of image on the retina.
6. Compare and contrast the eye and the camera.
7. State the defects of the eye and their causes.
8. Identify the type of lenses for correcting the various defects of the eye.
9. Trace the paths of light rays through simple and compound microscopes and telescopes.

14.1 The Simple Camera and Projector

a. The Simple Lens Camera

The simple lens camera consists of a light-proof box with a converging lens in front and a light sensitive film at the back. The function of the lens is to focus images of the object to be photographed onto the light-sensitive plate or film. There is a provision for adjusting the distance between the lens and the film, so that objects in front of the lens can always be sharply focused on the film. Sometimes the convex lens is replaced by a system of lenses. The simple lens camera has a fixed distance between the lens and the film. This distance is equal to the focal length of the lens. Some cameras have *bellows* or focusing rings by which the distance between the lens and the film can be adjusted.

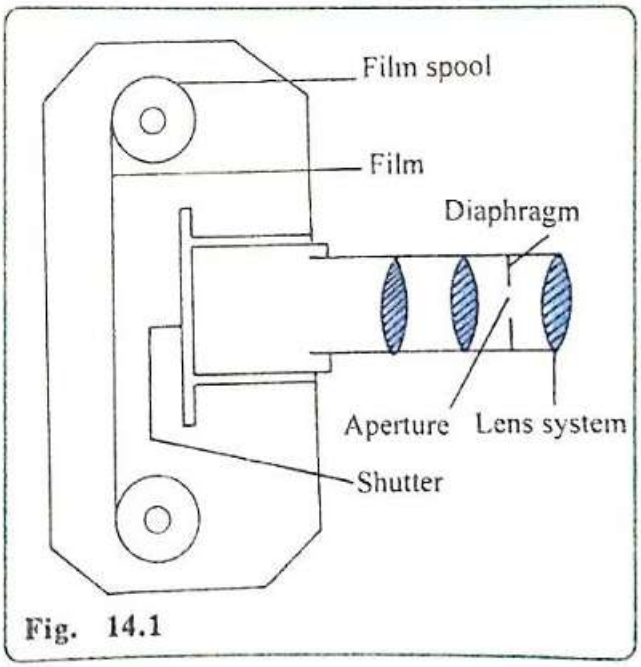
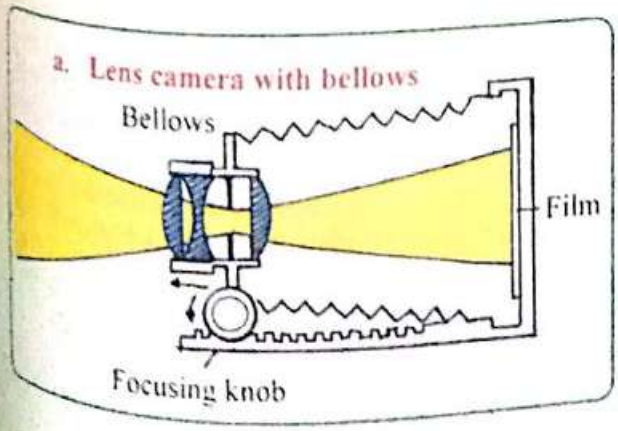


Fig. 14.1



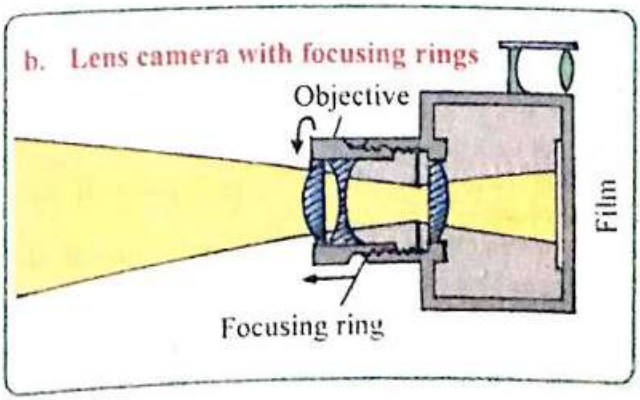


Fig. 14.2 The simple lens camera

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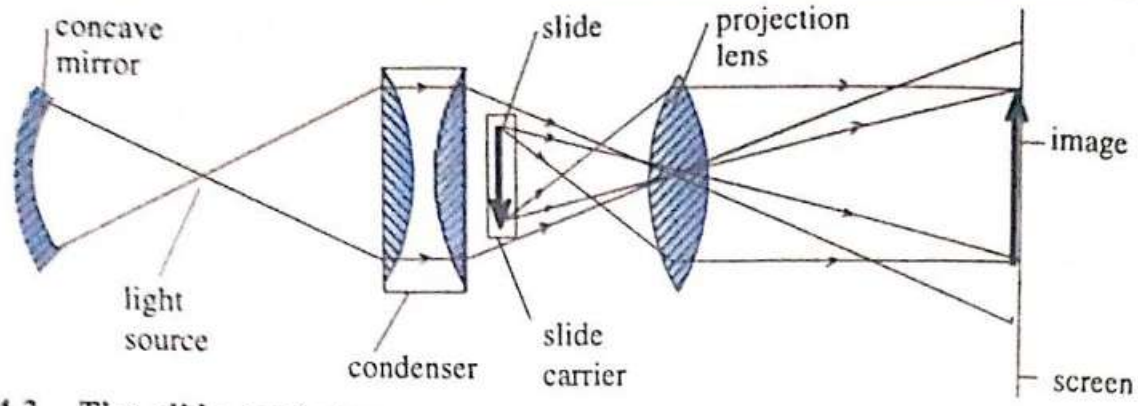


Fig. 14.3 The slide projector

The *diaphram* and the shutter control the amount of light entering the camera. The diaphram regulates the size of an aperture or opening which controls the amount of light energy sensitising the film. The shutter, of variable speed, between the lens and the film controls the time interval during which light is allowed to fall on the film. This time interval is called the *exposure time*.

The process of making a photograph begins and ends with light. Rays of light enter a camera and are focused into an image. The light exposes the film in the camera causing chemical changes on the film's surface. The exposed film is then treated with certain chemicals in a procedure called *developing*. Finally light is used to make a print by transferring the image from the film to a sheet of special paper.

To take a photograph, the camera lens is pointed at the object and the focusing ring is used to adjust the distance of the lens from the film until a sharp image is seen on the film. When the button is pressed, the shutter quickly opens and closes, exposing the film, for a brief period, to light from the object. After the photograph is taken the film is wound on between two spools until it is all used up.

b. The Projector

The projector is an instrument for projecting on a screen an enlarged image of a transparent object e.g. a slide. Fig. 14.3 shows the essential parts of a projector and their optical arrangements.

The functions of the various parts of the projector are:

1 The small but *powerful light source* to illuminate the object which is usually non-luminous. A converging mirror placed behind the light source helps to direct the light onto the film or slide.

2 The *condenser* consisting of two plano-convex lenses collects the light from the light source and concentrates it on the slide or transparency and illuminates every part of it strongly and evenly. Between the two lenses is usually an infra-red filter to prevent the passage of heat to the slide which might otherwise melt it.

3 The *slide carrier* is a framework in which the slide is placed upside down, so that it will appear erect on the screen.

4 The *projection lens* placed near the slide produces a real enlarged and inverted image of the slide and focuses it on to the screen. The lens is at a distance of between f and $2f$ from the object, where f is the focal length of the lens.

5 The *screen*, usually white, receives the image which is the right way up if the object is inserted upside down into the slide carrier.

The principle of the slide projector is used in cine-projectors, photographic enlargers and several other devices.

14.2 The Human Eye

Fig 14.4 shows a vertical section through the human eye. Like the camera the eye has a refracting system containing a converging lens which forms an image of an object in front of the eye on a light sensitive layer at the back of the eye. The essential parts of the eye and their various functions are:

1 The *Cornea* - the thick transparent bulge in the front part of eye. It serves as a protective covering in front of the lens and allows light into the eye. It also partly focuses the light entering the eye.

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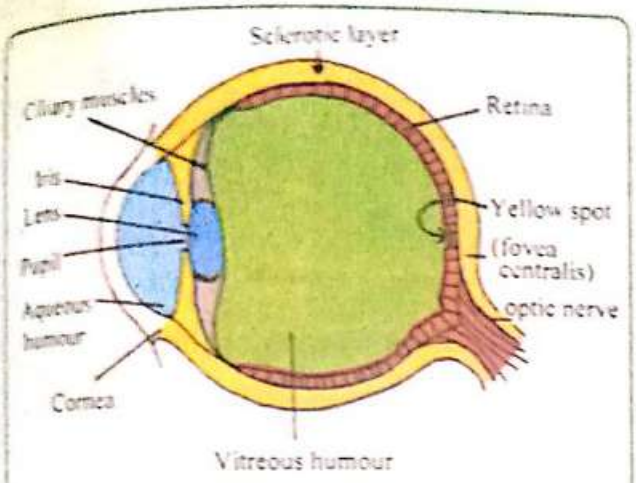


Fig. 14.4 Optical features of the eye

2 The *Aqueous humour* is the transparent liquid between the cornea and the lens and the vitreous humour is a jelly-like liquid between the lens and the rest of the eyeball. These liquids serve mainly to keep the eye in its spherical shape.

3 The *Iris* is behind the cornea. The iris is the part which gives the colour of the eye. It acts as a stop or diaphram of variable size. It has a tiny opening at its centre called the *pupil* which regulates the quantity of light entering the eye. The pupil looks black because the inner layer of the eye, the *choroid*, is black.

4 The *Crystalline lens* which focuses the light from the object on to the light sensitive retina.

5 The *Ciliary muscles* which support the lens and attach it to the wall of eye. These muscles by their contraction and expansion, alter the focal length of the lens changing its shape.

6 The *Retina* is the light sensitive area of cells at the back of the eye. The image is formed on the retina which consists of light sensitive nerve endings which are connected to the *optic nerve*. The *optic nerve* conveys the sensation of sight to the brain.

7 The *Yellow spot* is the most sensitive spot of the retina. It is here that light entering the eye are usually brought to a focus and here the clearest image is formed.

8 The *Blind spot* is the point of exit of the optic nerve from the retina. It is insensitive to light.

9 The *Sclerotic layer* is the outer covering of the eye.

The optical system of the eye consists of the cornea, the lens, and the aqueous and vitreous humour. They operate to form a real and inverted image of an external object on the retina. The retina transmits the impression created on it by this image to the brain through the optic nerve. The brain then interpretes the impressions.

Accommodation

The eye has an adjusting mechanism for focusing the image of an object on to the retina. It adjusts the focal length of the crystalline lens through the action of the ciliary muscles which control the curvature and hence the focal length of the lens. Thus through this mechanism both near and far objects can be focused at the retina. This ability of the human eye is known as *accommodation*.

Accommodation is the ability of the eye to alter or adjust the focal length of its lens so as to form clear images of objects at different distances on its retina. This adjustment is brought about through the action of the ciliary muscles.

To see objects close to the eye, the ciliary muscles increase the curvature of the lens, thereby shortening the focal length. The divergent rays from the near object are thus bent more so as to come to a focus on the retina.

Parallel rays from far objects are focused on the retina. Therefore the normal eye is much relaxed and the lens is slim when viewing far objects.

Binocular vision

Because our eyes are spaced slightly apart, each views an external object from a slightly different position. Therefore the images of the object falling on the retina are not exactly the same. The right eye sees more of the right side of the object, while the left eye sees more of the left side.

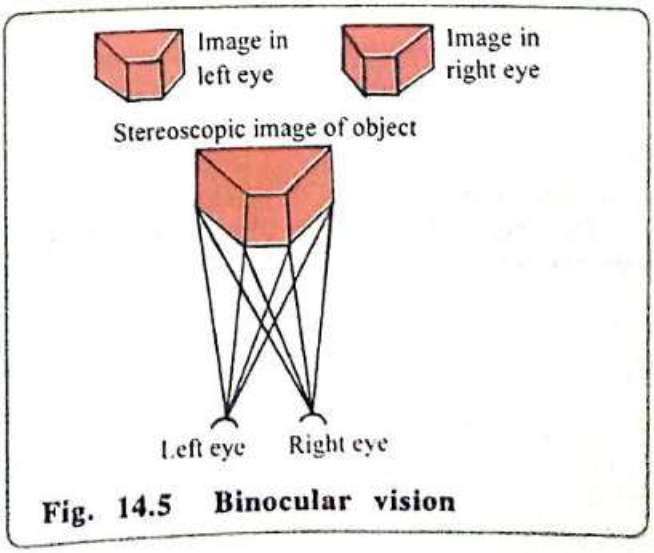


Fig. 14.5 Binocular vision

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When the two images are combined by the brain, the proper perspective is interpreted.

Binocular vision is the overlapping of the two images formed by both eyes which gives an impression of depth and solidity and makes it possible for us to see things in relief. Such a proper perspective of an object is impossible with one eye.

Binocular vision is possible only if the fields of view of the two eyes overlap. This is not possible in some birds because of the location of the eyes.

Persistence of Vision

When light from an object falls on the retina, it excites it and the clear image formed on the retina leaves a sensation of vision which persists for a short but definite length of time even after the light is removed. This retention of the vision of the image is known as *persistence of vision*. If a light flickers at very short intervals of about ten times per second, it appears to the eye as a continuous source of light owing to the persistence of vision.

This phenomenon is made use of in the *showing* and production of motion pictures, which are a rapid succession of still pictures which are cast on the screen. Persistence of vision makes the still pictures appear to have continuity.

Normal Vision and the Defects of Vision

Near Point and Far Point The nearest point at which an object can be clearly seen is called the *near point*. The farthest point at which an object can be clearly seen is called the *far point*.

For a normal eye, the near point is about 25 cm from the eye. This is the limit to which the ciliary muscles can operate, for at this point they are strained to a maximum, and the accommodation is, therefore the greatest. The distance from the near point to the eye is known as the *least distance of distinct vision*.

For the normal eye, the far point is at infinity. At this point the ciliary muscles are most relaxed and the lens is slim.

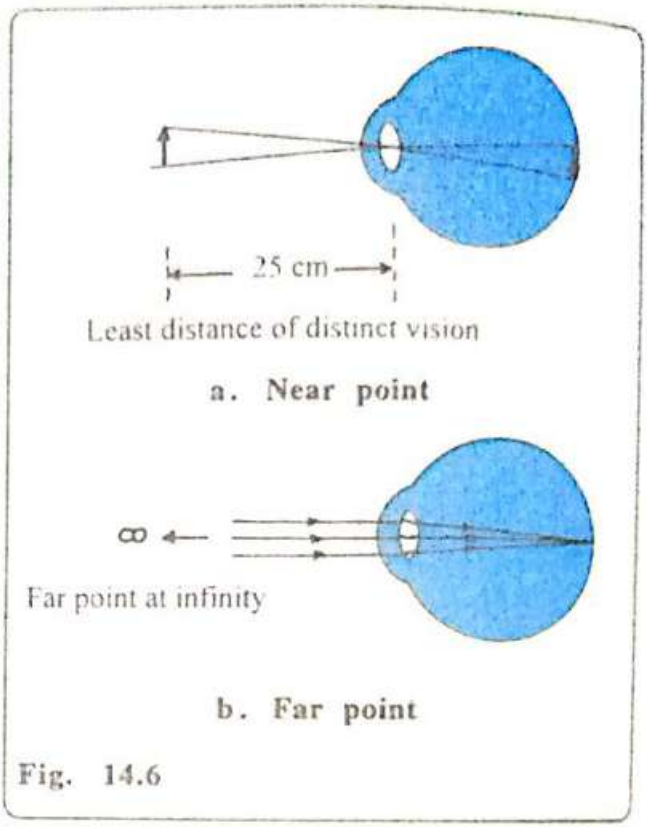


Fig. 14.6

The accommodation is also the least.

Some people whose eyes are not functioning normally are said to have some *defects of vision*. Two of the most common types are the *long sight* and the *short sight*. Both defects arise when the eye lens is unable to accommodate effectively.

Long sight (hypermetropia) and its Correction

A long sighted person can see far object but cannot see near objects clearly. His near point is farther away than 25 cm, the normal near point. An object at the normal near point N cannot be clearly seen by this eye because the rays from the object are brought to a focus at a certain distance behind the retina (fig 14.7a) This is caused by the eyeball being too short or the crystalline lens not being sufficiently convergent.

The defect is corrected with the aid of a suitable converging lens placed in front of the eye (fig 14.7b). The converging lens reduces the divergence of the rays entering the eye from an object just sufficiently to make the rays appear as if they originated from B, the near point of the defective eye. Hence a clear image of the object is formed on the retina.

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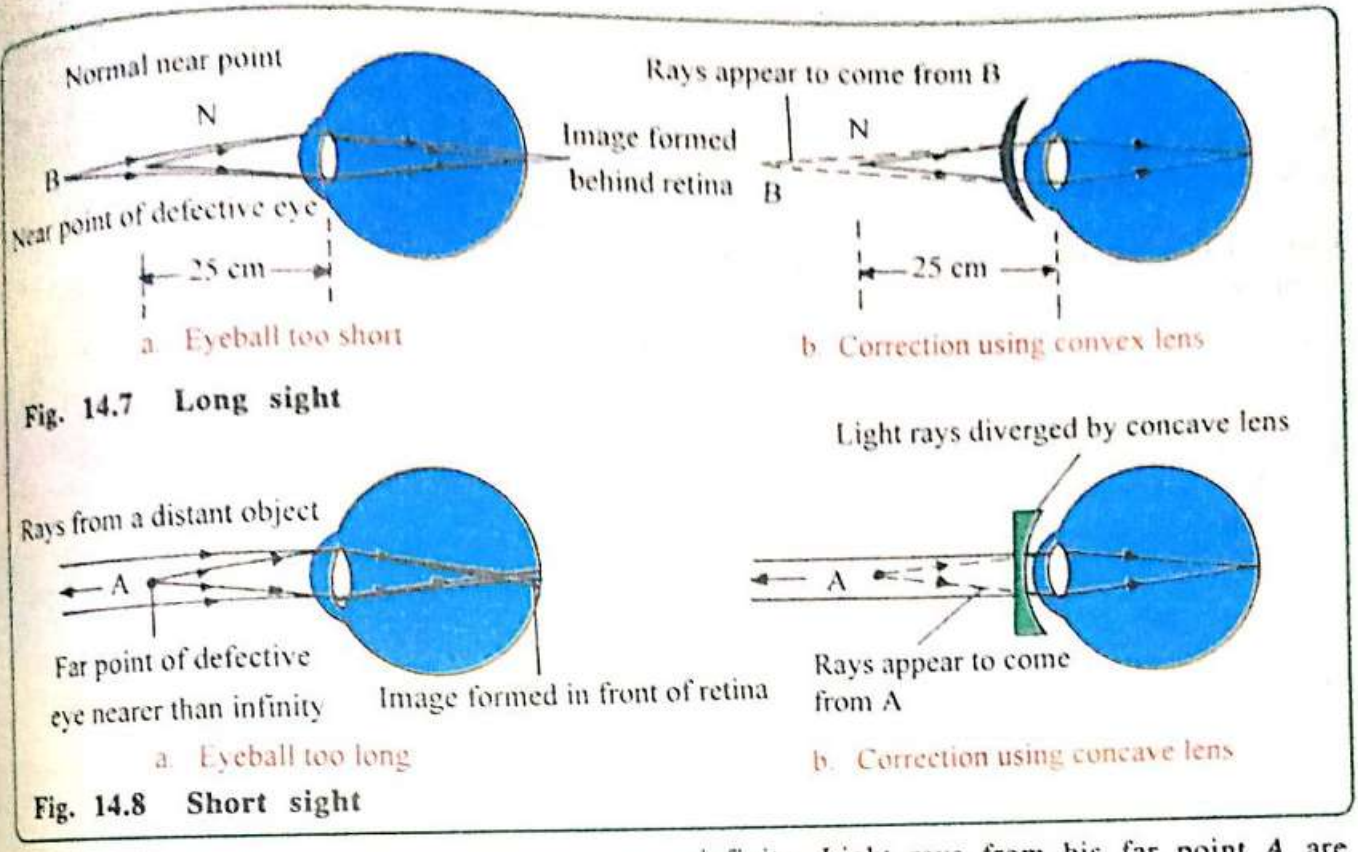


Fig. 14.7 Long sight

Fig. 14.8 Short sight

Example 14.1

Find the focal length of the lens required to correct the defect of vision of a man whose near point is 50 cm. Assume the least distance of vision of the normal eye to be 25 cm.

Solution

For this man to see objects clearly at 25 cm from the eye, the image must *appear* to be at 50 cm.

Hence $u = 25$ cm, $v = -50$ cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{-50} + \frac{1}{25}$$

$$\frac{1}{f} = \frac{1}{25} - \frac{1}{50} = \frac{1}{50}$$

$f = 50$ cm

He needs a convex lens of focal length 50 cm.

Short Sight (myopia) and its correction

A short sighted person cannot see distant objects clearly. This is because his eyeballs are too long or the crystalline lens is too converging that rays from distant objects are brought to a focus in front of the retina instead of at the retina. His far point at A is thus less than the normal far point which is at infinity. Light rays from his far point A are brought to focus at the retina. Therefore objects at A can be clearly seen, by the defective eye.

Short sight can be corrected by placing a suitable concave or diverging lens in front of the eye. The action of this lens is to diverge light rays entering the eye from a distant object and make them appear to be coming from the person's far point A (fig 14.8b).

Example 14.2

A short sighted man cannot see clearly objects beyond 90 cm from his eye. How can this defect be corrected?

Solution

The defect of short sight is corrected by the use of diverging lens. The image of a distant object must be made to *appear* to be at 90 cm in order to be seen.

Hence $u = \infty$ (infinity)

$v = -90$ cm

Using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{-90} + \frac{1}{\infty}$

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Since $\frac{1}{\infty} = 0$, we have $\frac{1}{f} = \frac{1}{-90}$

$f = -90$ cm

Correction is done with the use of a diverging lens of focal length 90 cm

Comparison of the Human Eye and the Camera

From our previous discussions we note the following similarities and differences between the human eye and the camera.

A. Similarities

(i) The camera includes light-tight box painted black inside. The human eye is impregnated with black pigments within.

(ii) The pupil in the eye serves the same purpose as the aperture of the camera.

(iii) Both the eye and the camera have converging lens to focus rays from an external object.

(iv) Both have light sensitive material on which the rays are focused, the retina in the eye and the film in the camera.

(v) The iris in the eye serves the same function as the diaphram in the camera, to regulate the amount of light entering the equipment.

B. Differences

(i) The Human eye has the power of *accommodation*, the ability of the lens to focus points at different distances on the retina by altering the focal length of the lens. Hence the eye lens has a variable focal length but the focal length of the camera lens is fixed.

(ii) The distance between the lens and the retina is fixed in the human eye. In the camera the distance between the lens and film can be varied.

(iii) The eye can suffer from defects of vision unlike the camera.

(iv) The eye is a biological organ while the camera is a mechanical device, leading to other differences.

14.3 Microscopes and Telescopes

a. The Simple Microscope (or Magnifying Glass)

The magnifying glass or simple microscope is a convex lens which is used to produce magnified images of small object. It is used for reading small print and for studying biological specimens.

When a converging lens is used as a simple microscope, the object is placed between the principal focus and the optical centre of the lens as shown in fig. 14.9. With this arrangement, a magnified, virtual and erect image is formed.

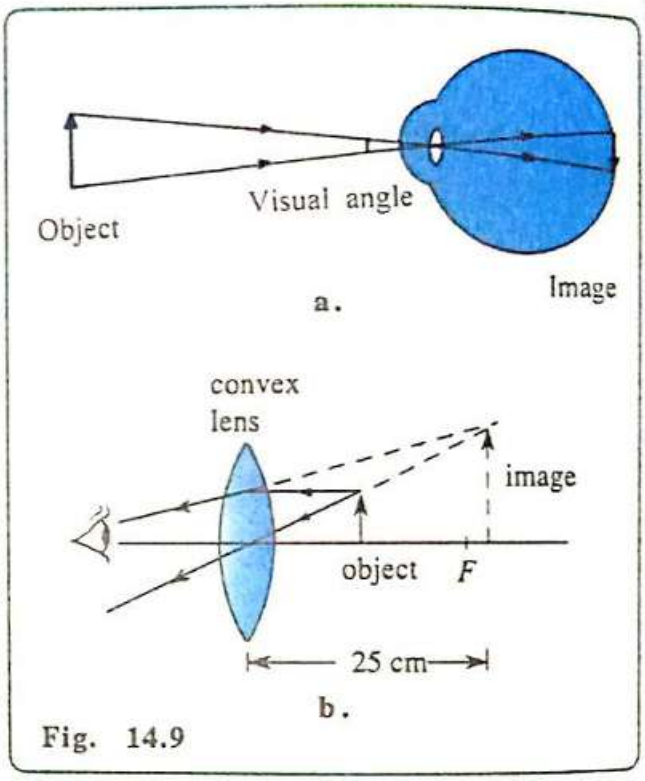


Fig. 14.9

The lens is moved until the image is seen distinctly at 25 cm from the normal eye. This image subtends a larger visual angle than that produced when the object itself is placed 25 cm from the unaided eye, and therefore appears bigger.

b. The Compound Microscope

To produce a higher magnification than that obtained from a simple microscope, a combination of two convex lenses is used in an arrangement called the compound microscope.

The compound microscope consists of two converging lenses of short focal lengths. The lens (O) nearer this object is known as the *objective* lens and the lens (E) through which the final image is seen is known as the *eyepiece*. The eyepiece has a comparatively larger focal length than the objective lens. The objective lens produces at I_1 a magnified, inverted and real image of an object AB placed at a distance u greater than its focal length (f_0). The eyepiece is adjusted until I_1 falls at a distance nearer to it than its principal focus, F_e.

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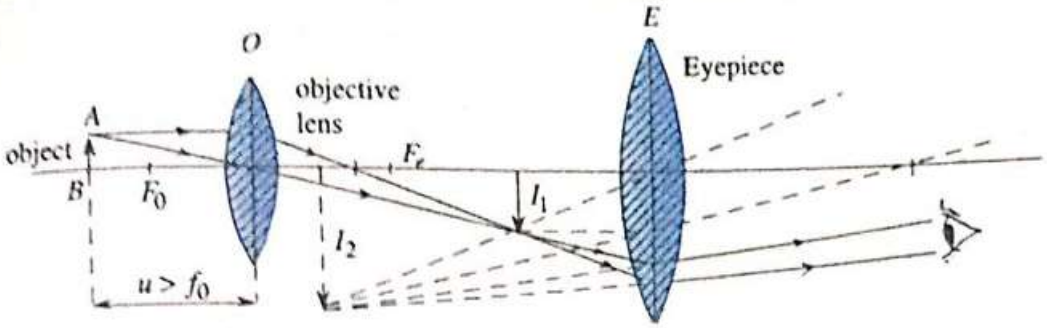


Fig. 14.10 Compound microscope

A magnified but virtual image I_2 is thus formed at I_2 with the eyepiece lens acting as a magnifying glass for an object at I_1.

Hence the action of two lenses produces a magnified but inverted image of an object.

The observer adjusts the eyepiece so that the final image is at the near point (about 25 cm for a normal eye).

The *magnifying power M* of the compound microscope is the product of the magnifications due to the two lenses. Thus $M = M_1 \times M_2$ where M_1 is the magnification produced by the objective lens and M_2 is the magnification produced by the eyepiece.

c. The Astronomical Telescope

The telescope is used in viewing distant objects, such as the stars and the planets. The simplest telescope is the *astronomical telescope*.

The astronomical telescope uses two converging lenses, the objective lens and the eyepiece respectively. The objective lens has a long focal length whereas the eyepiece has a short focal length.

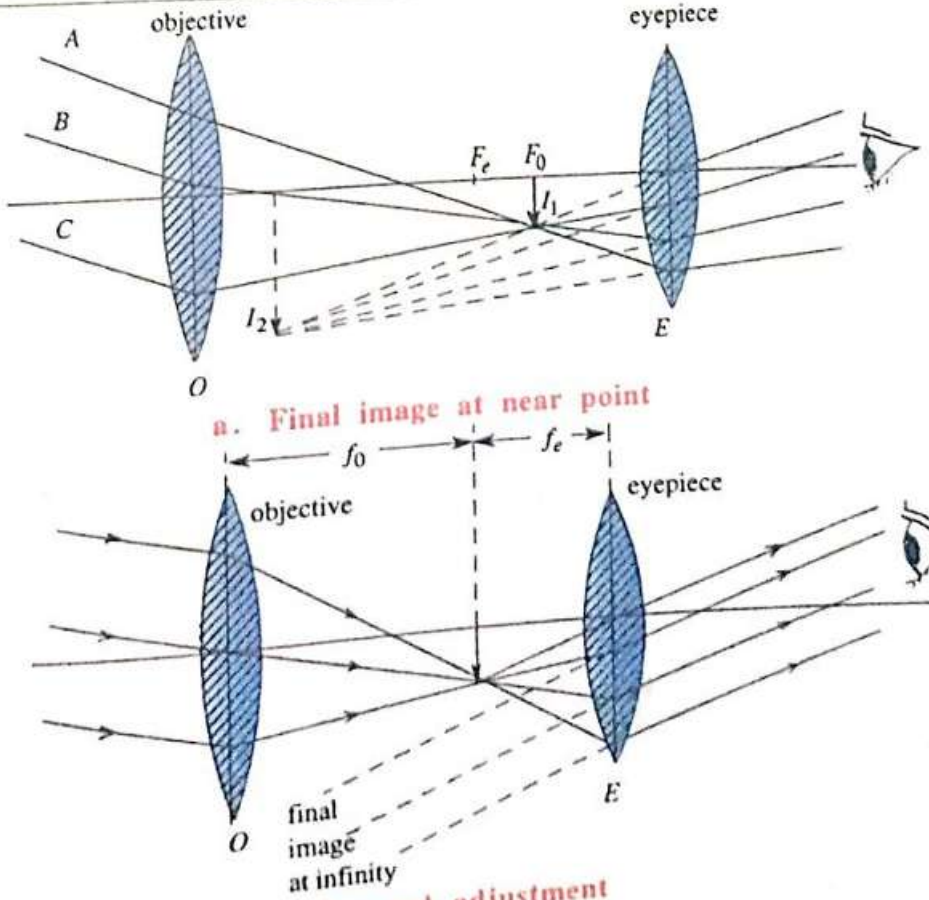


Fig. 14.11 Astronomical telescope

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The objective lens collects parallel rays from a distant object and forms a real inverted and diminished image I_1 at the principal focus (F_0) of the lens. The eyepiece is adjusted so that the image I_1 of the object falls at a point nearer the eyepiece lens than its focus F_e. The eyepiece then acts as a magnifying glass and forms a magnified and virtual image of I_1 at I_2. (fig 14.11a) The lenses can be arranged so that the final image is at infinity. In such an arrangement the principal focus of the objective must coincide with that of the eyepiece (fig 14.11b). In this arrangement the telescope is said to be in normal adjustment. Thus in normal adjustment, the image I_1 is made to fall at the principal focus F_e of the eyepiece which also corresponds with F_0 of the objective.

The astronomical telescope produces an inverted image. This can be tolerated when looking at stars, but is a disadvantage on the earth. The telescope owes its magnifying power to the fact that the angle subtended at the eye by the final image at infinity is very much greater than that subtended by the distant object.

d The Galilean Telescope

This type of telescope uses a convex lens of long focal length as the objective and a concave lens of short focal length as the eyepiece. The two lenses are mounted so that they have a common axis with the distance between them equal to the difference between their focal length (fig. 14.12). This is the Galilean telescope in normal adjustment.

The objective collects parallel rays from a distant object to form a real inverted image at its principal focus. However, before these rays actually converge to this principal focus, they are intercepted by the diverging lens which causes the rays to diverge and emerge as a parallel beam. A virtual, erect and magnified image is therefore formed at infinity.

The magnifying power of the telescope is given by

$$M = \frac{\text{focal length of objective}}{\text{focal length of eyepiece}} = \frac{f_0}{f_e}$$

For high magnification therefore we need an objective of long focal length and an eyepiece of short focal length, as in the case of the astronomical telescope.

The Galilean telescope can also be adjusted so that the final image forms at the near point (fig. 14.13). At this instance the telescope is not in normal adjustment.

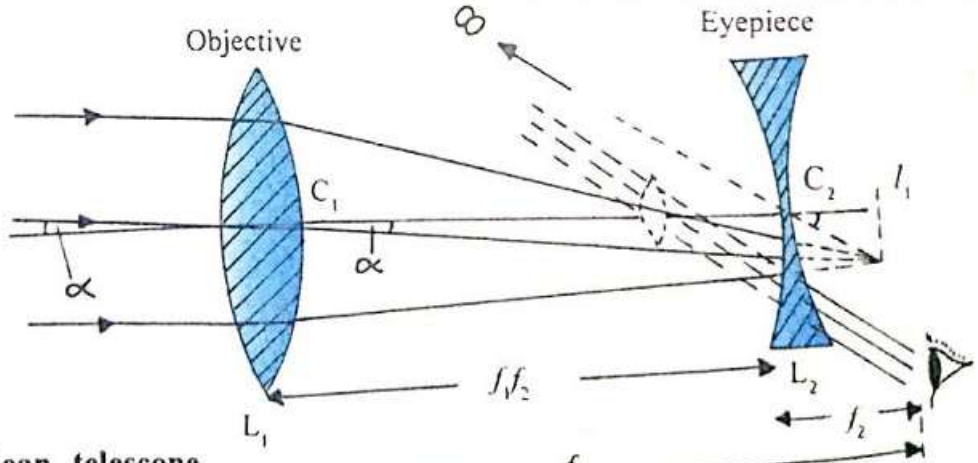


Fig. 14.12 Galilean telescope

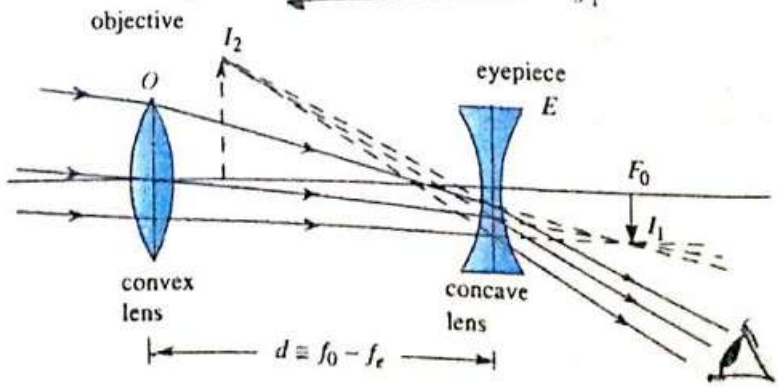


Fig. 14.13

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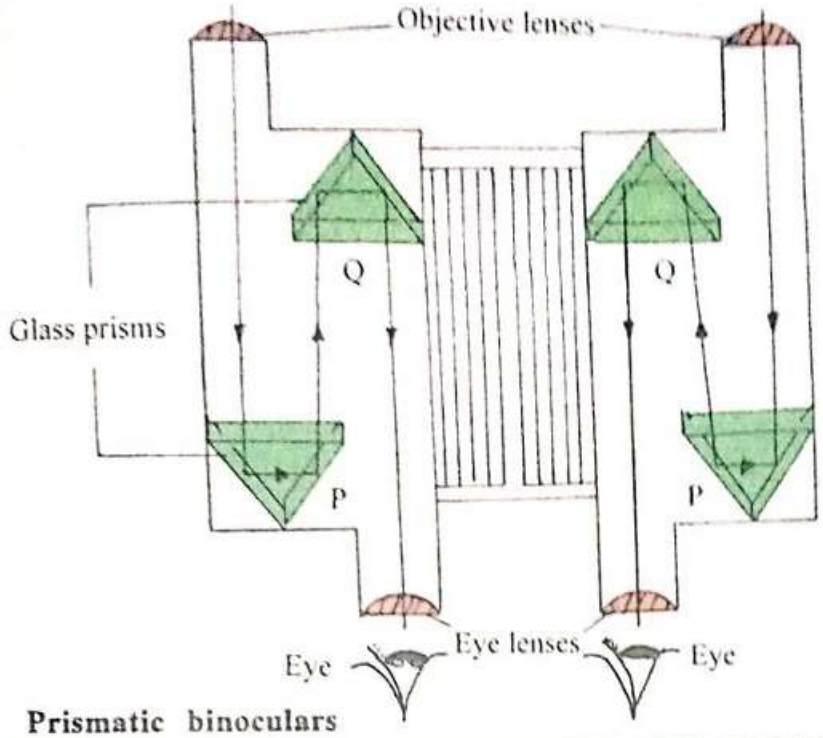


Fig. 14.14 Prismatic binoculars

Prism Binoculars

They are essentially a pair of astronomical telescopes each of which is fitted with two right angled isosceles prisms. The light passing through the objective is reflected from one prism P to another Q and then emerges through the eyepiece as shown in fig. 14.14.

The function of the prisms is thus to re-invert the image formed by the objective and present an erect image of the object to the eyepiece which subsequently magnifies it. An erect and magnified image is thus presented to the viewer.

The prism binoculars as well as presenting an upright image are also very compact since they reduce the optical path of the light ray from the objective to the eyepiece, very appreciably. Also because the binoculars are usually made in pairs (fig. 14.14:) they have an additional advantage of bringing the rays from two telescopes inwards to match the distance apart of the two eyes.

Summary

1 The simple lens camera consists of a light-tight box, a converging lens which focuses light from an object on to a light-sensitive film. The diaphram and the shutter control the amount of light entering the camera.

2 The human eye has a converging crystalline lens which focuses light from an object on to a light sensitive retina. The focal length of this lens can be adjusted by the action of the ciliary muscles.

3 *Power of accommodation* is the ability of the eye lens to focus points at different distances at the retina through the action of the ciliary muscles.

4 The vision afforded by the overlapping of images formed by both eyes is called *Binocular vision*. Such a vision enables us to have a proper perspective of an object or to see an object in relief.

5 The *Persistence of vision* is the persistence of a sensation of vision even after light from an object has been switched off.

6 A short-sighted person cannot see distant objects clearly. It is corrected by a suitable diverging lens. A long sighted person cannot see near object clearly. It is corrected by using a suitable converging lens.

7 The simple microscope has a single converging lens of short focal length. The compound microscope consist of two converging lenses of short focal length. The eyepiece has a longer focal length than the objective.

8 The astronomical telescope consists of a converging objective lens of long focal length and a converging eyepiece of short focal length. At normal adjustment the image is formed at infinity.

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9 The Prism Binoculars are in reality a pair of twin telescopes mounted side by side, one for each of the two eyes. Each binocular consists of two converging lenses and a pair of total reflecting prisms which invert the rays to give erect images.

Exercise 14

1 For correcting long sight defects in the human eye we require a
A converging lens B diverging lens
C microscope D periscope
E plain glass sheet.
(JAMB)

2 Which of the following optical instruments does not make use of a lens?
A Projector B Periscope
C Eye D Microscope
E Telescope
(SC/GCE)

3 Which of the following lens combinations is the most suitable for the construction of an astronomical telescope?
A The objective and eyepiece should be converging lenses, each with small focal length.
B The objective and eyepiece should be diverging lenses each with large focal length.
C The objective should have a larger focal length than the eyepiece and both should be converging.
D The objective should have ~ small focal length than the eyepiece and both should be diverging.
E The objective should be converging and have a larger focal length than the eyepiece which should be diverging.

4 Binocular vision
A restricts the field of view
B enables a person to see further
C enables objects to be seen in relief
D enables objects to be seen more clearly
E enables vision of object to persist after the light is switched off.

5 Which of the following is a correct statement about a compound microscope
A The magnification which can be produced is unlimited
B The image produced by the eyepiece is real
C The eyepiece has a shorter focal length than the objective
D The image produced by the objective lens lies inside the focal length of the eyepiece
E It uses a converging and a diverging lens.

6 The ability of the eye to focus objects at different distances is called
A Power
B Accommodation
C Normal vision
D Persistence of vision
E long sight.

7 What part of the camera corresponds to the iris of the eye?
A Shutter B Film C Lens
D Diaphram E Focusing ring.

8 At which of the following distances from the lens should a slide be placed in a slide projector if f is the focal length of the projection lens?
A Less than f B Greater than $2f$.
C Greater than f but less than $2f$
D Equal to f E Equal to $2f$.

9 Draw a labelled diagram of the vertical section of the human eye. Compare and contrast the structure and mode of operation of the human eye and the camera.

10 Draw a ray diagram to show how two suitable lenses may be arranged to form (a) a compound microscope (b) an astronomical telescope. Draw the paths of rays from the object to the image in both cases. State the relationship between the focal lengths of the two lenses in each case.

11 With the aid of a ray diagram, illustrate the action of a compound microscope. Rays should be clearly distinguished from construction lines.

A compound microscope has lenses of focal length 1.0 cm and 5.0 cm. An object is placed 1.2 cm from the objective lens. If a virtual image is formed 25 cm from the eyepiece, calculate the distance between the two lenses.

12 a. What is meant by the terms *short sight* and *long sight*, *least distance of distinct vision*.
b. Illustrate with the aid of diagrams how these defects can be remedied by the use of suitable lenses.
c. A longsighted person cannot see clearly objects closer to the eye than 75 cm. Recommend a suitable correcting lens for him to enable him read a book at a distance of 25 cm. What is the power of such a lens in dioptres.
d. A certain shortsighted person cannot see clearly objects beyond 50 cm from the

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eye. What are the focal length and the power of the lens he requires. What type of lens is it?

13 Explain with the aid of a diagram how a projector forms an erect image of a film on a distant screen. What adjustment is made to obtain clear pictures when the distance of the screen from the projector is increased?

An optical projector is used to project an image of a slide 50 mm square on to a screen placed 1.2 m from the lens. The image formed on the screen is 600 mm square. Find the focal length of the projecting lens of the projector.

Explain why (a) the slide is placed in the holder upside down (b) a condensed bright source of light must be used.

14 a. Draw the optical arrangement of an astronomical telescope in normal adjustment showing the positions of the principal foci of the lenses used and the paths of two rays from the top of a distant object through the instrument to the observer's eye.

b. Draw a ray diagram to show how two suitable lenses may be arranged to form a compound microscope. A microscope has an objective of focal length 1.0 cm and an eyepiece of focal length 8.0 cm. Both lenses are separated by a distance of 12.0 cm. A tiny object 1 mm long is placed 1.2 cm from the objective. Find by graphical or analytical method, the position and size of the final image.

15 a. Explain the following terms:
Binocular vision, Persistence of vision, Accommodation, Near point, Far point.

b. How does the human eye adjust itself to deal with
(i) Light of different intensity
(ii) objects at varying distances?

c. Draw a ray diagram to show how a converging lens placed near the eye may be used as a magnifying glass.

A magnifying glass gives a 4 times enlarged image at a distant of 25 cm from the lens. Find the focal length of the lens.

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Sound is a form of energy produced by vibrating bodies. For example, when we strike a tunning fork, its prongs vibrate. These vibrations give rise to sound waves.

This chapter will discuss the production and transmission of sound, characteristics of sound, forced vibration, speed of sound in different media, the measurement of the velocity of sound and the principle and types of musical instruments.

At the end of the chapter the student should be able to:

1 Identify the vibrating sources when sound is produced.
2 Distinguish between (a) noise and music (b) intensity and loudness (c) pitch and frequency as applied to sound.
3 Explain forced vibration and explain how it is used to amplify a sound.
4 Use the relationship $V = f\lambda$ in solving numerical problems.
5 Explain and discuss the production of standing waves in stretched strings.
6 Use resonance tube to determine the velocity of sound in air.
7 Classify musical instruments into (a) wind instrument (b) percussion instrument, (c) string instrument.
8 Explain the physical principles involved in the use of wind, string and percussion instruments.

15.1 Production and Transmission of Sound

Sound is produced by vibrating bodies. The vibrations cause the surrounding air to vibrate also producing a disturbance of the air. This disturbance travels out from the source of vibration in the form of *longitudinal waves*.

Almost any object can vibrate and hence be a source of sound. Later in the chapter we shall discuss some simple sources of sound, particularly musical instruments. In such instruments the source is set into vibration by striking, plucking, bowing, or blowing. These vibrations produce the sound from these instruments.

We can demonstrate the production of sound in the laboratory using a tunning fork. The tunning fork has two steel prongs which vibrate when struck with a hard rubber giving some sound. During the vibration the prongs of the tunning fork present a hazy appearance due to their rapid to and fro movements. If the vibrating prongs are dipped into a beaker of water, the water is seen to be violently agitated. Also when a small rubber ball suspended by a thread is brought near the vibrating prongs, the ball is seen to be immediately kicked aside as shown in fig 15.1.

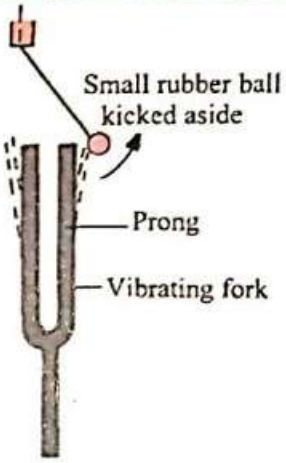


Fig. 15.1 Showing vibration of a tuning fork

An important aspect of sound transmission is that a material medium is always required for the passage of sound as we can demonstrate in the following experiment.

Experiment: To show that sound cannot travel through a vacuum but requires a transmitting medium

The apparatus is as shown in fig. 15.2.

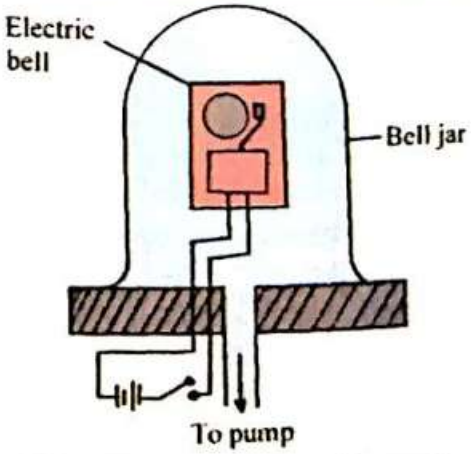


Fig. 15.2 Transmission of sound

It consists of an electric bell covered with a bell jar placed on the plate of a vacuum pump. The jar is made airtight by applying thick grease round the

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base of the jar. Electrical connections make the bell to ring and produce some sound which is clearly audible. The pump is then switched on to remove the air gradually. As more and more air is pumped out from the bell jar, the sound of the bell becomes fainter and fainter though the clapper is still seen to be striking the bell. No sound is heard when all the air has been pumped out leaving a vacuum in the bell jar. The clapper is still seen to be vibrating even though no sound is heard, showing that light waves can travel through a vacuum but sound waves cannot travel through a vacuum.

Sound is again heard when air is reintroduced in the bell jar.

Because there is little atmosphere to transmit sound in the moon, astronauts on the moon talk to themselves by radio even when they are close together.

Propagation of Sound Waves in Air

Sound waves are propagated through air by the displacement of air molecules. The molecules vibrate in a direction parallel to the direction of travel of the waves. As already discussed in chapter 11, waves of this nature are known as *longitudinal waves*.

Let us consider the propagation of sound waves produced by the vibration of the prongs of a tunning fork.

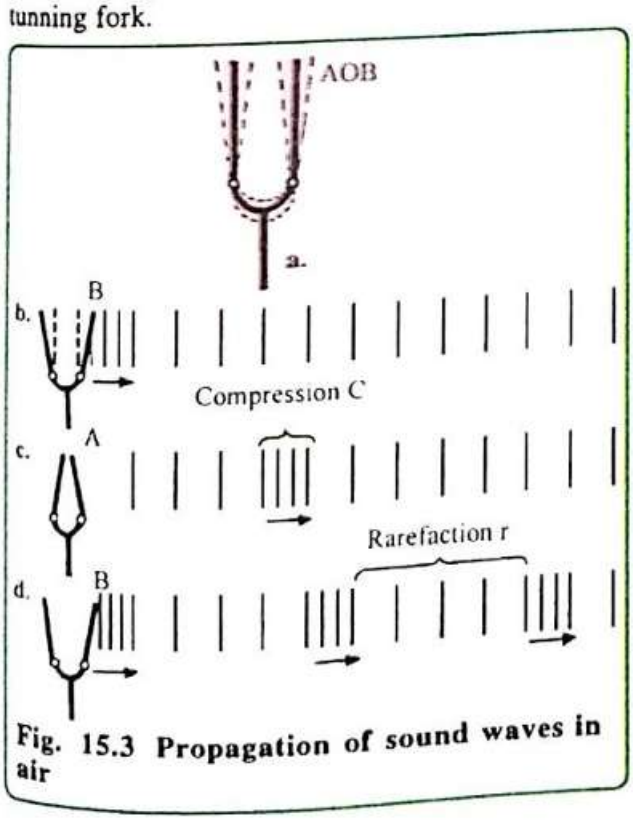


Fig. 15.3 Propagation of sound waves in air

As shown in fig. 15.3, when the vibrating prong moves to the right (position *B*) it compresses the air molecules nearest to it. This disturbance is then transmitted from particle to particle through the air. In this way, a pulse of *compression* (air pressure above normal) moving outwards from the prong is produced. Similarly when the prong moves to the left (position *A*), the pressure on the nearest layer of air is reduced and the air layer expands to the left producing a *rarefaction* (air pressure below normal). As a result of these movements, a series of alternate compressions and rarefactions travels outwards and propagates through the air.

It is important to emphasize that the air particles do not move along with the disturbance. While they vibrate about their mean position, their kinetic energy is transferred from one air layer to the next.

Apart from transmission through air and gases in general, sound can also be transmitted through solids and liquids. For example, if you press one end of a long metal or wooden rod to the ear, and have someone scratch or tap at the other end, the sound produced will be clearly heard. Sound does not readily pass from one medium to another, however. This is due to the difference in the densities of the respective media. Hence we do not readily hear out heart-beats except by pressing the ear to the chest or by using a *stethoscope* which provides a solid connection between the ear and the chest.

Velocity of Sound Waves

As already stated in chapter 11, the velocity, the frequency and the wavelength of sound waves are related by the formula.

$$V = f\lambda \qquad 15.1$$

where V = velocity of sound waves

f = frequency of sound waves

λ = wavelength of sound waves.

The frequency as already stated in chapter 11 is the number of waves per second passing a given point or the number of cycles or complete oscillations per second of the vibrating particles. The unit is in *Hertz*, (Hz). The wavelength is the distance between two successive crests or successive troughs. (See fig. 11.12). Thus when a note of sound of frequency 1800 Hz is played, the wavelength is given by

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$$\lambda = \frac{V}{f}$$

and assuming the velocity of sound in air is about 330 ms^{-1}, we have that

$$\lambda = \frac{330}{1800} = 0.183 \text{ m}$$

We see a flash of lightning before the accompanying sound of thunder because the velocity of light (about 3×10^8 ms^{-1}) is far greater than the velocity of sound (about 330 ms^{-1}). Thus suppose we hear the sound of thunder 10 seconds after we have seen the lightning flash, then the distance between us and the storm-centre, or the point of generation of the lightning and the thunder is given by

$S = Vt = (330 \times 10)$ m = 3. 3 km

The speed of sound varies from one medium to another. It is about 332 ms^{-1} in air at 0°C, about 1500 ms^{-1} in water, and about 5000 ms^{-1} in a steel rod. It is also found to vary with the square root of the absolute temperature, increasing in value with increasing temperature. Wind direction also affects the speed of sound. If sound waves travel in an opposite direction to that of the wind, the observed velocity of sound would be less than its actual value. On the other hand, the observed sound velocity would be greater than its actual value if the sound waves travel in the same direction as that of the wind. *Pitch* and *loudness* have no effect on sound velocity. Also the pressure of the surrounding air has no effect on sound speed.

Example 15.1

What is the wavelength of the sound from a tuning fork which vibrates at a frequency of 250 Hz in air? Take the velocity of sound as 330 ms^{-1}.

Solution

$V = f\lambda$

$\lambda = \frac{V}{f}$

$= \frac{330}{250}$ (m)

$= \underline{1.32 \text{ m}}$

Example 15.2

A source of sound produces waves in air of wavelength 1.65 m. If the speed of sound in air is 330 ms^{-1}, find the period of vibration in seconds.

Solution

$V = f\lambda$

$f = \frac{V}{\lambda} = \frac{330}{1.65}$

$T = \frac{1}{f} = \frac{1.65}{330}$

$= \underline{0.005 \text{ sec.}}$

Reflection of Sound. Echoes

Like other waves, sound waves can be reflected when they strike a plane surface. Reflection of sound obeys the laws of reflection in the same way as light waves (see chapter 11).

Reflection of sound gives rise to *echoes*.

An **echo** is a sound heard after the reflection of sound waves from a plane surface

When one stands in front of a wall and claps the hands, the time which elapses before the echo is heard depends on the distance away of the wall. The sensation of sound persists in the ear for about 0.1 second. So in order to hear the echo separately from the original clap, it must arrive at least 0.1 second later. Otherwise the ear will not be able to distinguish between the incident or original sound and its reflection or echo. Suppose the reflecting wall producing the echo is d metres away, the time taken to hear the echo is given by

$$\text{Time} = \frac{\text{total distance travelled by sound}}{\text{velocity of sound}}$$

Since the sound travels from the person to the wall and returns as an echo to the person, the total distance travelled is $2d$ metres.

Hence

$$t = \frac{2d}{330} \text{ sec.}$$

Since t must be greater than 0.1 sec, it follows that

$$\frac{2d}{330} > 0.1$$

or $d > 16.5$ m.

Therefore the minimum distance away of the wall for distinct echo is about 17 metres.

Echoes can sometimes be a nuisance in an auditorium if not controlled. The nuisance is brought about by the production of multiple echoes by walls, the floor and ceiling of the auditorium. The listener hears an echo each time the reflected sound passes him. These echoes together with the original sound, give rise to long drawn sound which confuses the listener. This continuous occurrence of echoes due to multiple reflection of the original sound is called *reverberation*.

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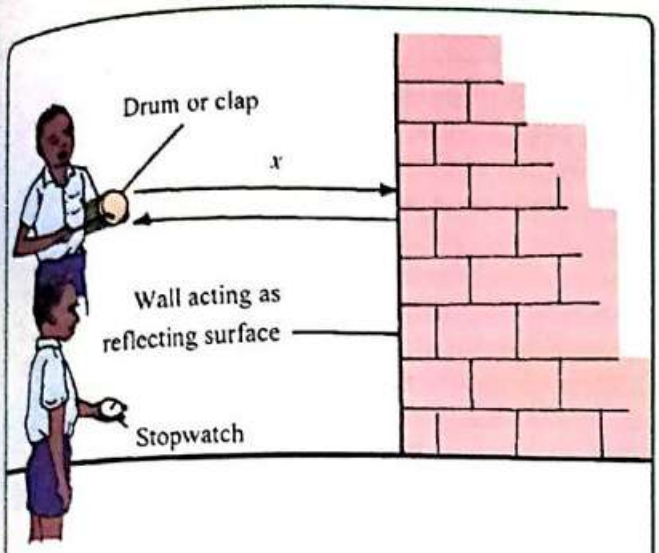


Fig. 15.4 Measuring the speed of sound

Reverberation is the perseverance of the sound after the source ceases. An auditorium or hall where reverberation occurs is said to have poor *acoustical properties*. Reverberation is reduced in halls by covering the ceiling and walls with soft perforated boards which can minimise the reflection of sound waves by absorbing them quickly. Other methods of controlling reverberation are by hanging curtains round the hall, and by having more openings in the walls.

Applications of Echoes

(i) Determination of speed of sound in air

We can use an echo method to determine the speed of sound in air by directing a sound signal to a wall and measuring the time taken for a sound to travel to the wall and back again.

Activity 15.1

1 Stand some distance directly in front of a vertical wall. Clap two hinged wooden boards sharply and listen for the echo from the wall.

2 Clap repeatedly and adjust the rate of clapping until each next clap coincides with the echo heard from the previous clap. The time interval between successive claps is then equal to the time taken for sound to travel twice the distance between the observer and the wall.

3 When the correct rate of clapping has been achieved, let another student or observer use a stop watch and measure the time for about 30 clap intervals. Find the average time interval between two successive claps, by dividing total time interval by 30.

4 Measure the perpendicular distance from where the clapper stands to the wall.

The velocity of sound is then calculated as follows:

distance from the wall = d (m).

Average time between successive claps = $\frac{t}{n} = T$

where t is the time interval for n clap intervals. Hence the sound travels a total distance = $2d$ (m) in time T seconds.

$\therefore$ Velocity of sound = $\frac{\text{distance}}{\text{time}}$

$$V = \frac{2d}{T} = \frac{2nd}{t} \qquad 15.2$$

(ii) Echo Sounding

Echo sounding method is used to determine the depth of a sea-bed from a ship.

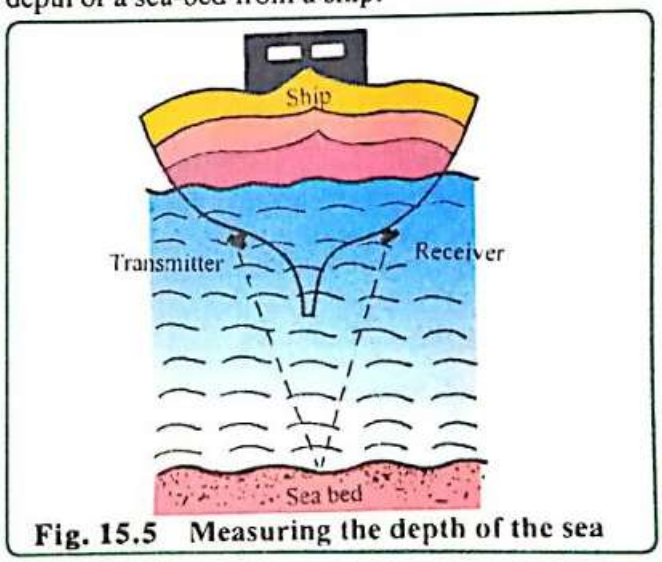


Fig. 15.5 Measuring the depth of the sea

The principle involves sending a sound impulse down into the sea-bed and detecting with a receiver the impulse reflected back after striking the sea bed. The time interval (t) taken to receive the echo of the transmitted impulse after reflection at the sea bed is noted. If the velocity of sound in water is V ms^{-1} then the depth of the sea is given by

$$d = \frac{V \times t}{2} \text{ (m)}$$

Example 15.3

A sound wave emitted from the bottom of a ship travels at 4500 ms^{-1} vertically downwards through water to an ocean bed 750 m deep, and is reflected upwards. What is the time interval between the instant the sound is emitted and the instant the echo is received?

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Solution

Let the time interval be t secs. In travelling to the ocean bed and back for the echo to be heard, the sound travels a total distance of 2×750 m $= 1500$ m.

$$V = \frac{\text{distance}}{\text{time}}$$

$$t = \frac{\text{distance}}{\text{velocity}} = \frac{1500}{4500}$$

$$= \underline{0.33 \text{ sec.}}$$

(iii) Exploration for gas and oil

Echoes are also used in the exploration for gas and oil. For such explorations, the geophysicists set off a small explosion on or just below the earth's surface. The resulting sound waves bounce off underground layers of rock. The nature of each echo and the time it takes for the reflected waves to reach the surface indicate the type and thickness of each rock layer present. By interpreting these they can locate possible mineral or oil-bearing rock formations.

(iv) Detection of Submarine

Underwater sound systems or devices called *Sonar* use sound waves to detect underwater objects. Warships can use *Sonar* to locate enemy submarine, and fishing boats can also use sonar system to locate schools of fish.

Refraction of Sound

Like light waves, sound waves are refracted when they pass from one medium to another of slightly different density (e.g. from air to carbon dioxide).

The refraction of sound explains why it is easier to hear sound coming from a distance at night than at day-time. In the day-time, the upper layers of air are colder than the layers near the earth. Since sound travels faster at higher temperatures, the sound waves are refracted in a direction away from the earth. The intensity of the sound waves thus diminishes. At night the reverse process takes place. The layers of air near the earth are colder than those higher up. Sound waves are therefore refracted towards the earth with a consequent increase in intensity.

Diffraction of Sound

The phenomenon of diffraction was studied in chapter 11. Sound waves also show the diffraction phenomenon. When a person speaks loudly in a room, the sound he produces can be heard round a corner without the speaker being seen. The sound waves are diffracted through the doors and windows which are openings of about 0.5 m–1 m. This shows that the wavelength of sound waves is of this order.

Interference of Sound

Interference effects are produced when two close sources emit waves of the same frequency and phase. In chapter 11 we distinguished between constructive and destructive interference. When two sources of sound waves interfere constructively, a loud sound is produced because sound from both sources reinforce each other. When they interfere destructively, silence or no sound is observed because the crest from one source overlaps the troughs of equal amplitude from the other source.

Polarization of waves occurs only in transverse waves like light. Since sound waves are longitudinal waves, polarization cannot be produced with sound waves.

Beats

When two wave trains, slightly differing in frequencies (e.g. from two tuning forks of nearly equal frequencies) travel along the same straight line in the same direction, then the resultant amplitude maximum and minimum. Thus the intensity of the resultant sound produced which is proportional to the square of the amplitude rises and falls with time. This is technically know as **waxing** and **waning** of sound. These alterations in loudness with time or this waxing and waning of the sound with time are called beats. The frequency f of beats is the number of intense sound heard per second.

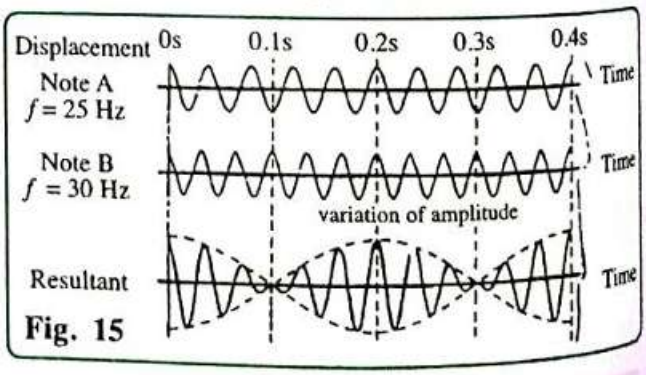


Fig. 15

The periodic variation in the loudness of a sound which is heard when two notes of almost the same frequency are played simultaneously is called beats.

If the frequencies of the sound waves are f_1 and f_2 it can be shown that the **beats frequency** f is given by $f = f_1 - f_2$

The beat frequency is the difference between the frequencies of the two waves.

Example 15

Two tuning forks of frequency 256Hz and 254Hz are sounded simultaneously. Calculate the frequency of the resultant beat.

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Solution

$$f = f_1 - f_2$$
$$= 265 - 254 = 2 \text{ Hz}$$

Use of beats

1 Beats are used in tuning of musical instrument, e.g. a piano. To tune a piano, a piano tuner listens for beats produced between the sound from a standard tuning fork and that from a particular string on the piano when the piano key is struck. The tension in the piano string is adjusted until the beats disappear, i.e. when the frequencies are the same.

2 Beats can be used to determine the frequency of a tuning fork. A tuning fork of unknown frequency f_1, is sounded together with a fork of know frequency f_2 to produce beats. The beat frequency f is measured and f_1 is calculated from $f = f_1 - f_2$

Inverse Square law in relation to transmission of sound waves

The intensity of sound at any point is inversely proportional to the square of the distance of the point from the source of the sound.
Thus

$$E = \frac{E_0}{d^2} \qquad 15.3$$

where E_0 = Intensity at source of sound $(d = 0)$
E = Intensity at a distance d from source.

15.2 Characteristics of Sound

A musical note possesses three characteristics: *pitch*, *loudness* and *quality*.

1 Pitch

The Pitch of a note is its position on the musical scale. It is a characteristic of a note which enables one to differentiate a high note from a low note. On a piano key board, the right hand side keys produce notes of high pitch and the left hand side keys produce notes of low pitch.

Pitch depends on the frequency of the sound wave. A low pitched note has a low frequency while a high pitched note has a high frequency. The pitch of a sound therefore increases as the frequency of the sound wave increases.

We can show that pitch depends on frequency by using (a) a disc siren or (b) a toothed wheel.

a. Method of Disc siren

A disc siren consists of a circular metal plate with holes punched at regular intervals as shown in fig 15.6.

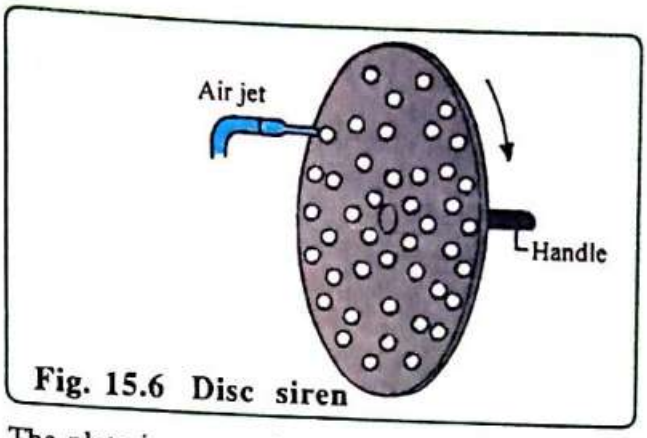


Fig. 15.6 Disc siren

The plate is mounted on a stand so that it can be rotated by hand or a motor. The number of revolutions of the disc is recorded by a counter. When the plate is rotated while a jet of air is directed onto the holes on the disc, a note is heard. The frequency of the waves produced is equal to the number of vibrations produced per second, i.e. it is equal to the number of holes passing across the jet of air per second. Thus frequency (f) is given by

$$f = \frac{\text{no. of revolutions} \times \text{no. of holes in the ring}}{\text{time of revolution in seconds}}$$

As the disc is rotated faster and faster, the note emitted is found to be of a higher and higher pitch. At the same speed of rotation of the disc, but with the air jet blowing harder through the holes, a louder note will be heard but its pitch remains the same. Thus the pitch of a note depends only on frequency.

b. Method of toothed wheel

Fig. 15.7 shows a toothed wheel mounted on a stand so that it can rotate freely.

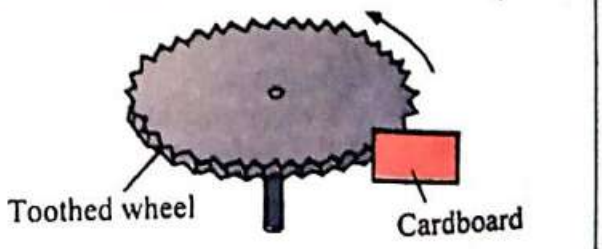


Fig. 15.7 The pitch of a note depends on its frequency

A piece of cardboard is held against the rotating wheel as shown. The cardboard vibrates as the wheel rotates and produces a note of frequency (f) given by

$$f = \frac{\text{no. of revolutions} \times \text{no. of teeth in the wheel}}{\text{time in seconds}}$$

As the wheel rotates faster and faster, the pitch of the emitted note is observed to be higher and higher. Thus we again observe that the pitch of a note depends on its frequency.

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Example 15.4

In an experiment to determine the frequency of a note, it was observed that the counter records a revolution of 500 for a time of 2.5 minutes. If the disc siren has 60 holes in a ring, what is the frequency of the note emitted?

Solution

$$f = \frac{\text{no. of revs} \times \text{no. of holes}}{\text{time in seconds}}$$

$$= \frac{500 \times 60}{2.5 \times 60 \text{ s}}$$

$$= \underline{200 \text{ vibrations per sec or } 200 \text{ Hz}}$$

Example 15.5

If in another experiment a note of the same pitch as in the above example is produced using a toothed wheel and it requires 400 revolutions of the wheel in 1.5 minutes to produce the note, calculate the number of teeth in the wheel.

Solution

Pitch is proportional to frequency. Since a note of the same pitch is produced, the frequency is 200 Hz.

$$\therefore \quad 200 = \frac{\text{no. of revs} \times \text{no. of teeth in wheel}}{\text{time in secs}}$$

$$200 = \frac{400 \times \text{no. of teeth}}{1.5 \times 60 \text{ secs}}$$

$$\therefore \quad \text{no. of teeth} = \frac{200 \times 1.5 \times 60}{400}$$

$$= \underline{\underline{45}}$$

2 Loudness and Intensity

Intensity of sound is the rate of flow of energy per second per unit area perpendicular to the direction of propagation of the sound wave.

Loudness is the magnitude of the sensation resulting from a sound reaching the ear. It depends on the intensity of the sound wave which reaches the listener's ear. It also depends on the square of the amplitude of the sound wave. A soft sound has a small amplitude but a loud sound has a large amplitude.

Loudness also depends on the mass of air which can be set into vibration. The greater the mass of air set into vibration, the louder will be the sound produced. This can be illustrated with a tuning fork or a dinner fork. If we set the prongs of the fork vibrating, it is noticed that the sound produced is faint or soft. The sound is observed to be remarkably loud when the end of the fork is placed on a table. This is because the large mass of air in contact with the surface of the table is set in motion when the fork is placed on the table.

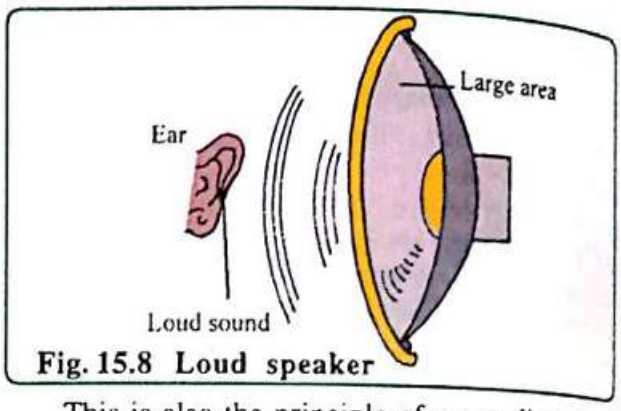


Fig. 15.8 Loud speaker

This is also the principle of a sounding board of a piano or the diaphram of a loud-speaker. The loudspeaker has a vibrating core with a large surface area. When in operation a large mass of air in contact with it is set into vibration to produce a loud sound.

3 Quality or tone (timbre)

Quality is a characteristic note of a musical instrument which distinguishes it from another note of the same pitch and loudness produced by another instrument. This is because a note played by a musical instrument is always a blend of the *fundamental frequency* and the *overtones* or *harmonics* of higher frequency. The quality of a note therefore depends on the overtones present in the note. Different musical instruments, produce different number and intensities of overtones together with the fundamental note, and so they produce notes of different quality.

Harmonics are frequencies which are multiples of the fundamental frequency which is the first harmonic.

Noise and Music

We can classify sound into noises or musical notes. *Noise* is due to vibrations of *irregular frequency* such as the rattling of a wheel on a rough road. Such irregular vibrations result in an unpleasant mixture of sounds.

Music is produced by vibrations of *regular* or *constant frequency*. A musical note possesses the characteristics of *pitch*, *loudness* and *quality*. When two musical notes have frequencies which are in the ratio of 2:1, the two notes are said to differ by an *octave*.

15.3 Forced Vibrations

All types of vibrating systems have their own natural frequency of vibration. This is the frequency with which the body or system will vibrate when it is left undisturbed after being set into oscillatory motion. For example the natural frequency (f) of a simple pendulum at any place depends only on the length of the pendulum according to the relation

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$T = 2\pi\sqrt{l/g}$ where $T = 1/f$.

If a vibrating body is placed in contact with another body, the later will also be forced to vibrate at the same frequency as the former. The vibration of the second body is called a *forced vibration.*

Forced vibrations are those vibrations that result from an external periodic force acting on a system and setting the system vibrating at the same frequency as the external periodic force.

An example of forced vibration is presented when a vibrating tuning fork is placed in contact with a table top. The sound is observed to become louder because the vibrating fork forces a larger surface (the table top) to vibrate, setting in motion a greater mass of air in contact with it.

Other common examples of forced vibrations are:

a. The vibrations of the cone of a loudspeaker caused by the fluctuations in the current flowing through the adjoining voice coil.
b. The vibrations of the diaphram of a telephone microphone or transmitter caused by the sound waves impinging on it. (p. 409)
c. The vibrating body of a violin caused by the vibrations of the strings of the instrument.

Resonance

Resonance is a special phenomenon of forced vibration. Whenever the frequency of a vibrating body (forcing frequency) acting on another body or system coincides with the natural frequency of the body or system, then the forced or induced vibration has a very large amplitude. This unique case of forced vibration is known as resonance.

Resonance is a phenomenon which occurs whenever a particular body or system is set in oscillation at its **own natural frequency** as a result of impulses or signals received from some other system or body which is vibrating with the same frequency.

Resonance can be illustrated by the following experiments:

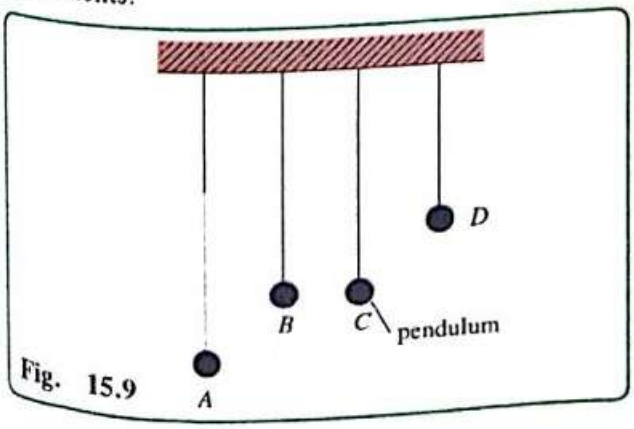


Fig. 15.9

1 Set up four pendulums A, B, C, D of different lengths each carrying a small mass. The pendulums should be hung from the same horizontal string. The lengths of B and C are made equal to each other while A and D are respectively longer and shorter than B as shown in fig 15.9.

If B is set oscillating at its natural frequency, in a short time it will be observed that the other pendulums will begin to oscillate with small swings except C which will swing rapidly in unison with B. All the pendulums will eventually swing in unison with B due to resonant vibration in the case of C but due to forced vibrations in the case of A and D. The resonance vibration between B and C is an example of mechanical resonance.

2 An example of *acoustical* resonance is illustrated with a tuning fork and a burette tube. A vibrating tuning fork is held over the top of the tube while the water filling the tube is slowly run out by means of a tap as shown in fig. 15.10. As the level of water is gradually lowered, there is an associated increase in the loudness of the sound from the air column of the tube until a certain stage when the sound produced is loudest.

This observation is explained as follows: the vibrations from the tuning fork prongs set the molecules of the air column into vibration which help to produce some sound.

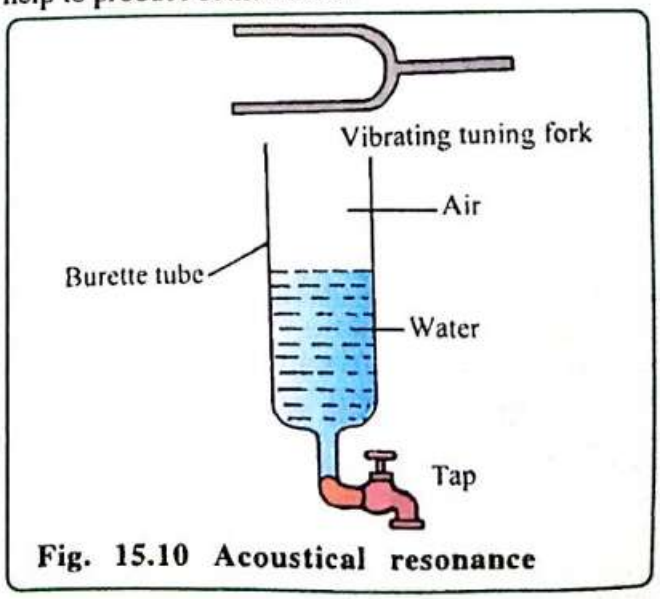


Fig. 15.10 Acoustical resonance

The sound continues to increase as more air particles are set in vibration. At the stage when the loudest sound is heard, *resonance* is said to occur. At this stage the air molecules in the air column of the tube are set at their own *natural frequency* as a result of the impulse or forced vibrations received

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from the prongs of the tuning fork which are vibrating at the same frequency.

Fig 15.11 shows how the amplitude of the forced vibration of the air molecules varies with the forcing frequency of the prongs of the tuning force.

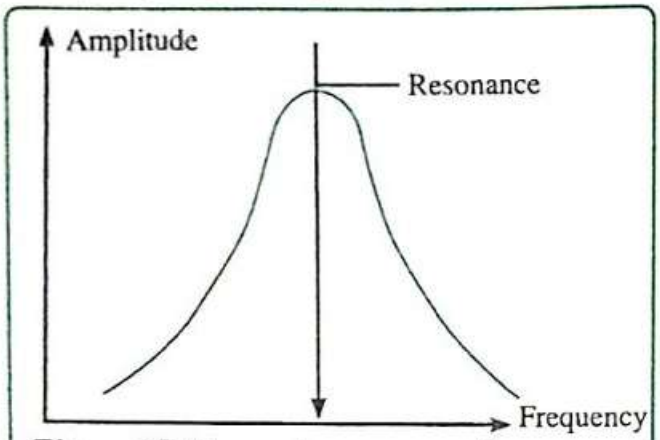


Fig. 15.11 f_0 = resonauce frequency
vibration

The diagram shows that if the natural frequency of the air column differs from that of the fork, only a feeble sound (low amplitude) will be heard, but when the frequency of the fork coincides with the natural frequency of the air column, a loud sound (large amplitude) can be heard.

Resonance finds an application in the tuning of a radio set to a broadcasting station. The radio is tuned to a station when the frequency of its receiver circuit is set in resonance by the incoming radio wave.

An example of a mechanical resonance is observed when a diver jumping up and down at one end of a diving board sets the board into resonance and so gain a considerable uplift before diving.

15.4 Vibrations in Strings and Pipes.

a. Vibration of Strings. Transverse Waves

Consider a stretched string or wire fixed at both ends B and C as in fig. 15.12. When it is plucked gently in the middle, a transverse wave travels along the vibrating string. At the fixed ends the wave is reflected back and we then have two progressive waves travelling in opposite directions along the string. Due to the superposition of the incident and reflected waves, interference takes place, and a *transverse stationary wave* is set up in the string or wire.

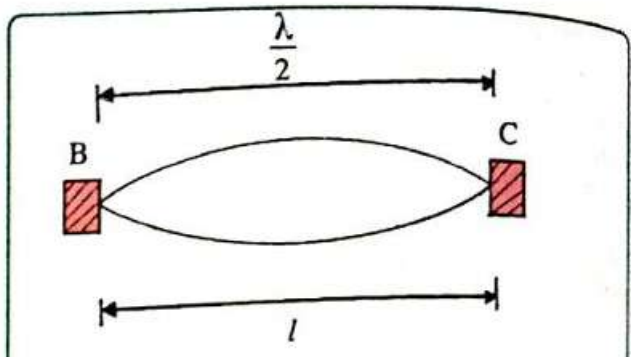


Fig. 15.12 Fundametal mode of vibration

In its simplest mode of vibration the wire vibrates in a single segment or loop as shown in fig 15.12, with its *nodes* at B and C because these points cannot move and an *antinode* at the centre where the string is most free to move. In this mode of vibration the vibrating wire produces a sound of the lowest possible note whose frequency is called the *fundamental frequency*. The mode of vibration giving rise to the fundamental frequency is known as the *fundamental mode of vibration.*

The distance between the two consecutive nodes is $\frac{\lambda}{2}$ and this is equal to the length of the string l.

$$l = \frac{\lambda}{2}$$

or $\lambda = 2l$ 15.4

For any wave we have that $v = f\lambda$ where v is the velocity, f, the frequency and λ the wavelength. Putting this in equation 15.4, we have that

$$f = \frac{v}{\lambda} = \frac{v}{2l} \qquad 15.5$$

Therefore, the fundamental frequency f_0 produced by a length l is $\frac{v}{2l}$.

Harmonics and overtones in a stretched string

The lowest frequency obtained from a plucked string when the string vibrates in one loop is called the fundamental frequency, f_0. Higher frequencies which are integral or whole number multiples of the fundamental frequency can also be produced in the string. These are called the *Harmonics* or *overtones* of the fundamental, e.g., $2f_0$, $3f_0$, $4f_0$ etc.; f_0 is the first harmonic.

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In fig 15.13, a stretched wire or string is fixed at both ends B and C. To obtain the harmonics, we 'damp' or touch the wire lightly with a finger at the centre O and simultaneously pluck the wire at a point P such that $BP = \frac{l}{4}$ where l is the length of the wire. The wire vibrates in two loops or segments as shown in fig. 15.13a.

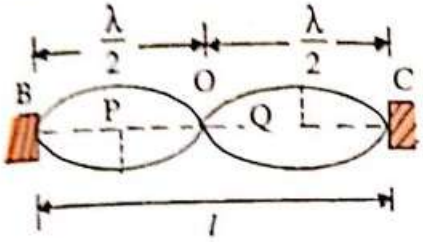


Fig. 15.13a First overtone

There are now three nodes (i.e. points that do not vibrate) at B, C and O, and two antinodes (points of maximum displacements) at P and Q. Since $BO = OC = \frac{\lambda}{2}$ we have that

$$l = \frac{\lambda}{2} + \frac{\lambda}{2} = \lambda$$

The note produced in this mode is called the *first overtone* or second harmonic and the frequency f_1 of this overtone is then given by

$$f_1 = \frac{v}{\lambda} = \frac{v}{l} \qquad 15.6$$

This is a higher note than the fundamental note. The wire can be made to vibrate in three loops or segments as shown in fig 15.13b.

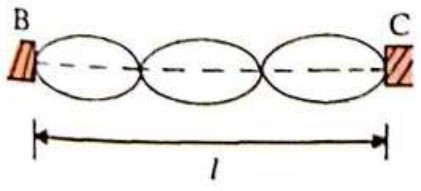


Fig. 15.13b Second overtone

This produces the next higher note called the *second overtone* or 3rd harmonic. As shown in the diagram

$$l = \frac{3\lambda}{2}$$

and from $v = f\lambda$, we have that

$$f_2 = \frac{3v}{2l} \qquad 15.7$$

Thus $f_0 = \frac{v}{2l}$; fundamental frequency or 1st harmonic

$f_1 = \frac{v}{l} = 2f_0$; 2nd harmonic or 1st overtone

$f_2 = \frac{3v}{2l} = 3f_0$; 3rd harmonic or 2nd overtone

and so on.

The fundamental frequency is the lowest frequency but the strongest audible frequency. The overtones are higher frequencies but are quieter or softer than the fundamental note.

We can distinguish clearly between *overtones* and *harmonics* of a vibrating body.

Overtones are the next higher frequencies or modes of vibration above the fundamental that a body can emit. Overtones which are whole number multiples of the fundamental frequency are known as Harmonics. If f_0 is the fundamental frequency, the harmonics are $2f_0$, $3f_0$, $4f_0$ etc. Overtones may or may not be harmonics of the fundamental. For example, the overtones of a bell are not harmonics at all, since their frequencies are not integral multiples of the fundamental frequency.

The Sonometer

The frequency of the note produced by a plucked string can be studied with a sonometer fig 15.14. It consists of a hollow *sounding-box* with a wire stretched along its length. The wire is attached to a peg at G. The other end is passed over a grooved wheel P and kept taut by attaching weights W at its free end. We can vary the length of the vibrating part of the wire by adjusting the movable bridge B.

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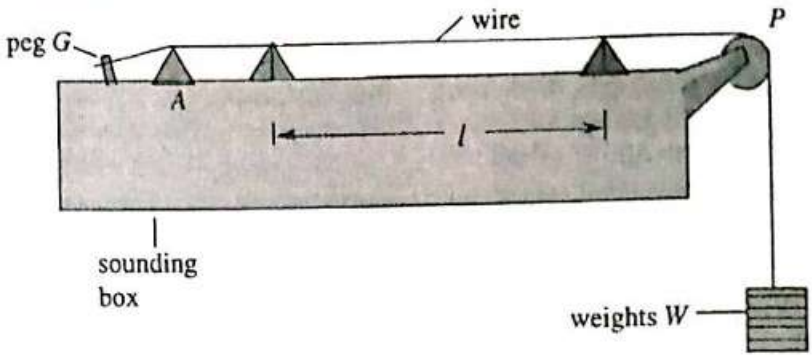


Fig. 15.14

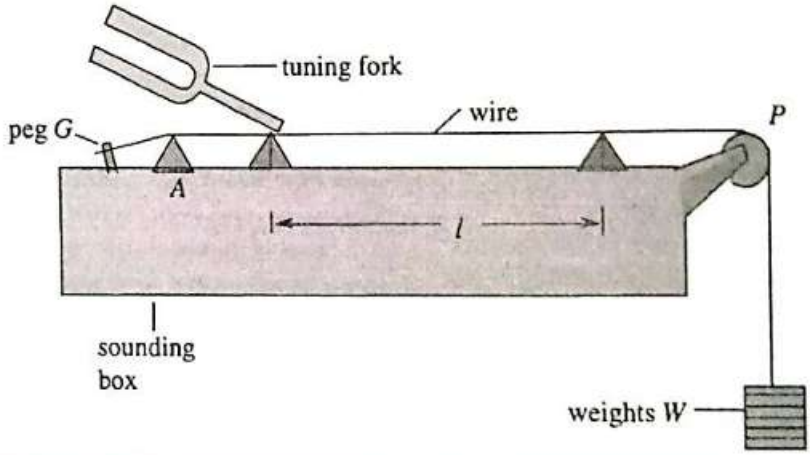


Fig. 15.15

Experiment to verify the variation of frequency with the length of a wire

1 Obtain 6 tuning forks of known frequencies. Keep the tension in the sonometer constant by attaching a known weight W to it.

2 Sound a tuning fork and adjust the movable bridge B so that when the string is plucked and the fork is sounding at the same time there is a throbbing note at a certain length l_1 of the vibrating wire. When nearing this position beats can be heard. Beats are heard when two notes of nearly equal frequencies are sounded together. The rise and fall or throbbing results from interference of the waves from the fork and the vibrating wire.

3 To test for resonance, place the shank of the vibrating fork on the bridge at B. A loud sound is heard if the natural frequency of the vibrating wire coincides with the natural frequency of the fork.

4 Repeat the procedure for the other tuning forks and tabulate your result as below

Frequency f	Length, l	$\frac{1}{l}$

5 Plot a graph of f against $\frac{1}{l}$.

A straight line graph passing through the origin as shown in fig. 15.16 is obtained.

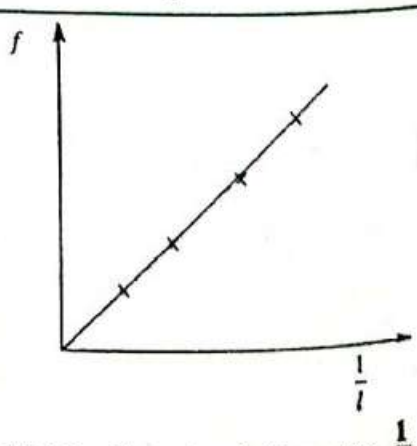


Fig. 15.16 Graph of f versus $\frac{1}{l}$

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This shows that the frequency of the vibrating wire is inversely proportional to its length, $f \propto \frac{1}{l}$ or $f = \frac{K}{l}$ where K is a constant.

$$f \propto \frac{1}{l}$$

Experiment to verify the variation of frequency with tension in a wire.

The last experiment is repeated but now the length of the wire is kept constant between the bridges of the sonometer, but the tension of the wire is altered by altering the tension (T) or weight W in the diagram. The results are recorded as follows:

Frequency f	Tension, T (or W)	$\sqrt{T}$

A graph of frequency f against the square root of the tension ($\sqrt{T}$) is a straight line graph passing through the origin.

$$f \propto \sqrt{T}$$

Thus the frequency of a vibrating wire is directly proportional to the square root of its tension, when the other factors are kept constant.

The experiment when repeated with wires of varying linear density or mass per unit length of wire, also shows that the frequency of a wire is inversely proportional to the square root of its mass per unit length, when the other factors are kept constant. Hence a thinner wire produces sound of higher frequency or pitch than a thicker wire.

$$f \propto \frac{1}{\sqrt{M}}$$

where M is mass per unit length.

Thus the frequency of a vibrating stretched string or wire depends on three factors; (a) length of string l, (b) tension T and (c) mass per unit length, M. The three important relations combine to

$$f \propto \frac{1}{l}\sqrt{\frac{T}{M}}$$

or
$$f = \frac{K}{l}\sqrt{\frac{T}{M}} \qquad 15.8$$

where K is a constant dependent on the mode of vibration.

For the fundamental mode of vibration, $K = \frac{1}{2}$ and equation 15.8 becomes:

$$f_0 = \frac{1}{2l}\sqrt{\frac{T}{M}} \qquad 15.9$$

where T is in Newtons and M in kilogrammes per metre and l is in metres, f is in Hertz.

Since $f_0 = \frac{1}{2l}\sqrt{\frac{T}{M}}$ is the fundamental frequency, the frequency of the first overtone is given $f_1 = \frac{1}{l}\sqrt{\frac{T}{M}} = 2f_0$. Similarly the second overtone f_2 is given $\frac{3}{2l}\sqrt{\frac{T}{M}} = 3f_0$. Thus the overtones of a stretched string are $2f_0$, $3f_0$, $4f_0$ and so on.

Example 15.6

A metal wire of mass 1 g and length 50 cm is under a tension of 80 N. Compute (a) the fundamental frequency (b) the first and second overtones (c) the speed of a transverse wave in the string.

Solution

From $f_0 = \frac{1}{2l}\sqrt{\frac{T}{M}}$ we have

$l = 0.5$ m, $T = 80$ N, $M = \frac{1\times10^{-3}}{0.5}$ kgm^{-1}

$$f_0 = \frac{1}{2\times0.5}\sqrt{\frac{80\times0.5}{1\times10^{-3}}}\ \text{sec}^{-1}$$

$= \underline{200 \text{ per sec}}$ or $\underline{200 \text{ Hz}}$

1st overtone $= 2 \times 200 = \underline{400 \text{ Hz}}$

2nd overtone $= 3 \times 200 = \underline{600 \text{ Hz}}$

From $v = f_0\lambda$ we have, since $\lambda = 2l$, that

$v = 200 \times 2 \times 0.5$

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$= \underline{200\ \text{ms}^{-1}}$

Example 15.7

Under a constant tension, a plucked string produces a note of 250 Hz when the length is 1. 0 m. At what length is the frequency 200 Hz? What frequency will be produced with a length of 0.5 m?

Solution

$f \propto \frac{1}{l}$ at constant tension

$f = \frac{K}{l}$

$250 = \frac{K}{1.0}$ or $K = 250$

When $f = 200$, $l = \frac{K}{f} = \frac{250}{200}$

$l = \underline{1.25\ \text{m}}$

When $l = 0.5$, $f = \frac{K}{l} = \frac{250}{0.5}$

$f = \underline{500\ \text{Hz}}$

b. Vibrations of air columns. Longitudinal Waves in Pipes

An air column is air contained in a tube or pipe. When both ends of the pipe are open, it is called an *open pipe*, but when one and is closed and one end is open, it is called a *closed pipe*.

When air vibrates inside a pipe, the waves produced are reflected at the ends producing a longitudinal stationary wave along the length of the pipe. Since the air particles at the closed end of the pipe cannot move, there is always a node at this end. At the open end the air particles are free to vibrate with maximum amplitude. An antinode is therefore always formed at the open end.

Vibrations produced in closed pipes

The simplest mode of vibration produced in a closed pipe is illustrated in fig 15.17a. The diagram shows the stationary wave formed in the air column when we blow gently at the open end. The disturbance created at this open end travels through the air column and is reflected at the closed end. There is therefore a node (N) at the closed end and an antinode (A) at the open end. Since the distance between a node and a consecutive antinode in a waveform is $\frac{\lambda}{4}$, we have from the diagram that the length of the tube,

$$l = \frac{\lambda}{4} \text{ or } \lambda = 4l$$

Therefore the fundamental frequency f_0 is given by

$$f_0 = \frac{V}{\lambda} = \frac{V}{4l} \qquad 15.10$$

where V is the velocity of the wave. By blowing harder across the open end of the pipe we can generate a note of higher frequency. For the first overtone (fig 15.17b) we have from the diagram that l is related to the wavelength of the vibrating air column by

$$l = \frac{3}{4}\lambda_1$$

$$\therefore \quad \lambda_1 = \frac{4}{3}l$$

Hence the frequency (f_1) of the first overtone is given by

$$f_1 = \frac{V}{\lambda_1} = \frac{3V}{4l} = 3f_0$$

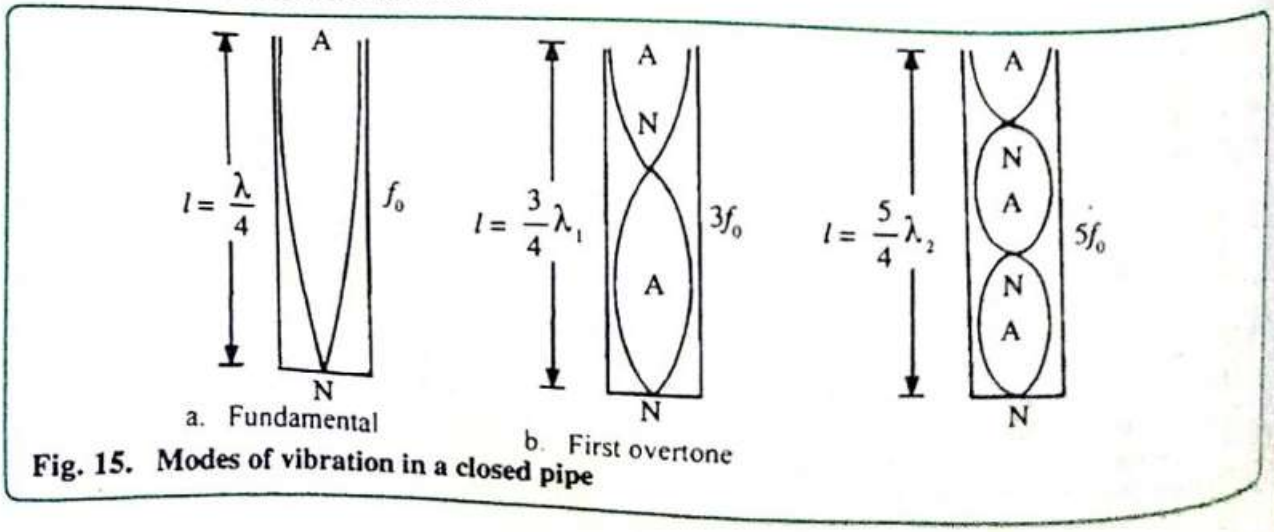
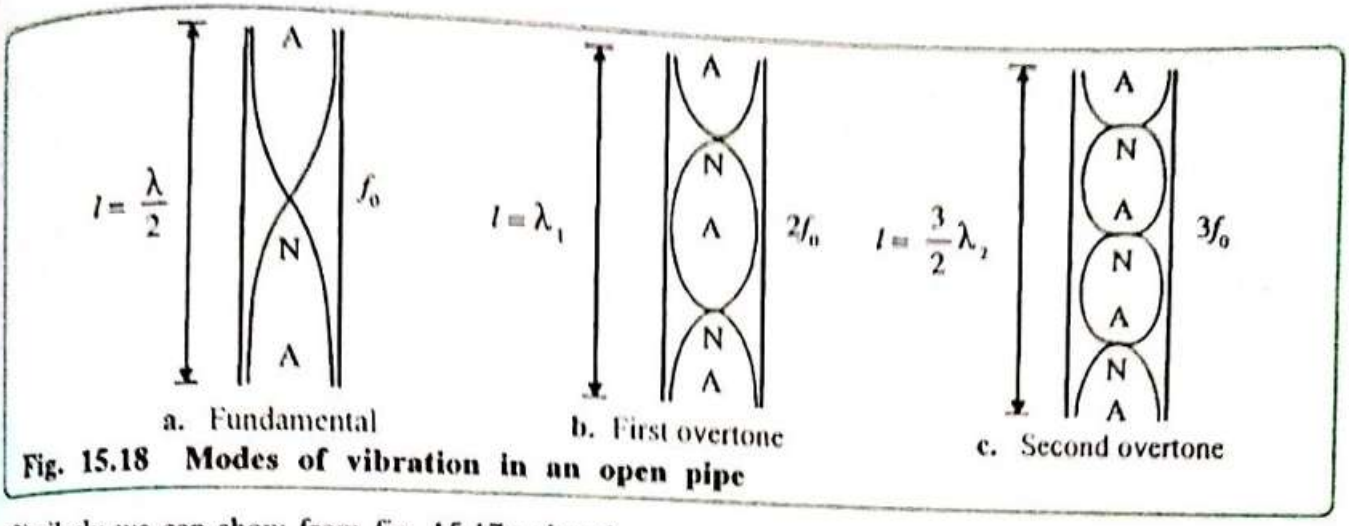


Fig. 15. Modes of vibration in a closed pipe

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a. Fundamental b. First overtone c. Second overtone

Fig. 15.18 Modes of vibration in an open pipe

Similarly we can show from fig. 15.17c. that the second overtone frequency f_2 is given by

$$f_2 = \frac{5V}{4l} = 5f_0$$

That of the third overtone is given by

$$f_3 = \frac{7V}{4l} = 7f_0 \text{ etc.}$$

Thus in a closed pipe, only odd numbers of harmonics are present as overtones accompanying the fundamental note. The possible harmonics are f_0, $3f_0$, $5f_0$, $7f_0$, etc.

Vibrations in open pipes

Stationary waves set up in an open pipe have antinodes at both ends. The fundamental or simplest mode of vibration is that in which the midpoint is a node (fig. 15.18).

Therefore $l = \frac{\lambda}{2}$ or $\lambda = 2l$.

$\therefore$ the fundamental frequency f_0 is given by

$$f_0 = \frac{V}{\lambda} = \frac{V}{2l} \qquad 15.11$$

The frequency (f_1) of the first overtone (fig 15.18b) is given by

$$f_1 = \frac{2V}{\lambda} = \frac{V}{l} = 2f_0$$

Similarly the frequency (f_2) of the second overtone is given by

$$f_2 = \frac{3V}{2l} = 3f_0$$

Therefore in an open pipe the harmonics present are $2f_0$, $3f_0$, $4f_0$, etc., that is, both odd and even harmonics are present as overtones.

Experiment: To determine the velocity of sound in air using a Resonance tube

We can determine the velocity of sound in air using a tuning fork of known frequency and a resonance tube. A resonance tube consists of a glass tube filled with water. The volume or length of air in the tube can be varied by gradually letting out the water through a rubber tubing.

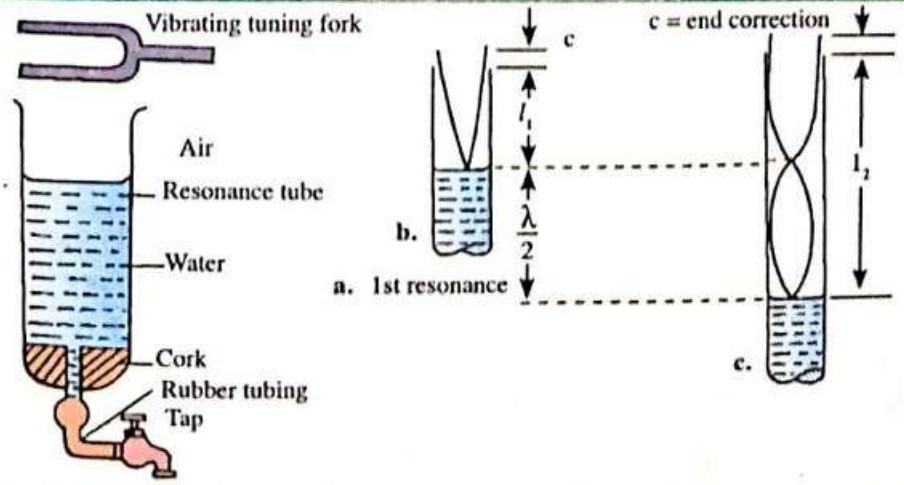


Fig. 15.19 Velocity of sound by the resonance tube method

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A suitable tuning fork of known frequency f is set into vibration and placed above the top of the tube full of water. Water is then run off slowly by means of the tap, T, until the first resonance, marked by a loud sound is heard. The length of the air column (l_1) when this occurs is measured.

By further increasing the length of the air column by letting out more water, a stage is reached where a second resonance sound is heard. The length of this second air column l_2 is also measured.

At first resonance, we have that

$$\frac{\lambda}{4} = l_1 + c \qquad 15.12$$

At second resonance

$$\frac{3\lambda}{4} = l_2 + c \qquad 15.13$$

where c is the *end correction* which arises from the fact that the antinode at the top does not exactly coincide with the top of the tube but projects slightly above it by an amount c known as the *end correction.*

Subtracting 15.12 from 15.13 we can eliminate the end correction to obtain

$$\frac{\lambda}{2} = l_2 - l_1$$

$$\lambda = 2(l_2 - l_1)$$

Substituting in the formula

$V = f\lambda$

we have

$$V = 2f(l_2 - l_1) \qquad 15.14$$

Thus the velocity of sound is obtained from the values of l_1, l_2 and the frequency f of the tuning fork.

Example 15.8

In an experiment to determine the velocity of sound in air, using a resonance tube, a student observes that the water level was 34.5 cm below the open end of the tube when a loud sound was obtained with a tuning fork of frequency 256 Hz. His next position of resonance was measured as 99. 7 cm using the same tuning fork. What is the velocity of sound as obtained from this experiment?

Solution

For 1st position of resonance,

$$l_1 + c = \frac{\lambda}{4}$$

For 2nd position of resonance,

$$l_2 + c = \frac{3\lambda}{4}$$

Hence $l_2 - l_1 = \frac{\lambda}{2}$

From $V = f\lambda$ or $f = \frac{V}{\lambda}$ we have that

$$V = 2f(l_2 - l_1)\ \text{ms}^{-1}$$

$l_2 = 99.7\ \text{cm} = 0.997\ \text{m};\ l_1 = 34.5\ \text{cm} = 0.345\ \text{m}$

$$V = 2 \times 256(0.997 - 0.345)\ \text{ms}^{-1}$$

$$= \underline{333.8\ \text{ms}^{-1}}$$

Example 15.9

The shortest length of the air column in a resonance tube closed at one end which resonates to a frequency of 560 Hz is found to be 20 cm. What is the wavelength of this sound in air? What assumption have you made?

Solution

At resonance $l + c = \frac{\lambda}{4}$ where c is the end correction. If we assume that the end correction c is negligible, then $l = \frac{\lambda}{4}$

$$\lambda = 4l = 4 \times \frac{20}{100} = \underline{\underline{0.8\ \text{m}}}$$

15.5 Application of sound waves in musical instruments

The source of any sound is a vibrating object. In musical instruments the source is set into vibration by striking, plucking, bowing or blowing. Standing waves are produced and the object vibrates at its natural resonant frequencies.

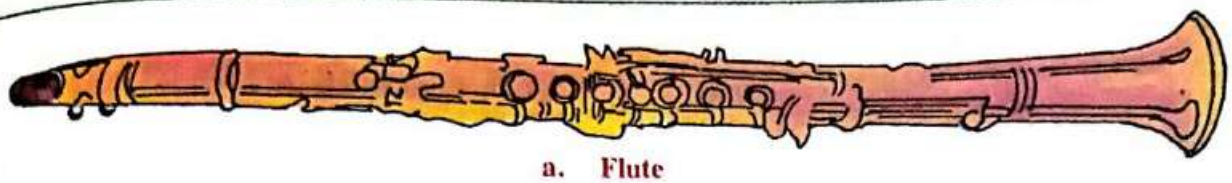
Musical instruments are classified into (a) wind instruments (b) stringed instruments and (c) percussion instruments.

a. Wind Instruments

These make sound through a vibrating column of air. Examples are flutes, trumpets, pipe organ, clarinets, and saxophones.

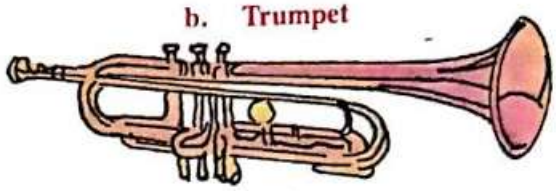
In each of these, the player blows through the open end causing a column of air to vibrate. By manipulating the keys of the instrument the player alters the length of the vibrating air column thereby

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a. Flute

Fig. 15.20 Wind instruments



b. Trumpet

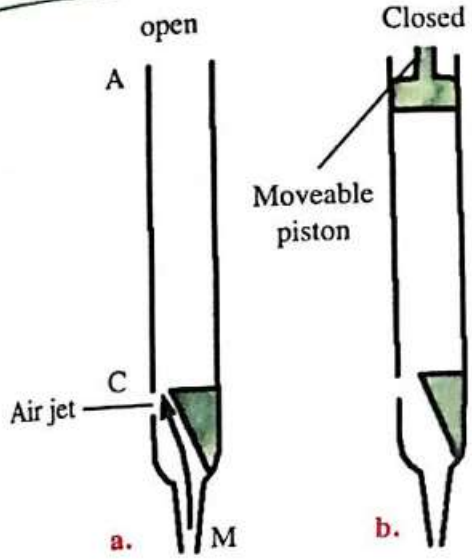
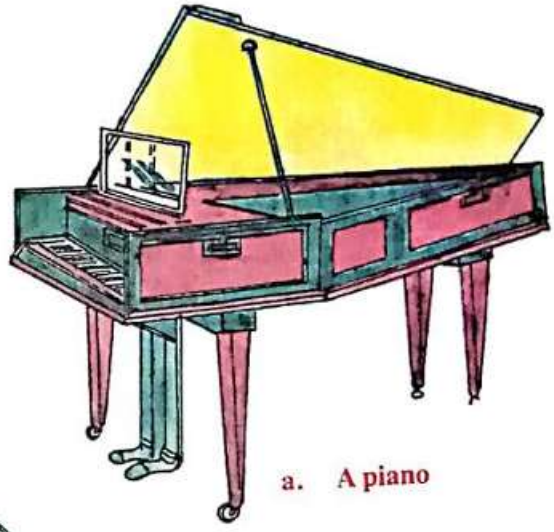


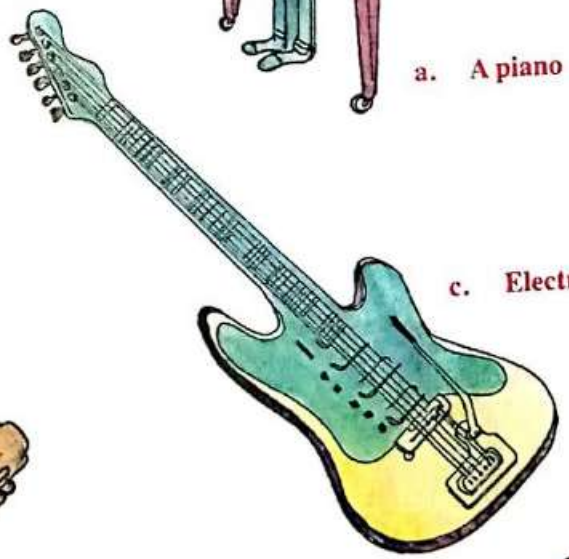
Fig. 15.21 Organ pipes



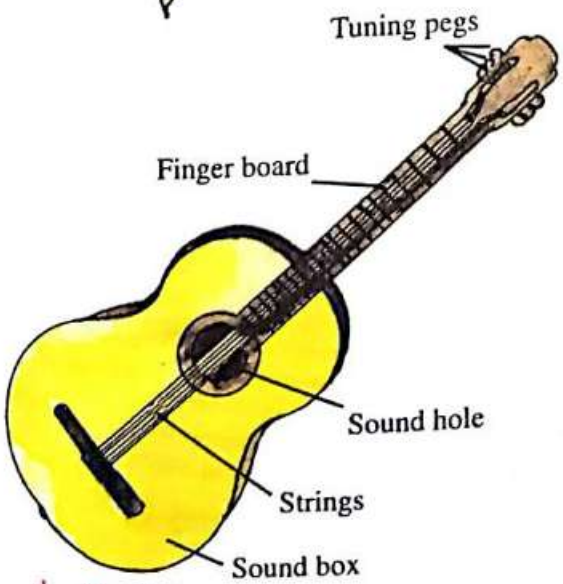
a. A piano



b. A violin



c. Electric guitar



d. A guitar



e. The strings of a violin are turned by being tightened or slackened. The player turns pegs to which they are attached while plucking or bowing the strings.

Fig. 15.22 Stringed instruments

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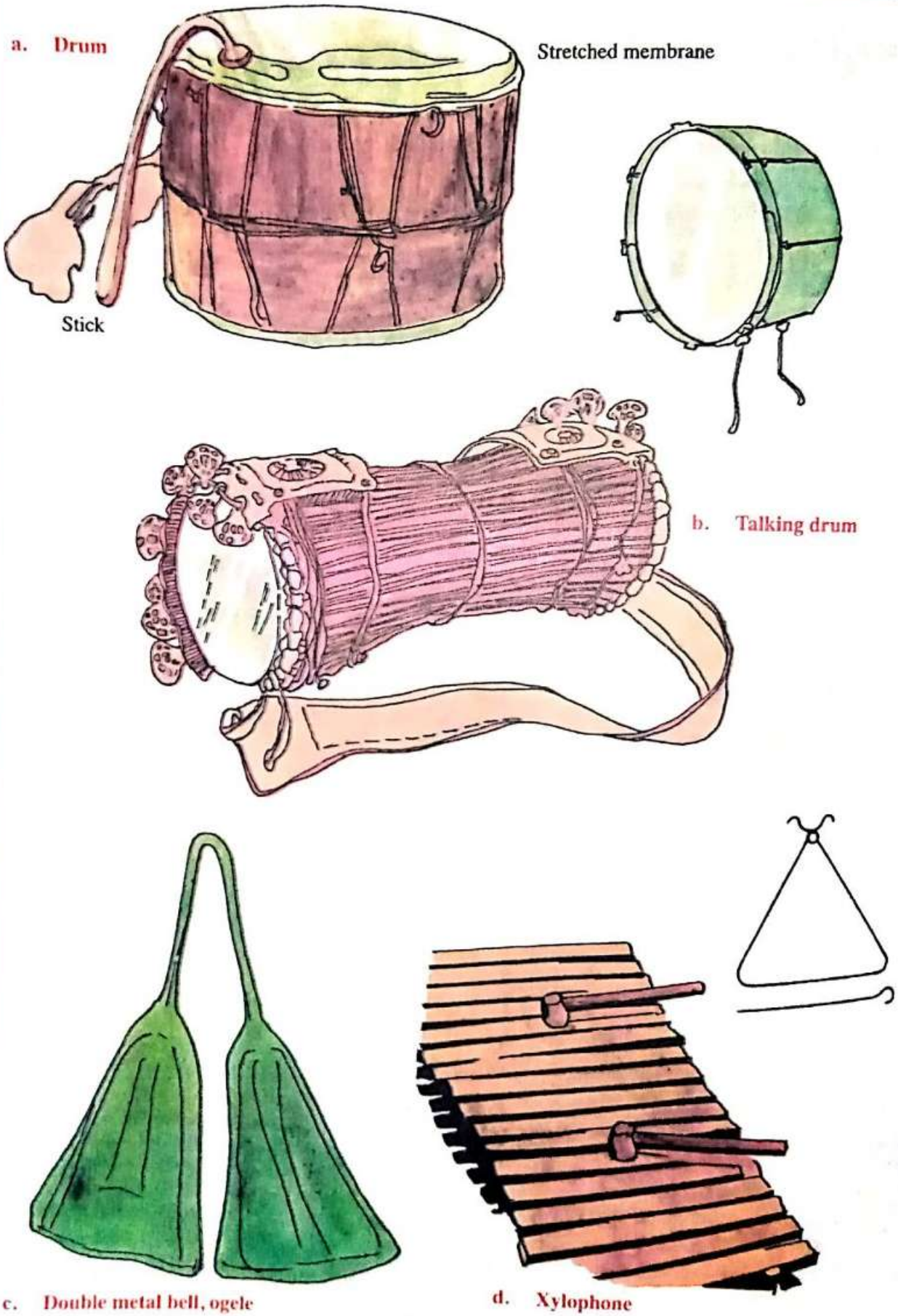


Fig. 15.23 Percussion instruments

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producing sound of different tone, frequency or pitch.

In some instruments, a vibrating reed or the vibrating lip of the player helps to set up vibrations of the air column. In organs, both open and closed pipes are used to produce different effects. In an open pipe (fig. 15.21), a jet of air is driven through the opening at M and escapes through C. This causes the vibration of the air column between AC which results in the production of even and odd harmonics. In the closed pipe (fig. 15.21b) one end is closed by a movable piston. The length of the air column is varied by the movement of the piston. The vibrations in the closed pipe produces only odd numbers of harmonics.

b. Stringed Instrument

In stringed instruments, use is made of the fact that the frequency of a vibrating string depends on its length, mass and tension in the string. A long, thick and loose wire or string produces a low frequency note but a short, thin, and taut string produces a high frequency note. The vibrations of the string produce both the fundamental frequency and the overtones. The various combinations of the fundamental frequency and the overtones produce musical sounds of various *quality*.

Examples of stringed instruments are guitars, sonometer, piano, violin. A violin has four strings of different densities but of equal lengths. The mass per unit length is therefore different for each string. The violinist can thus obtain a wide range of frequencies from the different strings.

In stringed instruments, the player varies the pitch of each string by pressing it at one point against a finger board – the shorter the length of the vibrating string, the higher the pitch of the sound. The heavier strings produce lower notes.

Stringed instruments would not be very loud if they relied on their vibrating strings alone to produce the sound waves since the strings are too thin to compress and expand much air. They therefore make use of a kind of a mechanical amplifier known as *sounding board* (piano) or *sounding box* (guitar, violin) which acts to amplify the sound by putting a greater surface area in contact with the air. When the strings are put into vibration, the sounding board or box also vibrates in unison. Because of the greater surface area of air in contact with the board or box, a louder sound is produced. The sounding box is not so important in electric guitars since the vibrations of the strings are amplified electrically.

c. Percussion Instrument

A percussion instrument produces sound when it is hit or struck. They have taut skin, membranes, rods or plates which vibrate when struck. The note produced is usually of short duration.

Example of such instruments are talking drums, bells, gongs, xylophones, tuning forks. A drum has a stretched membrane that vibrates when hit with a stick. Cords around its rim enable the player to change the pitch of the note emitted by the drum. Xylophones have metal or wooden bars that can be set into vibration. The wooden bars of the Xylophone tuned to various notes and arranged like a piano-keyboard are struck with mallets. Bells, cymbals and gongs also make use of a vibrating metal.

15.6 Hearing Aids

Hearing is the sense concerned with the perception of sound. We hear with the ear and see with the Eye. As with the eye, the ear may also have some defects or impairments. (For the structure of the ear, the student should consult any good biology textbook).

Just as we need spectacles or eye-glasses to improve, or enhance our ability to see very clearly, we also do need **hearing aids** to hear more loudly and distinctly.

Hearing aid is a device used by a person with impaired or defective hearing ability to improve his or her ability to hear properly. The device increases the loudness of sound in the ear of the wearer. This sense of hearing depends on the ability of the listener to detect the pressure of sound waves in the air around him or her. The hearing aid helps the listener by acting as an amplifier. It does so by increasing the pressure of sound waves and delivering it into the ear at a point within the ear canal.

Types of hearing-aids

There are two principal types of hearing aids. These are

(i) air-conduction aids and.

(ii) bone-conduction aids.

An air-conduction aid amplifies sound and brings it directly into the ear. This is the hearing aid most people wear. The bone conduction aids are for these people in whom sound cannot be transmitted through the outer or middle ear. This type of aid therefore brings sound waves to the bony parts of the head behind the ear. The bone transmits the air vibrations to the auditory nerves of the cochlea. Hearing becomes possible when these nerves are stimulated.

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(a)



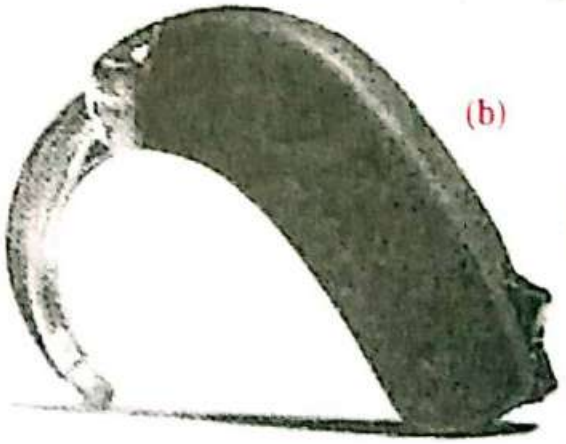
(b)

Fig. 15.24 CIC

Completely-in-the-canal hearing aid which fits completely in the canal.



(a)



(b)

Fig. 15.25 Behind the Ear

Behind-the-ear hearing aid. The hearing sets behind and over the ear. An earmold is placed into the ear canal with tubing connecting the two pieces.

Parts of hearing-aids

Modern hearing aids are electronically operated. They are made up of five major parts:

1) The Microphone, that picks up the sound and changes it into electrical signals.
2) the Amplifier that increases the strength of the signals.
3) A Receiver or ear-piece that changes the signals back to amplified sound that can be understood by the listener.
4) The Battery which supplies power for the operation of the equipment.
5) The ear-mold that secures the receiver in the ear canal.

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Summary

1 Sound is produced by vibrating bodies. The vibrating bodies produce the sound waves through the *compression* (pressure above normal) and *rarefaction* (pressure below normal) of the air. Sound waves are longitudinal waves.

2 Sound requires a material medium for transmission

3 Sound velocity (V) is related to its frequency and wavelength through the formula $V = f\lambda$.

4 Like light, sound waves can be reflected, refracted, diffracted and also can show the interference phenomenon.

5 Echoes are produced when sound waves are reflected by hard surfaces. *Reverberation* is the perseverance of the sound after the source ceases. It is due to the multiple reflection or echo of the original sound. Excessive reverberation in a concert hall is undesirable.

6 Echoes are used in (i) the determination of the velocity of sound (ii) the determination of the depth of sea beds (iii) in oil and gas exploration (iv) in sub-marines.

7 Sound of regular frequency is called a *tone* or musical note. A musical note possesses three characteristics: pitch, loudness and quality.

Pitch of a note is its position on the musical scale. It is the degree of highness or lowness of sound as perceived by a listener. Pitch depends on frequency.

Loudness is the magnitude of the sensation resulting from a sound reaching the ear. It depends on the amplitude of vibration of the body.

Intensity is the rate of flow of sound energy per second per unit area perpendicular to the direction of propagation of the sound waves.

Quality or *timbre* is a characteristic note of a musical instrument which distinguishes it from another note of the same pitch and loudness produced by another instrument. It depends on the overtones accompanying the fundamental note.

8 Hearing aid is a device used by a person with impaired or defective hearing ability to improve his or her ability to hear clearly or distinctly. Hearing aid is an amplifier of sound to the patient's ears.

There are two types of hearing aids:

a) air-conduction aids and

b) bone conducting aids

9 Noise is due to vibrations of irregular frequency, but music is due to vibrations of regular frequency.

10 The natural frequency of a body is the frequency with which a body will oscillate when it is left undisturbed after being set into vibration.

11 Forced vibrations result from an external periodic force acting on a system and setting it vibrating at the same frequency as the external force.

12 Resonance is a unique case of forced vibration which occurs when the frequency of a vibrating body acting on a system coincides with the natural frequency of the system. In this unique case, the free vibrations re-enforce the forced ones and a large amplitude of vibration is developed.

13 Velocity of sound in air increases with temperature. It is also affected by wind direction. It is independent of pressure of the surrounding air, and the pitch and loudness of the sound.

14 Stationary waves are produced when two waves with same frequency and amplitude travelling in opposite directions are superposed.

Nodes are points on a stationary waves which are at rest.

Antinodes are point where the amplitude of vibration is maximum.

The distances between successive nodes or antinodes is $\frac{\lambda}{2}$ but the distance between a node and the next antinode is $\frac{\lambda}{4}$ where λ is the wavelength.

15 The *fundamental frequency* (f_0) is the frequency at which a vibrating body produces sound of the lowest possible note.

Harmonics are whole number multiples of the fundamental frequency. f_0 is the first harmonic, $2f_0$, $3f_0$, etc. are the second and third harmonics etc.

Overtones are the next higher frequencies produced after the fundamental frequency by a vibrating body. Overtones may or may not be harmonics of the fundamental.

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16 The frequency f with which a stretched string vibrates depends on (i) length (l) (ii) Tension (T) and (iii) mass per unit length M. The fundamental note produced in a stretched string is given by

$$f_0 = \frac{V}{2l}\sqrt{\frac{T}{M}}$$

17 The frequency of the fundamental note produced by a stretched string is given by, $f_0 = \frac{V}{2l}$ where l is the length of the string. For a closed pipe, $f_0 = \frac{V}{4l}$ for an open pipe $f_0 = \frac{V}{2l}$ where V is the velocity of sound in air and l is the length of the pipe.

18 The velocity of sound in air can be measured by the resonance-tube experiment. The experiment must be repeated to eliminate end correction. $V_{solid} > V_{water} > V_{air}$.

19 Musical instruments can be classified into three types:

Wind instruments, e.g., flutes, trumpet, pipe organ, saxophone, etc.

Stringed instruments, e.g., pianos, guitars, violin, etc.

Percussion instruments, e.g., drums, bells, gongs, xylophones, etc.

20 **Beats** are periodic rise and fall in amplitude (or loudness) of the sound produced when two notes of nearly equal frequencies are sounded together.

Exercise 15

1 Sound waves differ from water waves in that
A energy transfer is involved
B they can be refracted and reflected
C no transfer of the medium is involved
D they are longitudinal waves
E they involve vibration.

2 A sounding tuning fork is brought near the end of a pipe containing an air column and the loudness of the sound increases. This effect is due to
A diffraction B an echo
C resonance D interference
E reverberation.
(SSCE)

3 A boy standing some distance from the foot of a tall cliff claps his hands and hears an echo 0.5 s later. If the speed of sound in air is 340 ms^{-1} how far is he from the cliff?
A 17 m B 34 m C 85 m
D 170 m E 680 m.
(SSCE)

4 The speed of sound travelling in various media increases in the following correct order:
A iron bar, air, water
B iron bar, water, air
C air, water, iron bar
D water, iron bar, air
E water, air, iron bar.
(SC/GCE)

5 Which of the following determine(s) the frequency of vibration of a stretched string?
I. The length of the string
II. The density of the material of the string
III. The tension in the string
A I only B II only C III only
D I and II only E I, II and III.
(SC/GCE)

6 A source of sound produces waves in air of wavelength 1.65 m. If the speed of sound in air is 330 ms^{-1}, the period of vibration in second is
A 200 B 2.0 C 0.5
D 0.02 E 0.005
(SSCE)

7 Why does the sound from an enclosed bell jar gradually fade away as the jar is gradually evacuated?
A The sound is forced out
B The pressure within the jar is reduced
C There is no more material medium
D The temperature is reduced
E The sound waves are absorbed by the inner walls.

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8 Which of the following statements is/are *not* true.
A The velocity of sound in air does not depend on its intensity
B The velocity of sound in air does not depend on the temperature.
C Sounds travel faster through a solid than through a gas.
D Sound does not require a material medium for transmission.
E The pitch of sound depends on its frequency.

9 A ship's echo sounder sends out a supersonic note which is received back at the ship 4 seconds after. If the velocity of sound through water is 1500 ms^{-1}. What is the depth of the seabed?
A 3000 m B 1500 m C 600 m
D 4500 m E 750 m.

10 What change(s) is/are observed in the sound produced by a siren when the speed of revolution is increased?
I. The pitch become higher
II. The wavelength of sound increases
III. The velocity remains constant.
A I only B II only C. III only
D I and II only E I and III only.
(SC/GCE)

11 In a stringed instrument like a guitar or violin, high pitched notes are produced by strings that are
A thin and short
B thick and short
C thin and long
D thick and long
E none of the above

12 When the length of a vibrating string is tripled, its frequency of vibration
A becomes six times the former value
B becomes one-third its former value
C becomes three times its former value
D becomes one-sixth the former value
E remains the same.

13 Which of the following statement(s) is/are false?
A In a flute, the air column is set into vibration by an air jet.
B In the reed-type organ pipe, the air column is caused to vibrate by the vibration of the reed.
C An open organ pipe will resonate to all harmonics.
D In a woodwind instrument such as flute, the effective length of the air column cannot be altered.
E In an instrument such as a trumpet, the vibrating air column is caused by the vibration of the player's lips.

14 a. Describe an experiment to show how the frequency of the note emitted by a vibrating string depends on the tension in the string.
b. Draw diagrams to show a vibrating string fixed at both ends emitting
(i) the fundamental frequency
(ii) the first overtone
(iii) the second overtone
c. How are the lengths of the waves in the three modes in (b) above related?

15 Explain the following terms: *Forced vibration*, *Echo*, *Reverberation*, *Resonance*, *End correction*.

Describe the resonance-tube method of measuring the wave-length in air of the sound emitted by a vibrating body. Why is the resonance tube so called?

If the shortest length of the tube for resonance is 0.1 m and the next resonant length is 0.35 m what is the frequency of vibrations? Assume the speed of sound in air is 340 ms^{-1}.

Why is it necessary to obtain more than one position of resonance?

16 What is a mechanical wave? Describe with the aid of a diagram, experiment to show that sound needs a material medium for transmission. State three characteristics of sound and mention the factor on which each depends.
(SSCE)

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17 Describe a simple experiment to determine the velocity of sound in air, stating two possible sources of error and the precautions that should be taken to reduce them.

Explain how echoes are formed. State one advantage and one application of echoes.

Two men are standing on the same side of a high wall and at the same distance from it such that they are 400 m apart. When one man fires a pistol, the other hears the first report 1.2 s after seeing the flash and the second report 0.5 s after the first. Explain why he hears two reports and calculate the

a. velocity of sound in air
b. perpendicular distance of the men from the wall.

(SC/GCE)

18 Explain the difference between
(i) a musical note and a noise
(ii) echo and reverberation (iii) intensity and loudness
(iv) free vibration and forced vibration
(v) overtones and harmonics.

A man, facing a high wall, notices that an echo is heard 4.0 seconds after he makes a sharp sound. After walking 200 m directly towards the cliff, an echo is heard after 3.4 seconds. Calculate (i) the speed of sound (ii) How far he was from the wall when he made his second observation of the echo.

19 Explain the meaning of the *frequency* of sound. Describe how a disc siren is used to produce sound.

A metal disc has 50 evenly spaced holes close to its rim. When it is rotated and an air jet plays on the holes, a note is heard. Explain this and calculate the speed of rotation which produces a note of frequency 250 Hz.Calculate the frequency of the note produced when a disc of 60 holes is rotated at the same rate.

20 a. If the speed of sound in air is 330 ms^{-1} at 0°C. What is the speed (i) at 15°C (ii) 100°C?

b. The length of the vibrating air column of a simple resonance tube can be altered by adjusting a water level. Resonance is found for a fork of frequency 440 Hz when the length of the air column is 18.8 cm and again when it is 57.3 cm. Explain how each of these positions occurs; calculate the speed of sound in the air in the tube and the position of the next resonance as more water is run out.

What would be the corresponding resonance distances if the tube were filled with a gas in which the speed of sound is 264 ms^{-1}?

21 Explain
(i) Why one is able to distinguish between a note produced by a violin and a guitar.
(ii) Why thunder is usually heard some seconds after a flash of lightning is seen.
(iii) How the depth of a seabed can be determined using the properties of sound.
(iv) The possible causes of poor acoustics in a concert hall and how one can rectify these.

22 Determine the pitch or frequency of the fundamental, and also of the first two overtones of (a) a closed pipe (b) an open pipe, if each pipe is 67 cm long and the temperature is 20°C. Take the speed of sound at 20°C as 343 ms^{-1}.

23 Write down the relationship between the frequency emitted by a string and its length, tension and mass per unit length.

A string of mass 3 grams has a length of 60 cm. What must be the tension in the string so that when vibrating transversely its first overtone has a frequency 200 per sec?

24 If each line in the following table refers to a definite frequency, calculate the values of *a* – *e*.

	Speed of sound in air in ms^{-1}	Wave-length in air in m	Speed of sound in material *M* in ms^{-1}	Wave-lenth in material *M* in m
1	330	0.15	*a*	0.60
2	350	0.06	4200	*b*
3	340	0.2	*c*	0.7
4	*d*	0.25	4200	3.1
5	350	*e*	1400	5

25 (i) What is meant by *resonance*?
(ii) Outline the necessary steps taken in a simple experiment to illustrate resonance.
(iii) Explain why a vibrating tuning fork sounds louder when its stem is pressed against a table top than when held in air.

(SSCE)

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26 a. (i) What is meant by *resonance*? Illustrate your answer with two examples of resonance.

(ii) Describe a laboratory experiment to determine the velocity of sound in air, stating one precaution taken in performing the experiment.

b. State *three* characteristies of a musical note and the factor on which each depends.

(SSCE)

27 (a) Illustrate, using labelled diagrams only, a sonometer wire of length l, vibrating and its

(i) fundamental

(ii) first overtone

(iii) second overtone.

(b) A tuning fork vibrating at a frequency of 512 Hz is held over the top of a jar filled with water and fitted with a tap at the bottom. If the jar is 60 cm tall and the speed of sound is 350 ms^{-1}, determine the possible resonance positions(s).

(SSCE)

28 (a) (i) What is an echo?

(ii) State two useful applications of echoes.

(iii) Why are the walls, floors and ceilings of a recording studio heavily padded?

(b) (i) Explain *timbre* and *overtones*.

(ii) What is *resonance*?

(c) As a ship approaches a cliff, its siren is sounded and the echo is heard in the ship after 12 seconds. 2.1 minutes later the siren was sounded again and the echo was heard 8 seconds later. If the speed of sound in air is 340 ms^{-1}, calculate the velocity at which the ship was approaching the cliff.

(SSCE)

29 (i) What are beats?

(ii) A sonometer wire has a frequency of 259 Hz. It is sounded alongside a tuning fork of frequency 256 Hz. Calculate the beat frequency.

(iii) The sonometer above is under a tension of 1200 N. If a metre of the wire has a mass of 0.03 kg, calculate the length *l* of the wire when it is vibrating in the fundamental mode.

(SSCE)

30 What are hearing aids? Outline the various parts of hearing-aids. How does hearing aid function?

31 (a) State the three characteristics of sound and the factor on which each of them depends.

(b) Explain resonance as applied to sound.

(c) What role dose echo play in the construction of a concert hall?

(d) The surface of an eardrum (assumed circular) has a radius of 2.1 mm. It resonates with an amplitude of 0.8×10^{-7} m as a result of impulses received from an external body vibrating at 2400 H_z. If the resulting pressure charge on the eardrum is 3.6×10^{-5} Nm^{-2}, calculate:

(i) the period of oscillation

(ii) the velocity

(iii) acceleration

(iv) force. ($\pi = 3.14$)

(SSCE)

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16 PRESSURE IN FLUIDS

The term *fluid* refers to any substance that can flow. Hence it applies to both liquids and gases.

This chapter will discuss the concept and definition of pressure, Pascal's principle and its application to hydraulic press and car brakes, and the dependence of pressure on the depth of a point below a liquid surface. We shall also discuss some simple instruments used to measure pressure and those whose operations depend on the effect of pressure. Finally we shall treat the determination of the relative density of liquids with U-tube and Hare's apparatus.

At the end of the chapter the student should be able to:

1 Explain the concept of pressure and give a concise definition of it.
2 State Pascal's principle and discuss its application to hydraulic press and car brakes.
3 Obtain the expression $p = h\rho g$ relating the pressure of a point below a liquid surface with the depth of the point below the liquid surface.
4 Describe the simple barometer, manometer, siphon, syringes and pumps.
5 Determine the relative density of liquids with U-tube and Hare's apparatus.

16.1 The Concept of Pressure

Suppose we apply a force F perpendicularly on an area A of the wall as shown in fig. 16.1, we say that a pressure P is exerted on the wall by the force F.

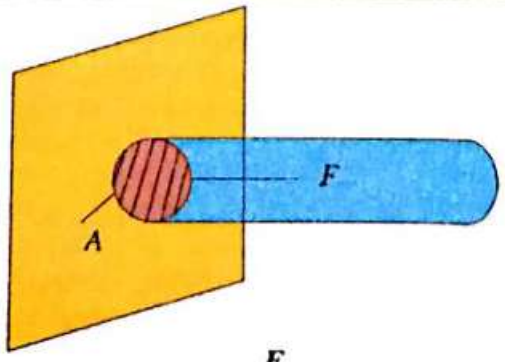


Fig. 16.1 Pressure is $\frac{F}{A}$

Thus force and area are involved in the definition of pressure but that force must act perpendicularly to the area under consideration.

Pressure is defined as the perpendicular force per unit area acting on a surface.

Symbolically we can write:

$$P = \frac{F}{A} \qquad 16.1$$

Since force F is measured in Newtons and the area A is measured in square metres, the S.I. unit of pressure P is Newtons per square metre (Nm^{-2}). Another unit of pressure is the Pascal (Pa).

$1\ pa = 1\ Nm^{-2}$

A third unit of pressure is the *bar*

$1\ bar = 10^5\ Nm^{-2} = 10^5\ pa$

Pressure is a scalar quantity, it has magnitude but no direction.

Example 16.1

Calculate the pressure exerted on the ground by a person weighing 70 kg whose two feet cover an area of 400 cm^2. If the man stands on one foot, what is the new pressure he exerts on the ground.

Solution

$$P = \frac{F}{A}$$

Force due to the weight of the person is given by

$F = mg$ N

$= 70 \times 10$ N $= 700$ N

where $g = 10\ ms^{-1}$.

Area covered by the two feet $= 400\ cm^2$

or $\frac{400}{100 \times 100} = 0.04\ m^2$

$$P = \frac{F}{A} = \frac{700}{0.04} = \underline{17.5 \times 10^3\ Nm^{-2}}$$

If the person stands on one foot, area is halved, i.e. $A = 0.02$

$$P = \frac{700}{0.02} = \underline{35 \times 10^3\ Nm^{-2}}$$

Thus the pressure is twice as much.

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Example 16.2

The pressure exerted by a force on a portion of a wall $0.5 m^2$ is 100 pa. Calculate the value force if it acts perpendicular to the wall.

Solution

$$P = \frac{F}{A}$$

$$F = PA = 100 \times 0.5\ (Nm^{-2} \times m^2)$$

$$= \underline{50\ N}$$

Example 16.1 shows that the less the area, the more the pressure exerted by a constant force. Hence the pressure exerted on the ground by a girl wearing a narrow heeled shoe is much greater than that exerted when the same girl wears flat heeled shoe. To cut a piece of meat or any object with a sharp knife requires less force than when we use a blunt knife. The area of the edge of a sharp knife is very small compared to that of a blunt knife. Therefore only a little effort needs to be exerted. It is for this reason that meat sellers in the markets are always sharpening their cutting knife. For the same reason the sharp point of a needle pierces the skin very easily.

16.2 Pressure in Liquids

Unlike solids which exert pressure only downwards, liquids exert pressure in all directions. Other important characteristics of liquid pressure are

(i) The pressure at all points at the same level within a liquid is the same.

(ii) The pressure in a liquid increases in direct proportion to the depth of the liquid.

(iii) The pressure in different liquids at the same depth varies directly with density.

Activity 16.1: To show that pressure exerted by a liquid increases with depth

(i) Make holes at different heights on the side of a tall vessel placed vertically on a laboratory table. Ensure the holes are slightly staggered to avoid collision of the jets of water with each other.

(ii) Plug the holes and fill the vessel with water

(iii) Remove the plugs and observe the jets of water from the holes.

(iv) Maintain the water level in the vessel with water from a tap.

It will be observed that the lowest hole *C* produces the most powerful jet. Water from *B* spurts out with greater force than that from *A* but with less force than that from *C* as shown in fig 16.2.

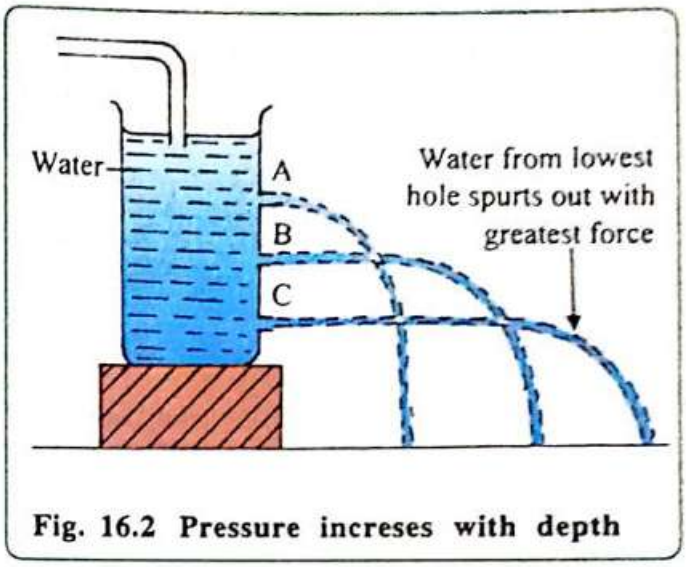


Fig. 16.2 Pressure increses with depth

This shows that the pressure due to the water increases with depth. It is for this reason that water rushes out with greater force from the taps in the lower floor than from the higher floors of a high-rise building. In the construction of dams, the engineers ensure that the bottom of the dam is made much thicker than the top since the pressure of the water increases with depth.

Activity 16.2: To investigate the pressure at different depths and directions in a liquid and the variation of pressure with liquid density.

(i) Tie a thin plastic membrane firmly over the mouth of a thistle funnel. Attach the thistle funnel to a U-tube by means of a stout rubber tubing as shown in fig. 16.3a. Pour a little coloured water in the U-tube.

(ii) Press the plastic membrane lightly with the finger and notice the corresponding change in the water levels of the U-tube. The greater the pressure on the plastic, the greater is the difference in the water levels in the U-tube. So, the U-tube with the attached thistle funnel acts as a convenient pressure gauge.

(iii) Place the funnel at a particular depth of the water in the glass vessel and note the difference (h) in the water levels in the U-tube. Move the mouth of the funnel so that it points in different directions but at the same depth and observe that the difference (h) in the water levels in the U-tube remains the same. This shows that the pressure in the liquid at a particular depth is the same in all directions.

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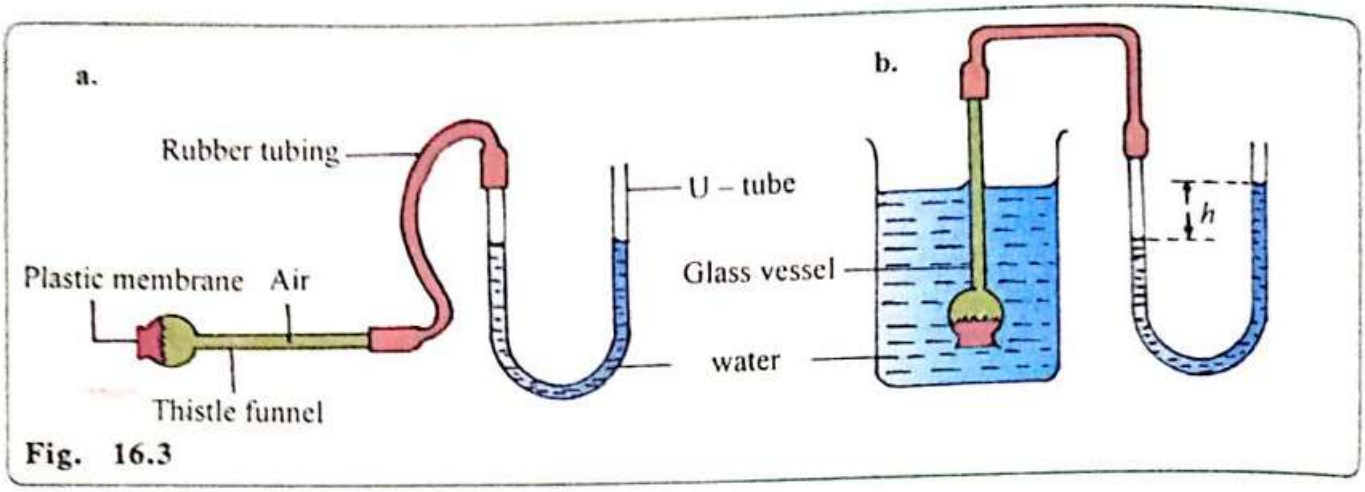


Fig. 16.3

(iv) Move the thistle tunnel to different depths in the liquid and note the variation of the height (h) in the U-tube with depth. It will be observed that the pressure of the water increases with depth.

(v) Now empty the vessel and pour a denser liquid (e.g. Copper sulphate solution, glycerine or paraffin) into the vessel. Repeat the experiments in section (ii) and (iii) and record your observations.
It will be found that at a particular depth, the pressure increases as the density (g) of the liquid increases.

The Pressure Formula: $p = h\rho g$

The pressure due to a liquid is known as hydrostatic pressure. We can derive a general formula for the hydrostatic pressure p at a depth h in a liquid of density ρ.

Consider an area A of a liquid at a depth h below the surface of the liquid. Imagine that a vertical cylinder of cross-sectional A is made of the given liquid and placed such that the centre of its base coincides with the point R of depth h below the liquid surface (fig. 16.4).

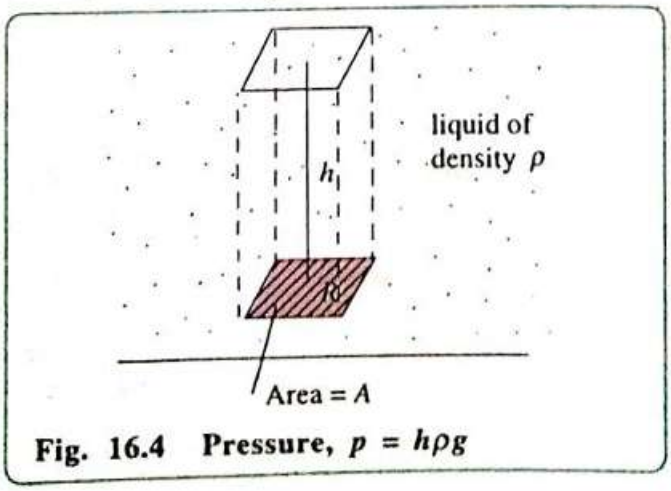


Fig. 16.4 Pressure, $p = h\rho g$

Volume of cylinder of liquid = Ah
Mass of liquid = $V\rho = Ah\rho$
Weight of liquid = $Ah\rho g$
where g is the acceleration due to gravity.

$$\text{Pressure, } p = \frac{F}{A} = \frac{Ah\rho g}{A}$$
$$= h\rho g$$

Hence $$p = h\rho g \qquad 16.2$$

When h is in metres, ρ in kgm^{-3}, and g is in ms^{-2}, then p is in Newton per square metre (Nm^{-2})

Equation 16.2 shows that pressure is directly proportional to h the depth below the liquid surface, and to the density ρ of the liquid. It is independent of the cross-sectional area of the containing vessel.

Example 16.3

Calculate the pressure of water at the bottom of a dam 200 m deep if the density of water is 1000 kgm^{-3}. Take $g = 10\ ms^{-2}$.

Solution

$$p = h\rho g = 200 \times 10^3 \times 10$$
$$= 2 \times 10^6\ Nm^{-2}$$

Example 16.4

A jar 40 m deep is full of liquid of relative density 1.3. At what depth below the surface of the liquid is its pressure equal to the pressure exerted by a column of 5 cm of mercury? (Relative density of mercury = 13.6, $g = 10\ ms^{-2}$).

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Solution

$p = h\rho g$

Pressure exerted by a column of liquid is given by

$$p_1 = h_1 \rho g = h_1 \times 1.3 \times 10$$

Pressure exerted by a column of mercury of height, 5 cm is given by

$$p_2 = h_2 \rho g = .05 \times 13.6 \times 10$$

When $p_1 = p_2$ we have

$$h_1 \times 1.3 \times 10 = 0.05 \times 13.6 \times 10$$

$$h_1 = \frac{0.05 \times 13.6 \times 10}{1.3 \times 10} \text{ m} = 0.523 \text{ m}$$

$$= \underline{52.3 \text{ cm}}$$

16.3 Atmospheric Pressure

The atmosphere is the whole volume of air surrounding us. The atmosphere consists a mixture of gases, some dust particles and water vapour. It extends to a height of a few hundred kilometres above sea level.

The atmosphere exerts some pressure on the surface of the earth, because its air has a lot of weight. The pressure is called *atmospheric pressure*. At sea level, it is about 10^5 Pa decreasing upwards with height. We do not feel this atmospheric pressure because every cell of our body maintains an internal pressure that just balances the external pressure.

In the same way the pressure inside a balloon balances the outside pressure of the atmosphere. An automobile tyre can however maintain pressure much greater than the external pressure because of its rigidity. Hence tyre gauges register the pressure over and above atmospheric pressure.

Experiments to show that air exerts pressure

a. Crushing can experiment

We can demonstrate that air exerts pressure by the following simple experiment. Pour a little water into a thin-walled can such as a kerosene tin. Boil the water for a few minutes so as to drive out the air in the can through the steam produced. Remove the flame and insert in the opening an air tight rubber stopper. Pour some cold water on the can and observe how the tin collapses. We can explain the collapse of the can in the following way: When cold water is poured on the can, the steam inside it condenses and leaves a partial vacuum in the can. Under this condition the external air pressure is no longer counterbalanced by the pressure within the can. The can is then crushed by this excess air pressure. The experiment thus shows that air exerts pressure.

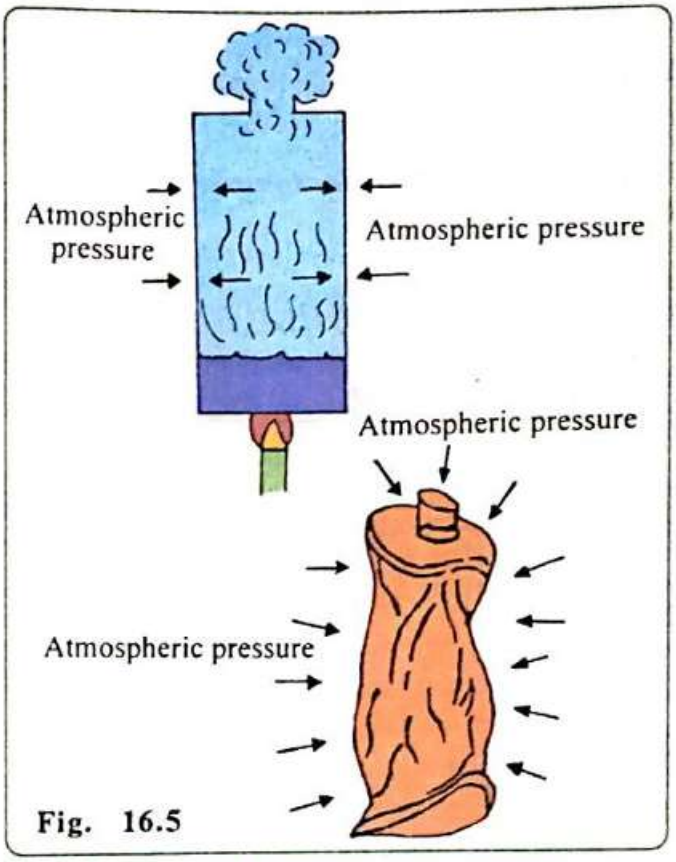


Fig. 16.5

b. The inverted beaker experiment

Fill a glass beaker with water and cover its mouth with a thin glass cover or a cardboard paper. Invert the beaker under a trough of water pressing the cardboard firmly to the rim of the beaker. Remove your hand from the cover and observe that the cardboard still remains in position, thus preventing the water from running out.

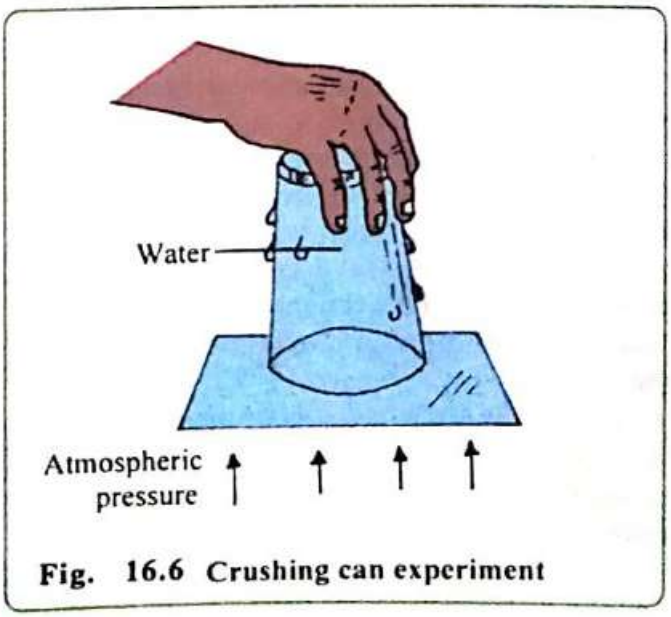


Fig. 16.6 Crushing can experiment

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The pressure due to the weight of the water in the beaker is counter-balanced by the upward pressure of the air on the cardboard.

The measurement of gas pressure and atmospheric pressure by barometers has been treated in chapter 10.

Applications of atmospheric and gas pressure

(i) The Siphon

The siphon is a device used to remove liquids from receptacles which cannot be lifted or easily reached, or cannot otherwise be emptied more conveniently. Liquids can only be transferred from points at higher levels to points at lower levels.

It consists of a bent flexible tube with one end dipping below the liquid surface in the vessel of liquid and the other end at a point of lower level. As long as the tube is filled with the liquid, there is a continuous flow from the higher point to the lower point.

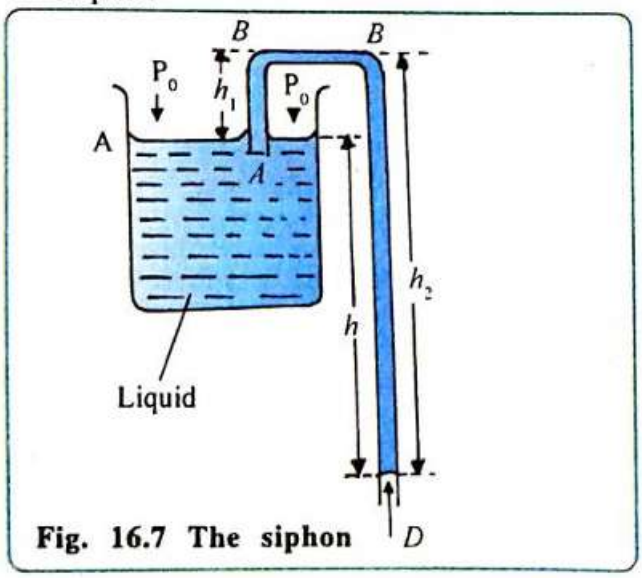


Fig. 16.7 The siphon

A simplified explanation of the action of the siphon is that liquid flows from A at higher level to D at lower level because of the excess pressure $h\rho g$ between A and D. The resultant pressure tending to force the liquid up to the tube at A is given by $p_0 - h_1\rho g$ where p_0 is the atmospheric pressure and h_1 is the height of the column of liquid BA. The resultant pressure tending to force the liquid up at D is given by $p_0 - h_2\rho g$, where h_2 is the height of the column of liquid BD. Since h_2 is greater than h_1, the resultant pressure acting upwards at A is much greater than that at D. The excess pressure driving the liquid from A to D is given by $h\rho g$ where $h = h_2 - h_1$ the difference in height between D and A.

For the siphon to function efficiently,

(i) the tube must be filled by the liquid,

(ii) D must be lower than A

(iii) the height h_1 must be less than the barometric height of the liquid (10.3 m for water).

The siphon can be used to empty a petrol or water tank.

(ii) The Syringe

The devise consists of a piston which fits tightly into a cylinder as shown in fig. 16.8. The cylinder has a nozzle at one end.

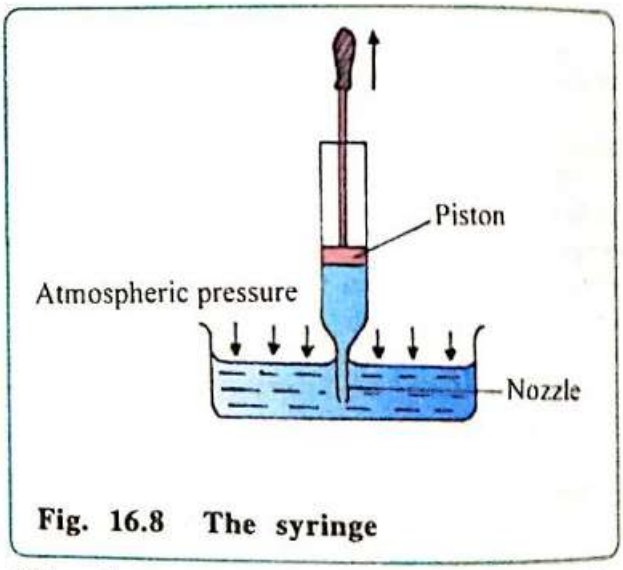


Fig. 16.8 The syringe

When the nozzle is dipped into a liquid and the piston is pulled up, atmospheric pressure forces the liquid up the cylinder. When the piston is pushed down, the liquid is forced out or ejected through the nozzle.

The syringe is commonly used for injecting liquid medicine into the body.

(iii) Bicycle Pump

This is a compression pump used to compress air from the atmosphere into a bicycle tyre. The component parts are as shown in fig. 16.9. It includes a piston which can move in and out of a cylindrical tube or barrel. At the end of this piston is a leather washer. The pump is connected to the bicycle tyre which usually has a rubber valve in it. When the piston is drawn out, the air pressure in the barrel is reduced below atmospheric pressure. Air from outside the pump then flows past the leather washer in the piston into the barrel with the tyre valve remaining closed. When the piston is

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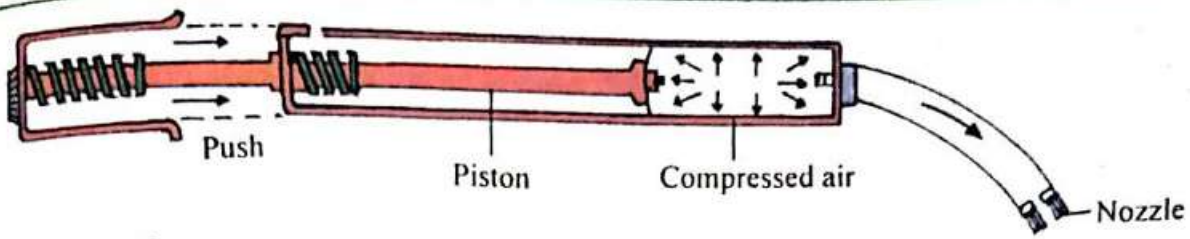


Fig. 16.9 Bicycle pump

pushed down, the pressure of the air inside the barrel increases forcing the washer against the sides of the barrel. As the air cannot escape, it is forced out of the pump through the now open valve into the inner tube of the tyre. In this way air from outside is forced into the tyre with each complete stroke of the piston.

(iv) Lift Pump or Common Pump

A pump is a device that can produce a pressure difference.

The lift pump is a device for raising water from wells, or for lifting kerosene from a tall tank. It consists of a piston that can move up and down a barrel. The piston fits tightly into the barrel. There are two valves V_1 and V_2 fixed on the piston and on the bottom of the barrel respectively. Both valves are fixed in such a way that they open upwards.

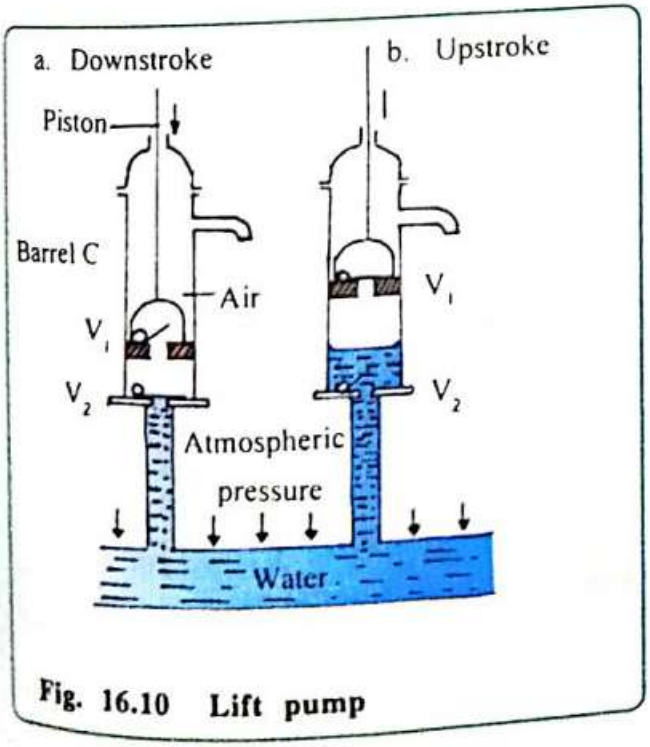


Fig. 16.10 Lift pump

The pump functions in four basic steps:

(i) When the piston moves downwards, the valve V_2 closes and V_1 opens to allow the air to move out through the piston.

(ii) When the piston moves upwards, the pressure inside the barrel decreases due to the increase in volume of the air inside it (Boyle's law). V_1 closes because of the air pressure above the piston, V_2 opens and atmospheric pressure forces water into the space above V_2.

(iii) In the next downward stroke, increase in pressure forces valve V_2 to close and V_1 to open, thus allowing water to flow above V_1.

(iv) The next upward stroke closes V_1 and the water is raised to the outlet pipe D where it escapes.

The lift pump cannot lift water above a height of about 10 metres because atmospheric pressure cannot support a column of water above 10 metres high.

(v) Force Pump

This pump is used when it is required to lift water continuously to a height above 10 m. Its component parts consists of a piston that can move up and down a barrel, and a side tube of the barrel connecting it to an air chamber. There are two valves V_1 and V_2 placed as shown in fig. 16.11 and an exit pipe E projecting into the air chamber.

(i) As the piston moves up, atmospheric pressure acting on the water surface opens valve V_1 and pushes water into the barrel. Valve V_2 is closed at this stage by the outside pressure acting on it. (fig. 16.11a)

(ii) As the piston moves down, it compresses water in the barrel closing valve V_1 and opening V_2, and letting in water into the side tube and the air chamber. The exit pipe E projecting into the air chamber traps in some

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air at the top of the chamber. This compressed air acts as a cushion, so that instead of coming through in bursts, the water comes through more evenly providing a steady flow of water.

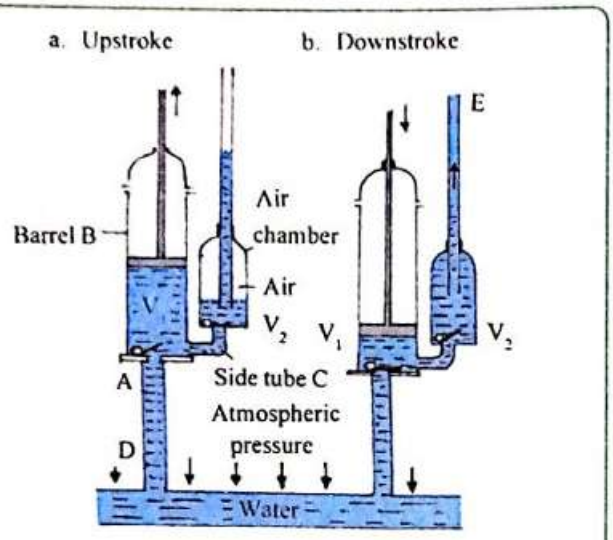


Fig. 16.11 Force pump

The maximum height to which the water can be raised depends on (a) the force available to work the pump (b) the ability of the pump and its working parts to withstand the pressure of the long column of water at the exit pipe.

16.4 Pascal's Principle

This principle deals with the transmission of pressure in fluids.

Pascal's principle states that the pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel

Thus when we apply an air pressure at a point of a bicycle tyre, the pressure in the tyre is increased equally at every other point along the tyre. There is an equal transmission of the applied pressure to all other points of the tyre.

Pascal's principle can be illustrated by a simple experiment using a piece of apparatus as shown in fig. 16.12. The apparatus consists of a bulb with holes of uniform size. Inside the bulb is a piston. The bulb is immersed in water and the piston is drawn up. Water rushes into the bulb through the holes. The bulb is now removed from the water and the piston is pushed hard so as to apply sufficient pressure to the water.

It will be observed that the water flows out from the holes with equal force, thus showing that the pressure applied to the piston has been transmitted uniformly throughout the water.

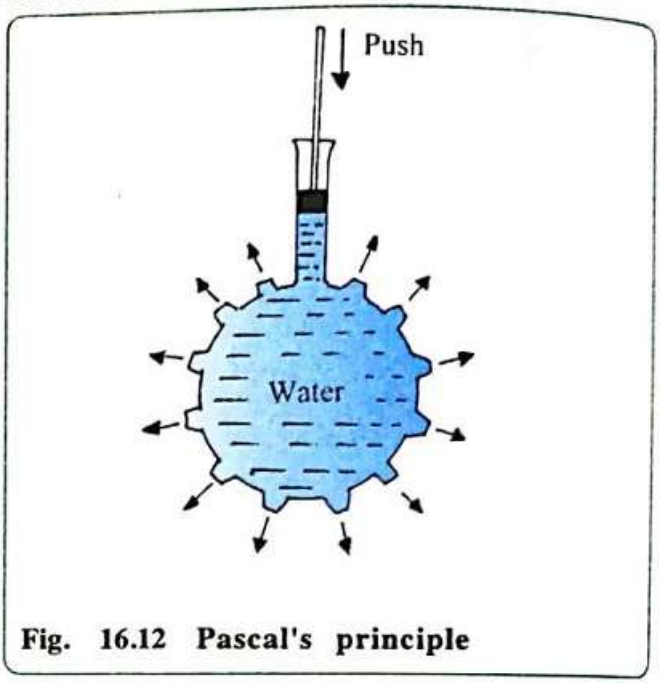


Fig. 16.12 Pascal's principle

Applications of Pascal's Principle

a. The Hydraulic press

Pascal's principle is illustrated by the operation of the hydraulic press. The press consists of two cylinders R and S of different diameters. The cylinders filled with water or oil are joined by a connecting tube as shown in fig 16.13.

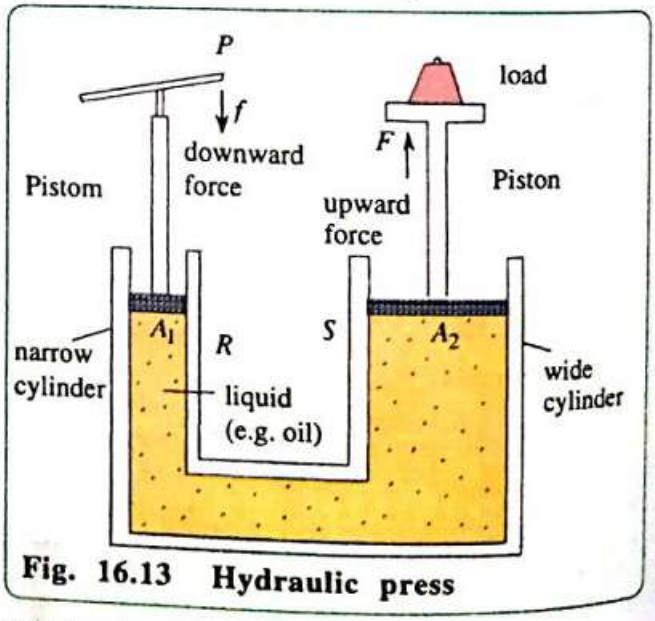


Fig. 16.13 Hydraulic press

A tight piston is fitted in each cylinder.

The piston or plunger of small cross-sectional area A_1 is used to exert a small force f directly on a liquid such as oil. The pressure $p = \frac{f}{A_1}$ is transmitted through the connecting pipe to the

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larger cylinder equipped with a larger piston of area A_2. Since the pressure is the same in both cylinders, we have:

$$p = \frac{f}{A_1} = \frac{F}{A_2} \qquad 16.3$$

or $$F = \frac{A_2 f}{A_1} \qquad 16.4$$

$$\frac{F}{f} = \frac{A_2}{A_1} \qquad 16.5$$

Since A_2 is very large compared to A_1, F will be very large compared with f. Hence we can apply a small input force(f) to obtain a large output force(F).

This large force can be used to raise a heavy car as in the hydraulic jack or to compress bags of cotton or wool as in a hydraulic press. Thus the mechanical advantage for a perfect hydraulic press is given by $\frac{F}{f}$ which from equation 16.5 is equal to $\frac{A_2}{A_1}$, the velocity ratio. Thus the greater the ratio of the cross-sectional areas of the pistons, the greater the mechanical advantage of the press.

To calculate the velocity ratio of the hydraulic press, let us suppose that when the smaller piston is depressed downward through a distance X, the larger piston moves up by a distance Y. Since the volume of liquid transferred is the same, we have that:

volume of liquid transferred

$$= X \cdot A_1 = Y \cdot A_2 \qquad 16.6$$

But velocity ratio $= \frac{\text{distance moved by effort}}{\text{distance moved by load}}$

$$= \frac{X}{Y} = \frac{A_2}{A_1} \qquad 16.7$$

Barber's chairs, dentist chairs, vehicle lifts and jacks and hydraulic brakes are all devices that make use of the principle of the hydraulic press.

b. Motor-Car foot brake

Car brakes use the hydraulic principle to transmit pressure from the foot-pedal to the brakes. Such a system slows down each wheel equally, thus reducing the danger of skidding. The device is as shown in fig. 16.14. When the brake pedal is pushed forward, the movement of the piston in the master cylinder compresses the oil inside the tubes and transmits the pressure to the brake shoes forcing each apart. The shoes now push on the revolving brake drums fixed to the wheels. This slows down the rotation of the wheels or stops them entirely depending on the force applied to the brake drum. When the pressure on the brake pedal is released, a return spring connected to the two brake shoes contracts and pulls them away from the brake drum. The wheel is then free again to rotate.

Example 16.5

The radius of the large cylinder of a hydraulic press is 20 cm and that of the small cylinder is 5 cm. If a force of 100 N is applied on the small piston, calculate the force on the large piston. If the piston of the small cylinder moves through a distance of 0.4 m through what distance can the load on the large piston be raised?

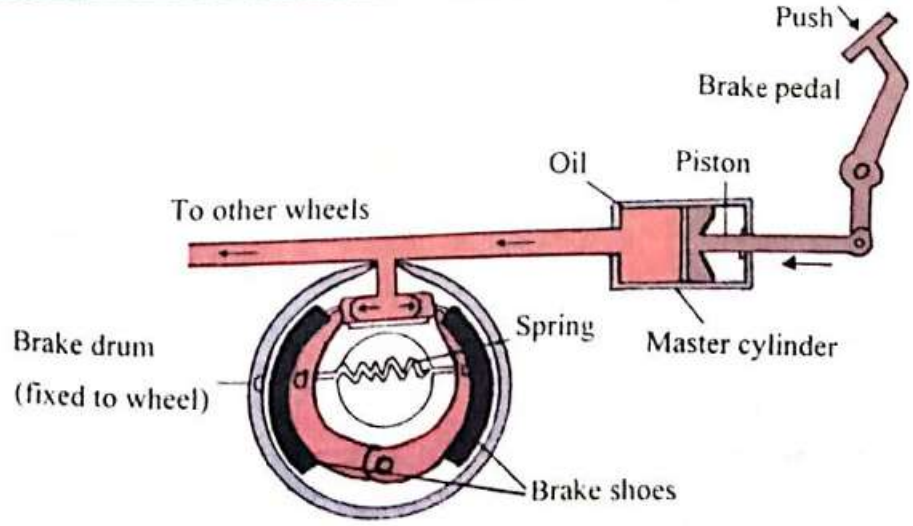


Fig. 16.14 Motor-car foot brake

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Solution

Area of smaller piston

$= \pi r_1^2 = (\pi \times 0.05 \times 0.05)\ m^2$

Area of large piston $= \pi r_2^2 = (\pi \times 0.2 \times 0.2)\ m^2$

Pressure applied to small piston

$$= \frac{F}{A} = \frac{100}{\pi \times 0.05 \times 0.05}\ Nm^{-2}$$

This pressure is transmitted to the large piston. If force on large piston $= F_1$ Newtons, then

$$\frac{F_1}{\pi \times 0.2 \times 0.2} = \frac{100}{\pi \times 0.05 \times 0.05}$$

$$F_1 = \frac{100 \times \pi \times 0.2 \times 0.2}{\pi \times 0.05 \times 0.05}$$

$= \underline{1600\ N}$

When the small piston moves through 0.4 m, volume of liquid moved = Area × distance

$= \pi \times 0.05 \times 0.05 \times 0.4\ m^3$

This volume is transferred to the large piston.

Distance moved by large piston $= \frac{\text{volume}}{\text{area}}$

$$= \frac{\pi \times 0.05 \times 0.05 \times 0.4}{\pi \times 0.2 \times 0.2}$$

$= \underline{2.5\ cm}$

16.5 To determine the relative density of liquids using U-tube and Hare's apparatus

These two methods of measuring the relative density of a liquid are based on the pressure exerted by a liquid column.

a. By U-tube

A little mercury is poured into the U-tube of uniform cross-section to cover the length of the bend. A liquid of known density e.g. water is poured into the right-hand limb and a liquid e.g. oil, of unknown density is poured into the left hand limb until the mercury has the same level in each limb as shown in fig. 16.15. The heights of the water and of the oil are measured and the density of the oil is calculated as follows:

Let height of water column above the common level C be h_w

Let height of the oil column above the common level C be h_l

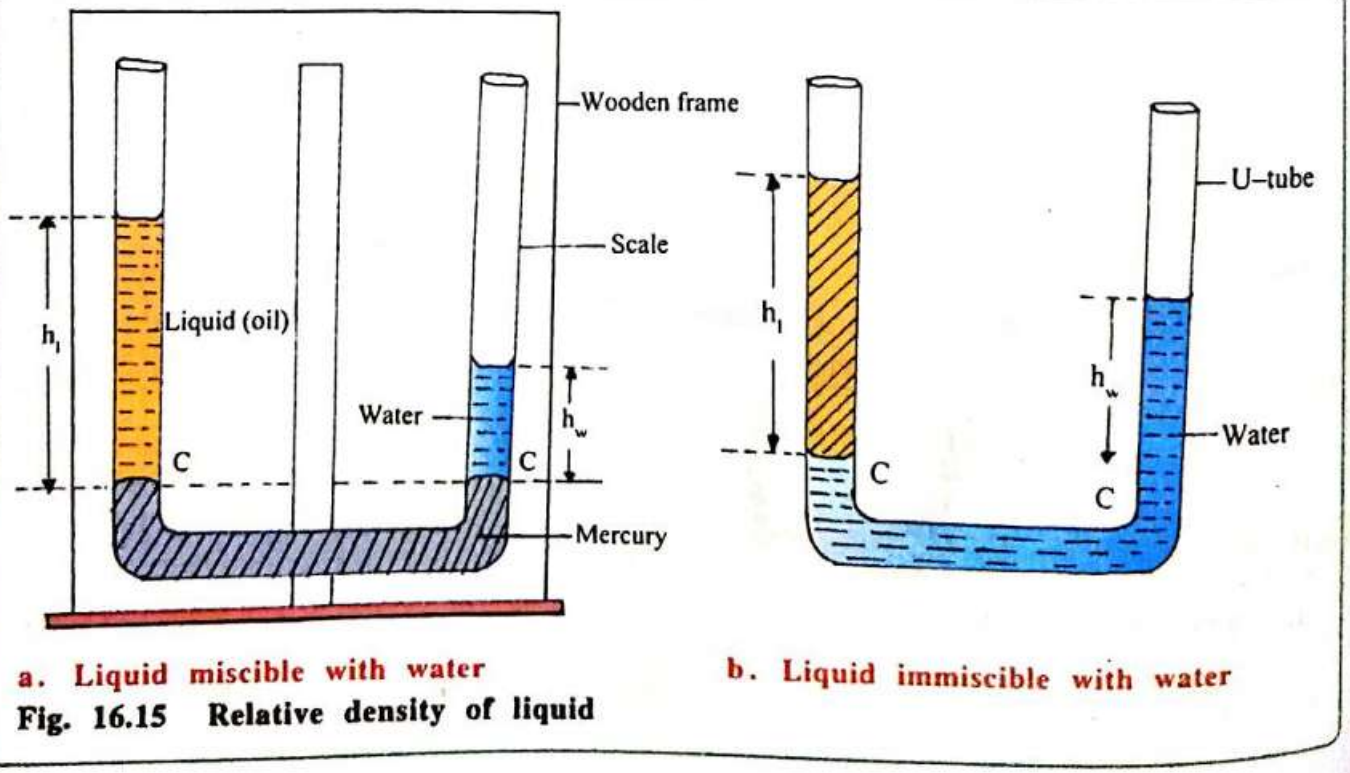
Pressure exerted at the common level C is the same in both limbs

$$\therefore \quad p = h_w \rho_w g = h_l \rho_l g \qquad 16.8$$

where ρ_l, ρ_w are the densities of the liquid and water respectively

Hence $h_w \rho_w = h_l \rho_l$

$$\frac{\rho_l}{\rho_w} = \frac{h_w}{h_l} = \frac{\text{height of water column}}{\text{height of liquid column}}$$



a. Liquid miscible with water

b. Liquid immiscible with water

Fig. 16.15 Relative density of liquid

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Hence relative density of the oil is

$$= \frac{\text{height of water column}}{\text{height of liquid column}}$$

By pouring a little more liquid, repeat the experiment five times to obtain a set of readings of h_w and h_l, then plot a graph of h_w against h_l. The slope of the graph gives a better estimate of the relative density.

If the liquid does not mix with water, we can dispense with the mercury in the experiment. We fill the U-tube with water to about one-third of the way up each limb. The liquid is then poured through one limb until the difference in the levels of water and liquid become clearly distinct. (fig. 16.15b). The heights h_l and h_w are measured as before from above the common level C. More liquid is poured and the experiment is repeated as before. The relative density of the liquid is calculated as already described.

b. Hare's Apparatus

The apparatus is as shown in fig. 16.16.

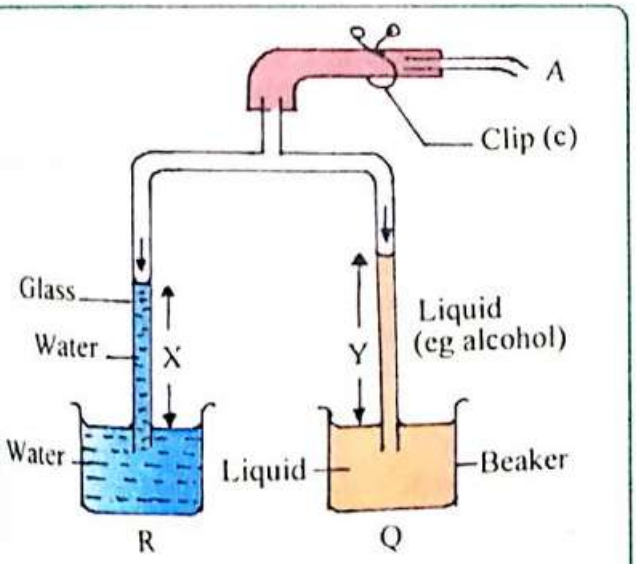


Fig. 16.16 Hare's apparatus

It consists of two vertical wide bore glass tubes connected at the top by a glass T-shaped tube bearing a clip C. The lower ends of the tube dip respectively into beakers R and Q of water and a liquid (e.g. alcohol) whose relative density is required. With the clip C open, air is sucked out of the tubes at A and the clip is then closed. Removal of air brings about a reduction in the pressure inside the tubes. As a result atmospheric pressure pushes the liquids up the tubes. The rise of liquids continues until the pressures exerted at the base of each column are equal to the atmospheric pressure. The heights X and Y, of the water and the liquid respectively are measured. The experiment is repeated about five more times allowing the liquids to rise higher in the tubes in each case.

The corresponding heights above the water and liquid surfaces $X_1, \ldots, X_6$ and $Y_1, \ldots, Y_6$ respectively are measured and recorded.

A graph of Y against X is plotted and the slope gives the relative density of the liquid.

The theory is as follows:

The pressure at the base of each column is the same.

Hence $p + x\rho_w g = p + y\rho_l g$

where ρ_w = density of water, ρ_l = density of liquid and p is the pressure of the air in the tube above each liquid.

$\therefore \quad x\rho_w g = y\rho_l g$

Relative density of liquid $= \frac{\rho_l}{\rho_w} = \frac{y}{x}$ = slope of graph of y against x

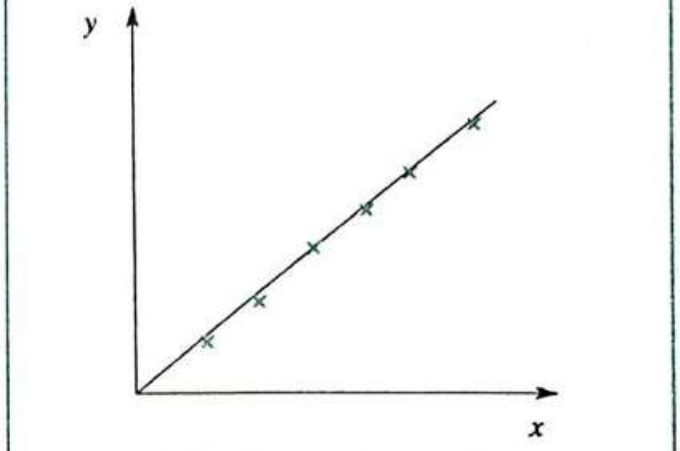


Fig. 16.17 Slope of graph = relative density

Precautions

1 Measure the height of the columns only when the levels are steady, i.e. there are no more movements of liquid levels.

2 Measure the heights from the base of the meniscus.

Hare's apparatus is used for comparing the densities of two miscible liquids. If the density of one liquid is known the other can be determined.

Summary

1 Pressure is the perpendicular force per unit area acting on a surface. It is a scalar quantity.

2 Liquids exert pressure in all directions.

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3 Pressure at a given point within a body of liquid, $p = h\rho g$, where ρ is the density of the liquid and h is the depth of the given point below the surface of the liquid.

4 We can demonstrate that air exerts pressure by the crushing can experiment and the inverted beaker experiment.

5 Pascal's principle states that the pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel.

6 The hydraulic press and the car brake system are applications of Pascal's principle.

7 We can measure the relative density of a liquid using the U-tube method and Hare's apparatus. Both methods depend on the pressure exerted by a liquid column ($p = h\rho g$).

Exercise 16

1 In a hydraulic press, a force of 20 N is applied to a piston of area 0.2 m^2. If the area of the other piston is 2 m^2, calculate the force exerted on it.

A 200 N B 50 N C 20 N
D 5 N E 2 N

(SC/GCE)

2 In an experiment to determine the density of a liquid using Hare's apparatus, the height of the water column is 12.0 cm and that of the liquid column is 16.0 cm. If the density of the water is 1.0 gcm^{-3}, calculate the density of the liquid.

A 0.75 gcm^{-3} B 1.33 gcm^{-3}
C 4.00 gcm^{-3} D 14.00 gcm^{-3}
E 28.00 gcm^{-3}

(SC/GCE)

3 Which of the following statements about the pressure in a liquid is *not* correct? The pressure

A varies with the depth of the liquid
B acts in all directions
C is affected by the density of the liquid
D changes with the temperature of the liquid
E changes with the material of the container

(SC/GCE)

4 What is the length of the liquid column in a barometer tube that would support an atmospheric pressure of 102000 Nm^{-2} if the density of the liquid is 2600 kgm^{-3}? ($g = 10ms^{-2}$)

A 10.75 m B 0.76 m C 3.92 m
D 39.23 m. (JAMB)

5 The areas of cross-section of two pistons in a hydraulic press are 2 and 150 cm^2. If a downward force of 20 N is exerted on the smaller piston, calculate the force on the larger one

A 112500 N B 1500 N
C 150 N D 15 N E 3.75 N.

(SC/GCE)

6 A pressure that can support a column of water 60 cm high supports a column of a given salt solution 50 cm high. The density of the salt solution is

A 2.4 gcm^{-3} B 1.2 gcm^{-3}
C 0.833 gcm^{-3} D 0.6 gcm^{-3}
E 3.6 gcm^{-3}

7 Explain the following

(i) The liquid from a siphon flows more quickly out of the siphon when the level of the mouth of the siphon is lower.
(ii) A lift pump cannot be used for lifting water to heights above 10 cm.
(iii) Modern vehicles use hydraulic breaking systems
(iv) The walls of a dam are much thicker at the base than at the top.
(v) Pins have sharp points
(vi) It is easier to cut meat with a sharp knife than with a blunt knife.

8 Define *pressure.*

a. What is the pressure exerted by a 50 kg girl as she places her weight on the heel of one shoe if the heel area is 0.002 m^2?

b. Calculate the pressure due to sea-water at a depth of 600 m. Take density of sea-water = 1030 kgm^{-3}.

9 State *Pascal's principle* of transmission of pressure in fluids. Describe an experiment to illustrate this law. Describe the application of this principle in the working of car-breaking system.

10 With the aid of a diagram explain the principle of the hydraulic press. Mention two uses of the hydraulic press.

A hydraulic press consists of two cylinders of cross-sectional area 0.2 m^2 and 5.0 m^2. The piston in the smaller cylinder is pushed down with a force of 100 N through a distance 0.2 m. Calculate

(i) the pressure transmitted by the fluid
(ii) the force exerted by the piston in the large cylinder

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11 Describe with the aid of a diagram how you would use the Hare's apparatus to measure the relative density of alcohol and show how the result is calculated.

If in the above experiment the lengths of the alcohol and water columns were found to be 20 cm and 15 cm respectively, calculate the relative density of the alcohol.

Explain the principle involved in drinking a liquid using a straw.

12 With the aid of a diagram, describe the action of a siphon.

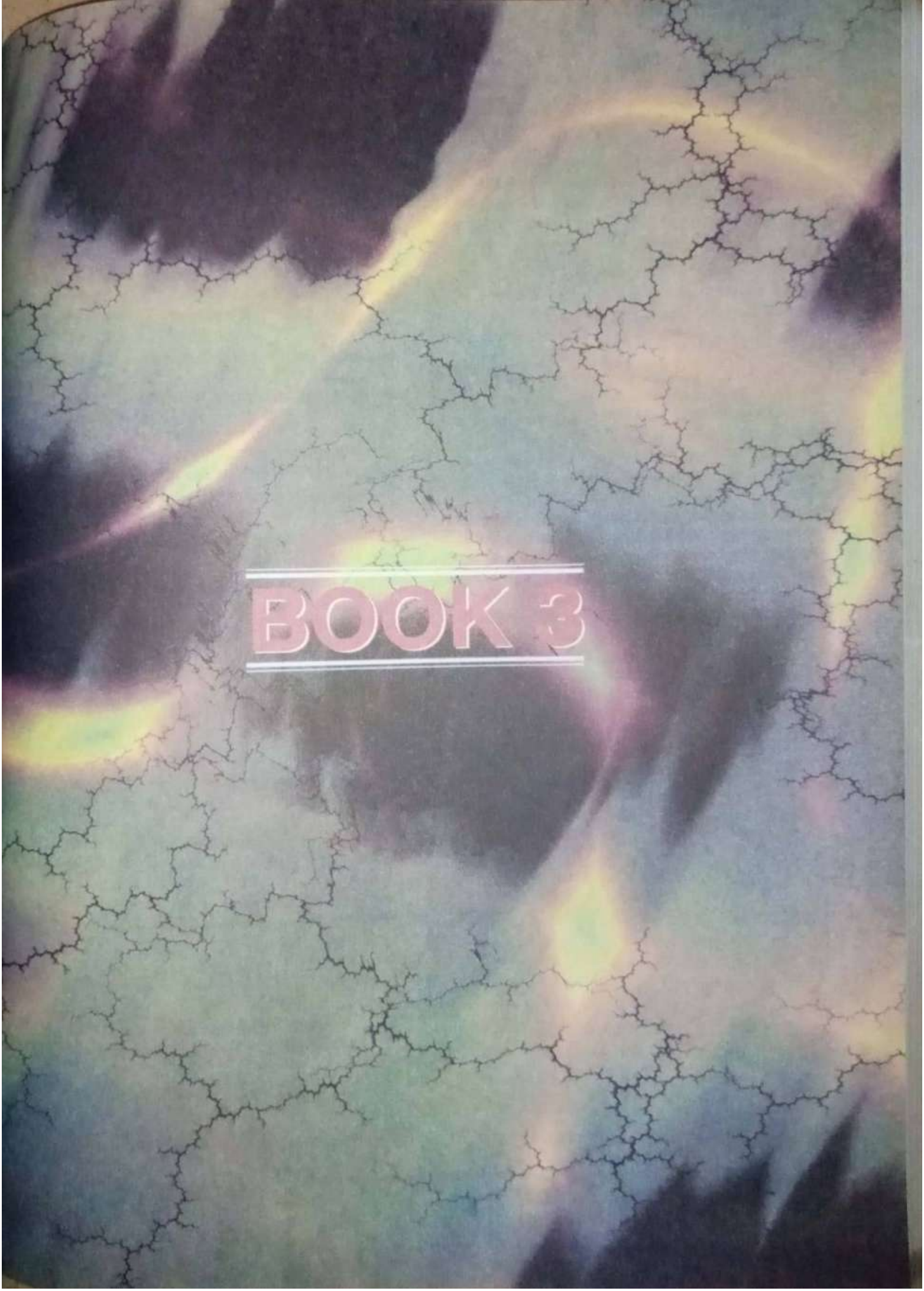
Explain why water in the bottom of a floating boat cannot be siphoned over the side.

13 Describe the construction and mode of operation of

(i) a lift pump

(ii) a force pump.

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1 ELECTOMAGNETIC WAVES

In Book 2 chapter 11, we studied the production and propagation of waves, different types of waves including sound and light, and the properties of waves. In this chapter we will study a unique type of waves which is made up of varying magnetic field.

At the end of this chapter the student should be able to:

1 Distinguish between electromagnetic waves and mechanical waves.
2 Compare and contrast in terms of frequency and wave length, six radiations in the electromagnetic spectrum.
3 State some uses of electromagnetic waves.
4 Apply the formula $V = f\lambda$ to solve simple problems relating to electromagnetic waves.

1.1 Introduction

We can distinguish between two general classes of waves: *mechanical waves* and *electromagnetic waves*.

Mechanical waves are waves that require a material medium for their propagation.

Examples of mechanical waves are water waves, sound waves, and waves on a rope. Water waves are due to vibrations of particles of water. Sound waves are due to vibrations of air particles. Water and air are the material media of propagation.

Electromagnetic waves are those that do not require a material medium for their propagation. They arise from the vibrations of electric (E) and magnetic (M) fields. The combination of the electric field and the magnetic field waves is called an Electromagnetic (E-M) wave.

In the E-M wave, the Electric and the Magnetic fields are mutually at right angles to one another and to the direction of motion of the wave as shown in fig. 1.1. Examples of E-M waves are light wave, X-rays and gamma rays.

Another important difference between mechanical and electromagnetic waves is in their velocities. Electromagnetic waves travel at the speed of light but mechanical waves travel at a speed less than that of light.

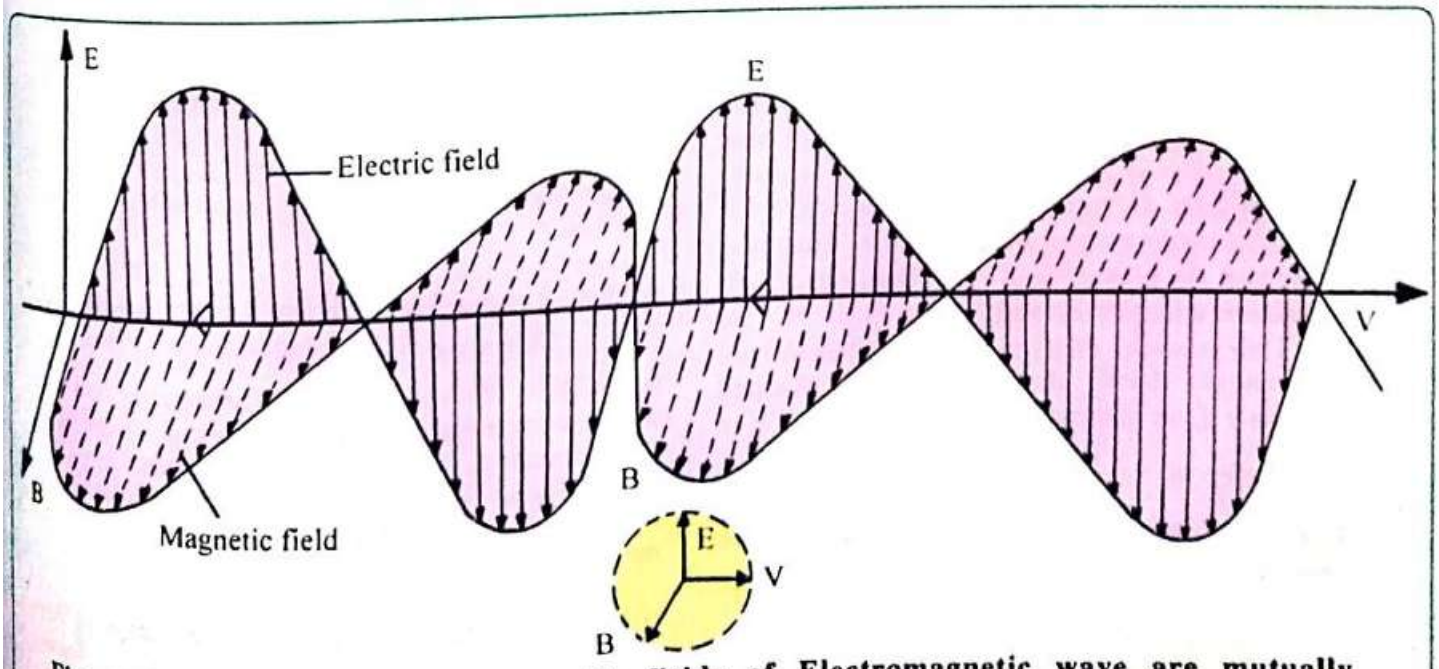


Fig. 1.1 The Electric and Magnetic fields of Electromagnetic wave are mutually perpendicular

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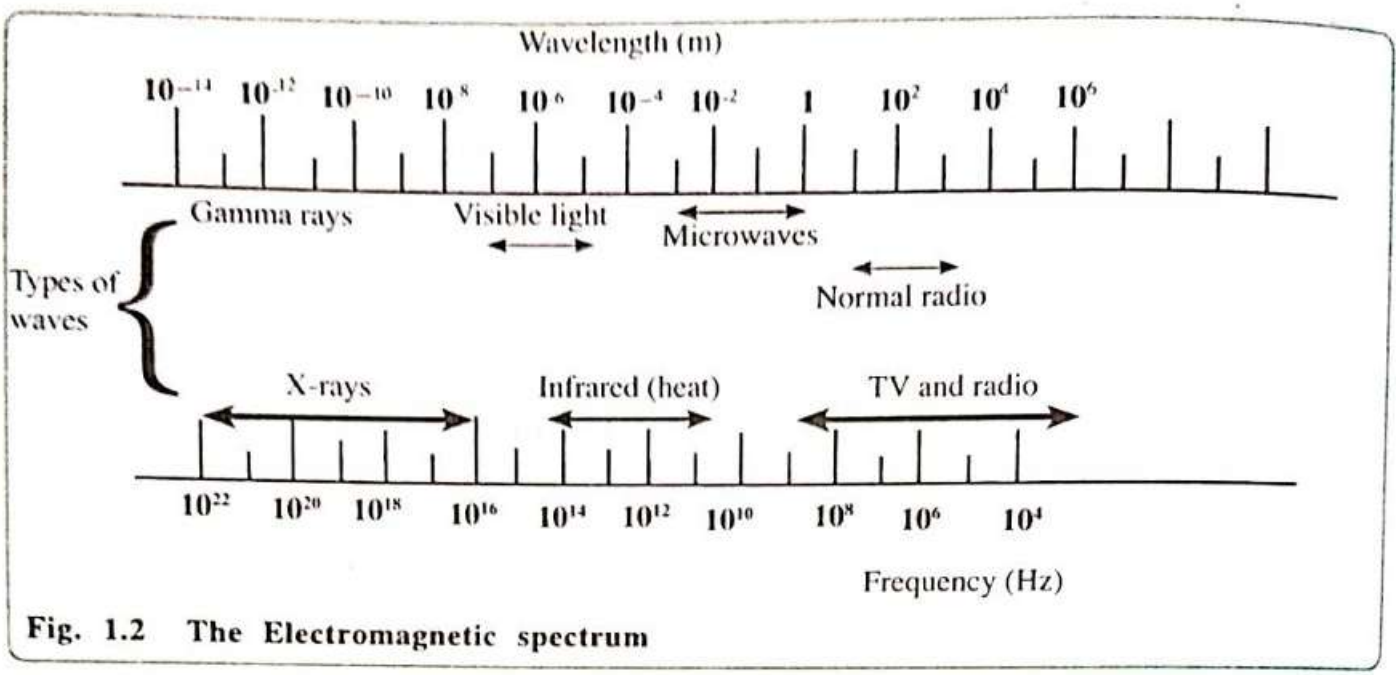


Fig. 1.2 The Electromagnetic spectrum

Thirdly, a mechanical wave may be transverse or longitudinal, but an electromagnetic wave is always transverse.

1.2 Types of Radiation

Electromagnetic waves were first predicted theoretically by the British physicist James Clerk Maxwell. He also proposed that light itself was electromagnetic radiation. The experimental evidence for the existence of E-M wave was provided by the German physicist, Heinrich Hertz.

Examples of electromagnetic waves arranged in order of increasing wave length are:

Gamma rays, X-rays, ultraviolet rays, visible light, infrared rays, micro-waves and radio waves.

The whole range of electromagnetic waves is known as the *electromagnetic spectrum*. The different rays that constitute the spectrum all have the same *basic property that they travel with the speed of light*. In vacuum this speed is 3.00×10^8 ms^{-1} and is represented by the letter c. They also all exhibit the properties of *reflection*, *refraction*, *interference* and *diffraction*.

They however differ in their wavelength (and frequency). Recall that wavelength (λ) and frequency (f) are connected by the relation

$$\lambda = \frac{v}{f} \quad \text{or} \quad v = f\lambda$$

Here $v = c$, the velocity of light. Hence we can write that

$$\lambda = \frac{c}{f} \quad \text{or} \quad c = f\lambda \qquad 1.1$$

The waves also differ in the way they interact with matter. This way, which depends on their frequency, influences how we can detect each type of wave. Hence the waves in the various parts of the electromagnetic spectrum differ in the means of production and detection.

Example 1.1

A certain radio station is assigned a frequency of 2000 KHz. Estimate the wavelength of its radio waves.

Take $c = 3 \times 10^8$ ms^{-1}.

Solution

$v = c = 3 \times 10^8$ ms^{-1}

$\lambda = \frac{c}{f}$

$f = 2000$ KHz $= 2000 \times 10^3$ Hz $= 2 \times 10^6$ Hz

$$\lambda = \frac{3 \times 10^8 \ (ms^{-1})}{2 \times 10^6 \ (s^{-1})}$$

$= \underline{150 \text{ m}}$

1.3 Uses of Electromagnetic Waves

(i) **Radio Waves** are electromagnetic waves whose wavelengths lie in the range of 1 mm to a few kilometres. They are commonly used in radio transmissions. Radio waves of shorter wavelength (~ 1 cm) are used in radar, that is radio detection and ranging.

(ii) **X-rays** – the range of the wavelengths of X-rays is from 10^{-10} cm to 10^{-7} cm. They have tremendous penetrating power, and are able to penetrate brick, bones etc. They are therefore used in hospitals to make X-ray pictures or radiographs of the bones and internal organs of the body. They are also used to detect flaws in various kinds of materials used in heavy industries. They are used to

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study the structure of crystals, the structure and properties of atoms and molecules and their arrangements in matter. They are also used at airports as X-ray scanners to detect concealed weapons in luggages.

(iii) **Gamma rays** - these are electromagnetic waves whose wavelengths are of the order of 10^{-8} cm. Their penetrating power is about 1000 times greater than that of X-rays. They are used in the medical field to sterilize medical supplies, like surgical equipment, drug, food, bandages etc. They are also used to treat cancers and tumours.

Summary

1 Mechanical waves require material medium for their propagation. Electromagnetic waves do not require material medium for their propagation.

2 Electromagnetic waves travel with the speed of light but mechanical waves travel with velocity less than speed of light.

3 Electromagnetic waves are always transverse but mechanical waves can be transverse or longitudinal.

4 Electromagnetic waves arranged in order of increasing wavelength are Gamma rays, X-rays, ultra-violet rays, visible light, infrared rays, microwaves and radio waves. These constitute the Electromagnetic Spectrum

5 Uses of electromagnetic waves
   (i) Radio waves in radio transmission and radar.
   (ii) X-rays in radiography and in X-ray diffraction.
   (iii) Gamma rays in medical fields for sterilization of medical equipment and in the treatment of cancers and tumours.

Exercise 1

1 Which of the following is not an electromagnetic radiation?
   A X-ray B Radio waves
   C Sunlight D Sound waves
   E Infrared radiations.
   (SSCE)

2 In which of the following groups are the radiations arranged in the increasing order of their wavelength?
   A Radio waves, gamma rays, X-rays
   B X-rays, gamma rays, radio waves
   C X-rays, radio waves, gamma rays
   D Gamma rays, radio waves, X-rays
   E Gamma rays, X-rays, radio waves.
   (SC/GCE)

3 Which of the following statements is not correct?
   A X-rays have a smaller wavelength than ultraviolet rays.
   B Gamma rays are less penetrating than X-rays
   C Ultraviolet rays have shorter wavelength than visible light.
   D Infrared rays can be detected by their heating effect.
   E Gamma rays travel in straight lines.
   (SC/GCE)

4 Distinguish between a mechanical and an electromagnetic wave. Give one example of each.

5 Arrange the following electromagnetic radiations in the order of increasing wavelength, ultraviolet, gamma rays, radio waves, infrared, visible light, X-rays. Which has the highest frequency?

6 What is the speed of electromagnetic radiation in free space? What frequency should a radio station have if its waves are to have a wavelength of 200 m?

7 Electromagnetic waves from the sun reach the earth even though the space between us and the sun is vacuum. Sound waves emitted by the sun cannot reach the earth. Why this difference?

8 Name five uses of electromagnetic radiations.

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2 GRAVITATIONAL FIELD

In Book 1, chapter 6 we introduced the concept of *Field*, and described a field as a *region of space under the influence of some physical agency such as gravitation, magnetism and electricity*. We also noted that the field concept is a convenient and informative method of describing the influence of one body over another body separated from the first by some distance.

In chapter 2, of Book 1 we had defined *force-fields as forces whose sources do not require contact with the body that they affect*. The effects of such forces are felt by bodies which are not in contact with them. Their actions are felt at a distance. Two classes of such force fields were also described:

(i) Scalar field – those that have only magnitude but no directions and

(ii) Vector fields – those that have both magnitude and direction e.g., gravitational, magnetic and electric fields.

The next four chapters will further deepen and expand our discussions and understanding of the field concept and the student should therefore review thoroughly the subject of *fields* as discussed in Book 1 before getting into this chapter and the other three chapters.

At the end of the current chapter, the student should be able to:

1 Calculate the gravitational force between two masses.
2 Calculate the gravitational force between two planets.
3 Explain the meaning of 'G' and show that 'g' is the force per unit mass on the earth's surface.
4 Relate Kepler's Laws to the motion of the solar system.
5 Distinguish between Natural and Artificial satellites.
6 Explain how artificial satellites are launched.
7 Explain the concept of Escape Velocity.
8 Use the inverse square law to calculate the escape velocity of an object from the earth' gravitational field.

2.1 Concept of Gravitational Field

There are two types of forces - contact forces and force fields. *Contact forces* are most common in everyday life. For example, you push or pull on a wheelbarrow, a tennis racket exerts a force on a tennis ball when they make contact, your foot exerts a force on a football when you kick it. Force fields, e.g. gravitational forces, act even when the two bodies are not in contact. The earth, for example exerts a force on a falling mango fruit. It also exerts a force on the moon which is about 385,000 km away. The sun itself exerts a force on the earth even though the earth is about 1.5×10^8 km distant from the sun.

In order to explain the observation of forces acting at a distance, it has been postulated that a gravitational field surrounds everybody that has mass, and this field fills up all of space. A second body at a particular location near the first body experiences a force because of the gravitational field that exists there. The gravitational field of the first body acts directly on the second body

Gravitational field is a region or space around a mass in which the gravitational force of the mass can be felt.

2.2 Gravitational Force between two masses

Gravitation is the force of attraction exerted by a body on all other bodies in the universe. Hence a gravitational force exists between a body and all other bodies around it. Gravitational forces act between all masses and hold together planets, stars and galaxies, Each mass has a gravitational field around it.

It was Sir Isaac Newton who first proposed the relationship between the gravitational force F, between two masses, m_1, m_2 and the distance, r, between these masses. He proposed his famous *Law of Universal gravitation* which we can state as follows:

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Every particle in the universe attracts every other particle with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. This force acts along the line joining the centre of the two particles.

Newton's law of universal gravitation an example of an *inverse square law*, can be stated in equation form as:

$$F \propto \frac{m_1 m_2}{r^2} \qquad 2.1$$

The magnitude of this force can be written as

$$F = \frac{G m_1 m_2}{r^2} \qquad 2.2$$

where m_1 and m_2 are the masses of the two particles, r is the distance between them and G is a *universal constant of gravitation* or simply *gravitational constant*.

Gravitational constant, G, has the same numerical value for all objects ($G = 6.67 \times 10^{-11}$ Nm^2kg^{-2})

Newton's law of gravitation refers to the force between two particles or bodies. Actually such a gravitational force is a pair of forces, an action - reaction pair. Although the masses of the particles may be different, forces of equal magnitude act on each, and the action line of both forces lies along the line joining the bodies. Mass m_1 attracts m_2 with a force given by equation 2.2. By Newton's third law of motion, action and reaction are equal and opposite. Therefore mass m_2 also attracts m_1 by an equal but opposite force.

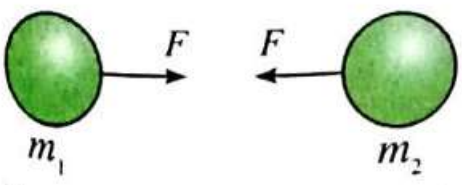


Fig. 2.1 Gravitational attraction between two blocks

Hence equation 2.2 gives the magnitude of two equal but opposite forces.

Gravitational attraction keeps the moon in its orbit around the earth and the earth in its orbit around the sun. *Gravitational forces are always those of attraction.*

Example 2.1

Two 5.0 kg spherical balls are placed so that their centres are 50.0 cm apart. What is the magnitude of the gravitational force between the two balls? ($G = 6.67 \times 10^{-11}$ Nm^2kg^{-2})

Solution

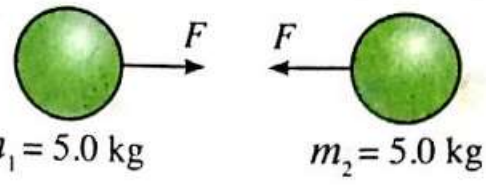


Fig. 2.2

$$F = \frac{G m_1 m_2}{r^2}$$

$$= \frac{6.67 \times 10^{-11} \times 5.0 \times 5.0}{0.5^2} \text{ N}$$

$$= \frac{6.67 \times 25 \times 10^{-11}}{0.5^2} \text{ N}$$

$$= \underline{6.67 \times 10^{-9} \text{ N}}$$

Because of the universal law of gravitation, there is a gravitational force of attraction between the sun and the planets, between earth and the moon, and also between other planets.

Example 2.2

Determine the force of attraction between the sun ($m_s = 1.99 \times 10^{30}$ kg) and the earth ($m_e = 5.98 \times 10^{24}$ kg). Assume the sun is 1.50×10^8 km from the earth

Solution

$$F = \frac{G m_s m_e}{r^2}$$

$$r = 1.5 \times 10^8 \text{ km} = 1.5 \times 10^8 \times 10^3 \text{ m}$$

$$\therefore \quad F = \frac{6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 5.98 \times 10^{24}}{(1.5 \times 10^8 \times 10^3)^2}$$

$$= \underline{3.53 \times 10^{22} \text{ N}}$$

Example 2.3

Calculate the gravitational force of attraction between the electron of a hydrogen atom and the proton when the electron orbits the proton at its average distance of 0.53×10^{-10} m. Take mass of electron as 9×10^{-31} kg, and mass of proton as 1.67×10^{-27} kg.

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Solution

$$F = \frac{Gm_1m_2}{r^2}$$

$$= \frac{6.67\times10^{-11}\times9\times10^{-31}\times1.67\times10^{-27}}{(10^{-10})^2}\ \text{N}$$

$$= \frac{6.67\times9\times1.67\times10^{-49}}{(0.53)^2}\ \text{N}$$

$$= \underline{3.57\times10^{-47}\ \text{N}}$$

Example 2.4

Calculate the gravitational force of attraction between two protons in a chemical compound. Take the mass of a proton as 1.67×10^{-27} kg and the distance between the protons in the chemical compound as 10^{-10} m.

Solution

$$F = \frac{Gm_1m_2}{r^2}$$

$$= \frac{6.67\times10^{-11}\times1.67\times10^{-27}\times1.67\times10^{-27}}{(10^{-10})^2}\ \text{N}$$

$$= 6.67\times(1.67)^2\times10^{-45}$$

$$= \underline{1.86\times10^{-44}\ \text{N}}$$

Examples 2.3 and 2.4 show how small the gravitation force of attraction between atomic masses are. We shall see in the next chapter that the electrical force of attraction between these masses are enormously greater than the gravitational force or attraction.

2.3 Relation between the Gravitational Constant 'G' and the acceleration of gravity at the earth surface 'g'

The earth is supposed to be a sphere of radius, r_e, with its mass m_e concentrated at the earth's centre. The distance of any object on the earth's surface to the centre of the earth is r_e the earth's radius. The gravitational force of attraction of the earth on any mass, m, on the earth's surface is given by

$$F = \frac{Gm_em}{r_e^2} \qquad 2.3$$

This is the force of gravity on the mass due to the earth, that is, the weight of the object, mg, where 'g' is the acceleration due to gravity.

Thus

$$F = \frac{Gm_em}{r_e^2} = mg$$

The force per unit mass, $\frac{F}{m}$, is given by

$$\frac{F}{m} = \frac{Gm_e}{r_e^2} = g$$

Hence

$$g = \frac{Gm_e}{r_e^2} \qquad 2.4$$

This means that *the acceleration due to gravity 'g' can be considered as the force per unit mass on the earth's surface*. According to equation 2.4, the acceleration of gravity, 'g' at the surface of the earth, is determined by m_e (the earth's mass), and r_e (the earth's radius). Hence from equation 2.4, we should expect 'g' to be slightly less on the tops of mountains than at sea level since r_e the distance from the earth's centre, is slightly greater at the top of the mountain than at sea level. This is what actually obtains in practice.

If 'g' and 'G' are accurately known, we can use equation 2.4 to calculate the mass of the earth, m_e.

$$m_e = \frac{gr_e^2}{G} \qquad 2.5$$

Example 2.5

Determine the mass of the earth if the radius of the earth is approximately 6.38×10^6 m, $G = 6.67 \times 10^{-11}\ \text{Nm}^2\text{kg}^{-2}$ and $g = 9.80\ \text{ms}^{-2}$.

Solution

$$m_e = \frac{gr_e^2}{G}$$

$$= \frac{9.80\times(6.38\times10^6)^2}{6.67\times10^{-11}}\ \text{kg}$$

$$= \underline{5.98\times10^{24}\ \text{kg}}$$

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Example 2.6

The radius of the moon is one-fourth, and its mass is one eighty-first that of the earth. If the acceleration due to gravity on the surface of the earth is 9.8 ms^{-2}, what is its value on the moon's surface.

Solution

The relation between g and G is given by

$$g = \frac{Gm_e}{r_e^2}$$

For the earth, $g_e = \frac{Gm_e}{r_e^2}$ (i)

For the moon, $g_m = \frac{Gm_m}{r_m^2}$ (ii)

Since G is the universal constant of gravitation it is constant for both equations.

$$r_m = \frac{r_e}{4} \quad \text{(iii)}$$

$$m_m = \frac{m_e}{81} \quad \text{(iv)}$$

From equation (i)

$$G = \frac{g_e r_e^2}{m_e} \quad \text{(v)}$$

Putting (v) in (ii) we have

$$g_m = \frac{g_e r_e^2}{m_e} \cdot \frac{m_m}{r_m^2}$$

$$= 9.8 \cdot \left(\frac{r_e}{r_m}\right)^2 \cdot \frac{m_m}{m_e} \quad \text{(vi)}$$

From (iii) $\frac{r_e}{r_m} = 4$

From (iv) $\frac{m_m}{m_e} = \frac{1}{81}$

Putting these in equation (vi) we have:

$$g_m = 9.8 \times (4)^2 \times \frac{1}{81} = 9.8 \times 16 \times \frac{1}{81}$$

$$= \underline{1.9\ ms^{-2}}$$

2.4 Gravitational Potential

The work done in raising a mass, m from the ground surface to a height, h above the ground is given by

$W = mgh$

The work has been done against the gravitational pull of the earth. This work appears as the gravitational potential energy (PE) of the body.

This potential energy depends on the height, h, or the relative position of the body from the ground or zero level where the P.E. is considered to be zero.

In general, points in any gravitational field possess *gravitational potential*. If free to move, a body will tend to move from a point at higher gravitational potential to points of lower gravitational potential

Gravitational Potential (G) at a point is defined as the work done in taking unit mass from infinity to that point. Unit is Jkg^{-1}

The gravitational potential V is given by

$$V = -\frac{Gm}{r} \qquad 2.6$$

where m is the mass producing the gravitational field and r is the distance of the point to the mass. The gravitational potential decreases as r increases and becomes zero where r is infinitely large. The negative sign in equation 2.6 indicates that the potential at infinity (zero) is higher than the potential close to the mass.

Example 2.7

Find the Gravitational Potential at a point on the earth's surface. Take mass of earth as 5.98×10^{24} kg, its radius as 6.38×10^6 m and $G = 6.67 \times 10^{-11}$ Nm^2kg^{-2}.

Solution

$$V = \frac{Gm}{r}$$

$$= \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{6.38 \times 10^6}$$

$$= \underline{6.25 \times 10^7 \text{ Joules kg}^{-1}}$$

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Solar System

The Solar System consists of the **Sun** and those celestial or heavenly bodied or objects that move around it, and bound to it by **gravity**. These objects that circle round the **Sun** are known as **Planets**.

to be growing brighter year by year.

Most of the planets in the Solar System possess secondary systems of their own, being orbited by planetary objects called **natural satellites** or **moons**.

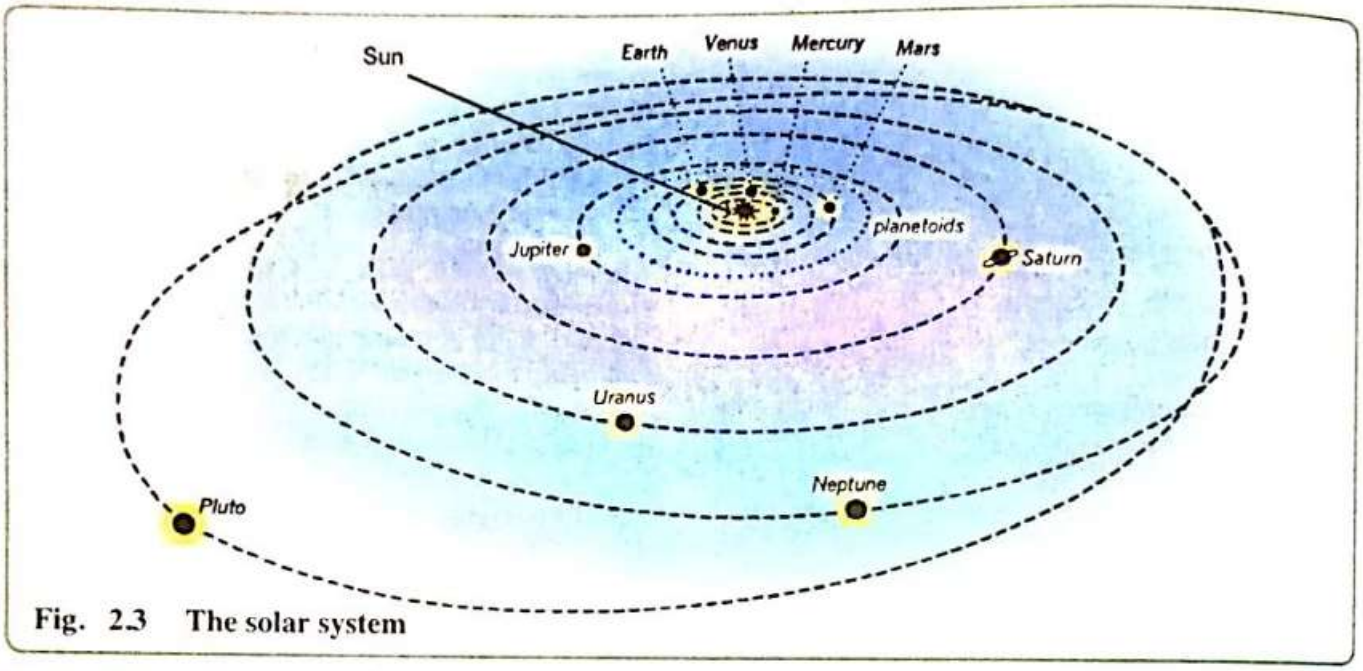


Fig. 2.3 The solar system

These planets move round the Sun in elliptical or egg shaped orbits with the Sun at the centre. The planets starting with the one nearest the Sun are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune and Pluto.

The four smaller inner planets, Mercury, Venus, Earth and Mars, are called ***terrestrial planets*** (because they resemble the earth), and are primarily composed of rock and metal. The four outer planets are called the ***Gas Giants*** and are substantially more massive than the terrestrial planets.

The two largest planets, Jupiter and Saturn, are composed mainly of hydrogen and helium.

The Solar System is also home to smaller objects called ***Asteriods, Comets, Meteriods,*** etc

The Sun is the principal component of the Solar System. It is a giant star and contains over 99 percent of the Solar System's known Mass and dominates it gravitationally.

The Sun is a big ball of hot glowing gas and glows all the time emitting its own light and heat. It is the only luminous object in the solar system. It has a diameter that is more than 100 times the diameter of the Earth. It has a large mass (about 332900 Earth mass) and produces temperatures and densities in its core great enough to sustain **Nuclear fusion**, which releases enormous amount of energy (Solar energy) mostly radiated into space as electromagnetic radiation. The Sun is reported

The following should be noted about planets:

(i) Each planet revolves around the Sun in its own orbit.

(ii) The nearer a planet is to the Sun, the faster is its speed of revolution.

(iii) All the planets revolve in one direction.

(iv) The orbit of the planets (except that of Pluto) all lie in the same plane.

2.5 Kepler's Laws

Kepler's laws of planetary motion describe the orbits of objects about the Sun. The laws can be stated as follows:

Law 1: The planets each travel along an ellipse with the Sun at one **focus**.

Law 2: The line joining the Sun and the planets sweeps out equal areas in equal times.

Law 3: The square of the period of revolution of the planets are proportional to the cubes of their mean distances from the Sun ($T^2 \alpha r^3$)

2.6 Natural and Artificial Satellites

A satellite is any object that moves round the earth or any body in space. The two types of satellites are:

(i) Natural satellites and

(ii) Artificial satellites

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A Natural Satellite or **Moon** is a **celestial body** that orbits a planet or smaller body, which is called the **primary**. Technically the term **natural satellite** could refer to a planet orbiting a star, but it is normally synonymous with moon. Natural satellites are used to identify non-artificial satellite.

The moon we see shinning in the night over the earth is an example of natural satellite, its diameter is about a quarter that of the earth, and is about 348300km away from the earth. The moon does not shine with its own light, instead it reflects the light from the Sun. The Earth's moon is the largest natural satellites in the solar system. **Artificial satellites** are man made or human built objects orbiting the earth and other planets in the solar system. Such satellites are used to study the earth and other planets, and to help us in communication. They are used by astronomers to observe the distant universe. (see fig. 14.7 of this book)

The first artificial satellite ***sputnik 1*** was launched in 1957 by the then ***Soviet Union***. Since then many such satellites have been launched into orbit. Some have broken down or expired and they are mere debris or junk satellites hovering in space.

How are satellites launched into space? To launch a satellite into space. It is first carried to the required height above the earth using rockets (see chapter 14: rockets and satellites). Then the satellite is launched in a direction parallel to the earth's surface with a definite velocity V as shown in fig 2.4. It is launched by giving it an impulse, by firing jets, to deflect it in a direction parallel to the tangent of the orbit. Its velocity is boosted to maintain it in the orbit. The satellite with the necessary electronic equipment inside rises vertically when launched.

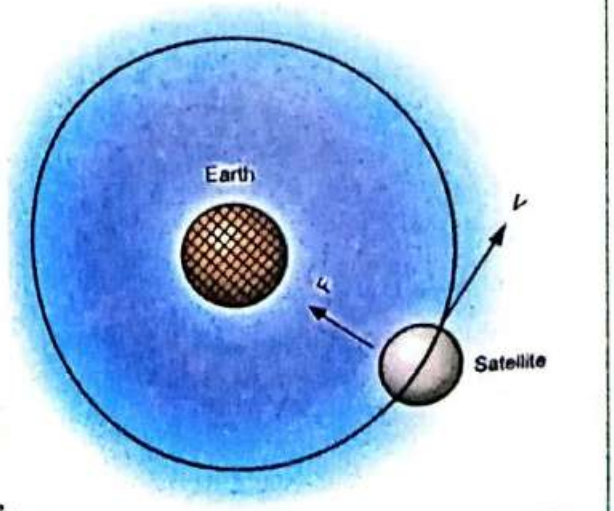


Fig. 2.4 Satellite launched with velocity V in a direction parallel to Earth's surface

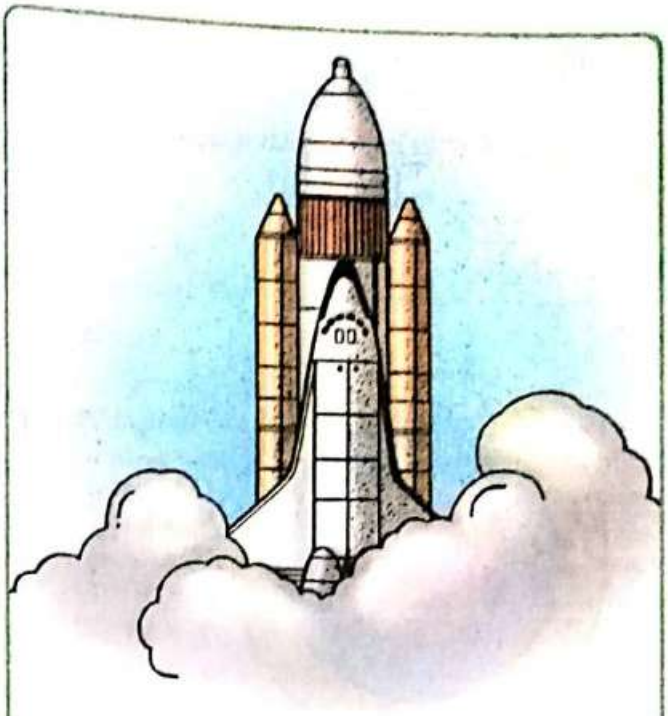


Fig. 2.5 Launching of artificial satellite

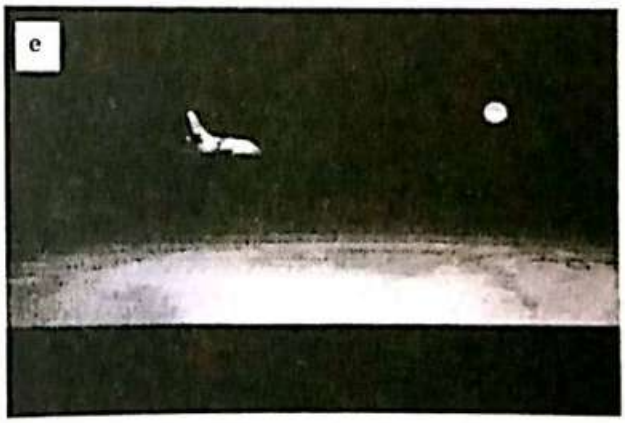


Fig. 2.6 Satellite moving freely in an orbit (the moon is seen at the background)

2.7 Escape Velocity

There are many man-made satellites that circle around the earth at the present time. One common feature of these bodies is that they are held in an approximately circular path by the earth's gravitational pull. It is this force that provides the needed centripetal force required to keep the satellites in their orbits. The velocity (v_s) of the satellite as it orbits round the earth is given by

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$$\frac{mv_s^2}{r_e} = F = \frac{Gmm_e}{r_e^2} \qquad 2.7$$

(centripetal force) = (Gravitational force)

Hence

$$v_s = \sqrt{\frac{Gm_e}{r_e}} \qquad 2.8$$

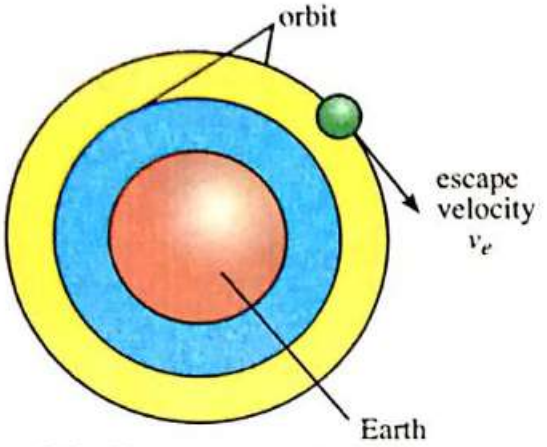


Fig. 2.3 Escape velocity

This is the velocity with which the satellite moves round the earth. Notice that the mass of the satellite does not enter into equation 2.8. All satellites in orbit with radius r_e must have the same speed. For a satellite to escape from the earth and never return, it must be launched with a velocity greater than that required to make it orbit.

We define the escape velocity (v_e) as the minimum velocity required for an object (e.g. satellite or rocket) to just escape or leave the gravitational influence or field of an astronomical body (e.g. the earth) permanently.

We can obtain the formula for the Escape velocity using the Newtons law of universal gravitation which is an inverse square law:

$$F = \frac{Gm_e m}{r^2} \qquad 2.2$$

Let m be the mass of the satellite, and m_e, the mass of the earth.

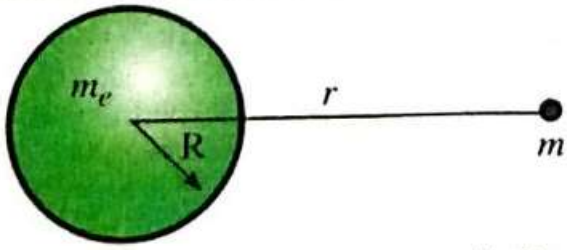


Fig. 2.6 Formula for escape velocity

The work done in carrying a mass m from a point at a distance r from the centre of the earth, to a distance so great that the gravitational field is negligibly weak is given by

$$W = F \times r \qquad 2.9$$

But from equation 2.2, $F = \frac{Gm_e m}{r^2}$

Hence, $W = \frac{Gm_e m}{r^2} \cdot r = \frac{Gm_e m}{r}$ 2.10

This work must equal the kinetic energy of the body of mass m at this point, having a velocity v_e. This kinetic energy is given by:

$$\text{KE} = \frac{1}{2}mv_e^2$$

$$\frac{1}{2}mv_e^2 = \frac{Gm_e m}{r}$$

$$v_e^2 = \frac{2Gm_e}{r}$$

$$v_e = \sqrt{\frac{2Gm_e}{r}} \qquad 2.11$$

If we launch the mass m from the earth's surface, where $r = R$, we then have that $v_e = \sqrt{\frac{2Gm_e}{R}}$

But from equation 2.5 $m_e = \frac{gr^2}{G}$

Hence $v_e = \sqrt{\frac{2G}{R} \cdot \frac{gR^2}{G}}$

$$v_e = \sqrt{2gR} \qquad 2.12$$

where R is the earth's radius.

Example 2.8

Calculate the escape velocity of a satellite from the earth's gravitational field.

$g = 9.8\ \text{ms}^{-2}$, radius of earth $= 6.4 \times 10^6$ m.

Solution

$$v = \sqrt{2gR}$$

$$= \sqrt{2 \times 9.8 \times 6.4 \times 10^6}\ \text{ms}^{-1}$$

$$= 11 \times 10^3\ \text{ms}^{-1}$$

$$= 11\ \text{kms}^{-1}$$

Thus to escape the gravitational influence of the earth, the satellite must be projected with a velocity greater than 11 kms^{-1}.

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Satellites are launched into different orbits depending on their mission. Synchronous satellites are satellites that move round the earth at the same rate as the rotation of the earth. That means their period of rotation is 24 hours. They thus appear stationary when seen from the earth. They are said to move in a ***geosynchronous orbit***. Such a satellite is thus in the same spot over the earth while the earth rotates, allowing for communication and television broadcasts to be transmitted continually from one part of the world to another. Such an orbit is thus also called a ***parking orbit***. (see fig. 14.9 of this book).

Satellite like the ***Space Shuttle*** and the ***Hubble Space Telescope*** used by astronomers move in a low-earth orbit. This orbit is only a few hundred kilometres above the earth but still outside the earth's atmosphere

Artificial Satellite can have a range of missions including (i) communication (ii) scientific research (iii) weather observation (iv) photograph and mapping (v) Geographic Information System (GIS) (vi) Global Positioning System (GPS) generally used in navigation. (See Chapter 14 for details).

Example 2.9

For a satellite which is close to the earth's surface its radius of orbit may be assumed to be equal to the earth's radius, r = 6.4 x 10^6. Estimate

(a) the speed of the satellite

(b) the period of rotation of the satellite

Solution

Gravitational attraction F on the satellite is given by

$$F = \frac{Gm_e}{r^2}$$

where m_e = mass of earth and m is mass of satellite. F is also equal to $\frac{mv^2}{r}$ where v is the velocity of the satellite.

$$V = \sqrt{\frac{Gm_e}{r^2}}$$

But = gr_e^2 (from equation 2.5)

Therefore, $V = \sqrt{\frac{Gr^2}{r}} = \sqrt{gr}$

$$= \sqrt{9.8 \times 6.4 \times 10^6}$$

$$= \mathbf{8.0 \times 10^3 ms^{-1}}$$

b) Period $T = \frac{2\pi}{w} = \frac{2\pi r}{v}$ (since $w = \frac{v}{r}$)

$$T = \frac{2\pi \times 6.4 \times 10^6}{8.0 \times 10^3}$$

$$= \mathbf{5 \times 10^3 secs = 83 minutes}$$

Summary

1 A gravitational field is a region or space around a mass in which the gravitational force of the mass can be felt.
Gravitational force of attraction exists between a mass and all other masses around it.

2 Newton's law of universal gravitation states that the gravitational force of attraction between any two masses, m_1, m_2 is directly proportional to the product of the masses, and inversely proportional to the square of the distance (r) between them.

$$F = \frac{Gm_1m_2}{r^2}$$

G is the gravitational constant (= 6.67×10^{-11} Nm^2kg^{-2})

3 Acceleration due to earth's gravity 'g' is related to the gravitational constant G by

$$g = \frac{Gm_e}{r_e^2}$$

where r_e is the earth's radius and m_e is the mass of the earth. The force per unit mass on the earth's surface is 'g'.

4 Escape velocity is the minimum velocity required for an object to just escape from the gravitational influence of the earth.

Escape velocity $v_e = \sqrt{2gR}$ where R is the radius of the earth.

5 Gravitational Potential (V) at any point in a gravitational field is defined as the work done in taking unit mass from infinity to the point.

$$V = \frac{Gm}{r}$$

Here m is the mass producing the gravitational field, and r is the distance of the point to the mass.

6 With a velocity of about 8 kms^{-1}, a satellite can move in a circular orbit close to the earth's surface. With a velocity greater than 8kms^{-1} but less than 11 kms^{-1}, a satellite describes an elliptical orbit round the earth. With a velocity above 11 kms^{-1}, the satellite completely escapes from the gravitational attraction of the earth.

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Exercise 2

1 What do you understand by a *field*, a *gravitational field*?

2 State Newton's law of universal gravitation. Write down the expression for the gravitational force between two masses. Explain the meaning of each term in your expression.

3 Calculate the force or attraction between two small objects of mass 10 kg and 50 kg respectively which are 10 cm apart. Take G as 6.67×10^{-11} Nm^2kg^{-2}.

4 The earth has a mass of 6.0×10^{24} kg, the moon a mass of 7.4×10^{22} kg. Calculate the force of attraction between the two if the distance between their centres is 3.84×10^{8} m.

5 State Newton's law of universal gravitation. What is the difference between, G, the universal gravitational constant and 'g', the acceleration due to gravity, Write down the relationship between them.

If g = 9.8 ms^{-2}, $G = 6.7\times10^{-11}$ Nm^2kg^{-2}, calculate the mass of the earth if the radius of the earth is taken as approximately 6400 km.

6 The average radius of Jupiter's orbit round the sun of mass 2×10^{30} kg is 7.8×10^{11} m. If the mass of Jupiter is 1.9×10^{27} kg, find the gravitational force the sun exerts on Jupiter. $G = 6.67\times10^{-11}$ Nm^2kg^{-2}.

7 Explain the meaning of escape velocity. Write down the expression for the escape velocity of a rocket from the earth. Explain the meaning of each term in the expression.

The escape velocity of a rocket from the earth is 11 kms^{-1}. Taking the acceleration due to gravity as 9.8 ms^{-2}, calculate the radius of the earth.

8 The radius of the planet Venus is nearly the same as that of the earth, but its mass is only 80% that of the earth. If an object weighs W_e on the earth, what does it weigh on Venus? Calculate the value of 'g' on Venus.

9 If the mass of a proton is 1.67×10^{-27} kg and the mass of an electron is 9.11×10^{-31} kg, calculate the force of gravitation between:
(i) a proton and an electron
(ii) two electrons
(iii) two protons.

Take $G = 6.67\times10^{-11}$ Nm^2kg^{-2}, distance between the protons = 4.0 m, distance between the electrons = 2×10^{-2} m, distance between the proton and the electron = 5.4×10^{-11} m.

10 Calculate the force per unit mass on the earth's surface. Take $G = 6.67\times10^{-11}$ Nm^2kg^{-2}, mass of the earth = 5.98×10^{24} kg, and radius of the earth = 6.38×10^{6} m.

11 What is the solar system? Distinguish between ***natural*** and ***artificial satellites***.

12 State keplers laws pertaining to the motion of solar system.

13 How are artificial satellites launched into orbits? What is a ***geosynchronous orbit***?

14 Outline the applications of artificial satellite in the service of man.

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3 ELECTRIC FIELD 1

The concept of an electric field was introduced in Book 1 Chapter 7. There we defined an electric field as a region or space where a charged body experiences an electric force. The student is advised to review the whole of Chapter 7 of Book 1 before studying the present chapter.

At the end of this chapter, the student should be able to:

1 Calculate the electric force between two charges.
2 Explain electric field intensity and electric potential.
3 Explain the term capacitance.
4 Calculate the equivalent capacitance for series and parallel arrangement of capacitors.
5 Determine the energy stored in capacitors.

3.1 Electric Force between point charges. Coulomb's Law

In the preceding chapter, we discussed the gravitational force of attraction between two masses and obtained the magnitude of that force. There is also an electrical force between two point charges. The French Physicist Charles Coulomb (1736–1806) investigated the nature and magnitude of this force. He showed that if the bodies were small compared with the distance between them, then the force between two such bodies was inversely proportional to the square of the distance, r, between them.

Hence

$$F \propto \frac{1}{r^2} \qquad 3.1$$

This is an *Inverse Square Law*. Coulomb's investigations showed that the force was also proportional to the product of the charges, q_1, q_2 on the two bodies, i.e.

$$F \propto q_1 q_2 \qquad 3.2$$

A combination of expressions 3.1 and 3.2 led to *Coulomb's Law* which is stated as follows:

The electric force between two point charges, q_1 and q_2 separated by a distance r is directly proportional to the product of the charges and inversely proportional to the square of the distance between the charges.

$$F \propto \frac{q_1 q_2}{r^2} \qquad 3.3$$

The magnitude of this force is given by

$$F = \frac{K q_1 q_2}{r^2} \qquad 3.4$$

K in the above equation is a constant of proportionality. In SI units, F is in Newtons, q_1, q_2 are in Coulombs(C), r is in metres. The constant K is usually taken as:

$$K = \frac{1}{4\pi\varepsilon_0} \qquad 3.5$$

where ε_0 is a constant known as *permittivity* of free space. Hence we can finally state the value of force F as:

$$F = \frac{1}{4\pi\varepsilon_0} \times \frac{q_1 q_2}{r^2} \qquad 3.6$$

The unit of ε_0 is Coulomb^2–Newton^{-1}–metre^{-2} ($C^2N^{-1}m^{-2}$). $K = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ Nm^2C^{-2}.

Equation 3.6 gives the magnitude of the force that either object exerts on the other. The direction of this force is along the line joining the two objects. If the two bodies have charges of the same sign, the force is that of repulsion, the force on each object is directed away from the other. If the two objects have opposite charges, the force is that of attraction, the force on one is directed towards the other. The force one charge exerts on the second is equal but opposite to that exerted by the second on the first. This is in accordance with Newton's third law. (Action and reaction are equal and opposite)

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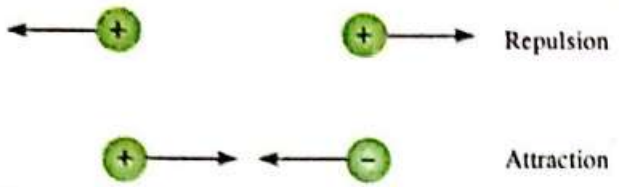


Fig 3.1 Direction of force on charges

Example 3.1

Find the force of repulsion between two protons of 1.6×10^{-19} C each if the distance between them is 5.3×10^{-11} m. (Take $\frac{1}{4\pi\varepsilon_0} = 9.0\times10^{9}\ \text{Nm}^2\text{C}^{-2}$)

Solution

$$F = \frac{1}{4\pi\varepsilon_0}\times\frac{q_1q_2}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2} \quad (\text{since } q_1 = q_2 = q)$$

$$= \frac{9\times10^{9}\ (\text{Nm}^2\text{C}^{-2})\times(1.6\times10^{-19}\ \text{C})^2}{(5.3\times10^{-11}\ \text{m})^2}$$

$$= \underline{8.2\times10^{-8}\ \text{N}}$$

We see from this example how large this force is compared to the gravitational force of Example 2.4.

Example 3.2

Three charges +10 C, −20 C and 16 C are distributed as shown in fig. 3.2. Find the resultant force acting on the 10 C charge.

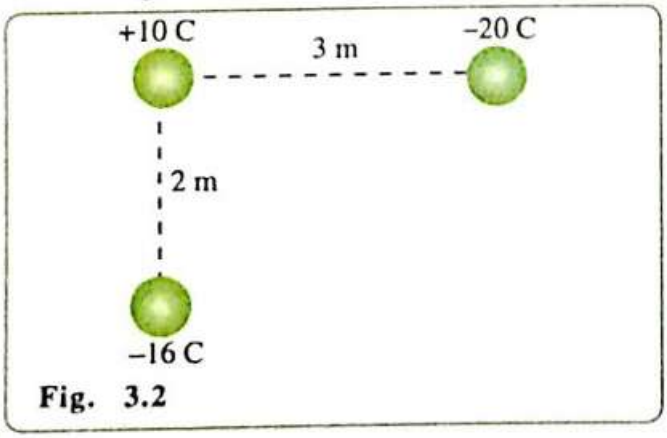


Fig. 3.2

Solution

Force of attraction between +10 C and −20 C is given by

$$F_1 = \frac{1}{4\pi\varepsilon_0}\times\frac{10\times20}{3^2}$$

$$= \frac{9\times10^{9}\times200}{9} = 2\times10^{11}\ \text{N}$$

Force of attraction between +10 C and −16 C is given by

$$F_2 = \frac{9\times10^{9}\times10\times16}{2^2}$$

$$= 36\times10^{10} = 3.6\times10^{11}\ \text{N}$$

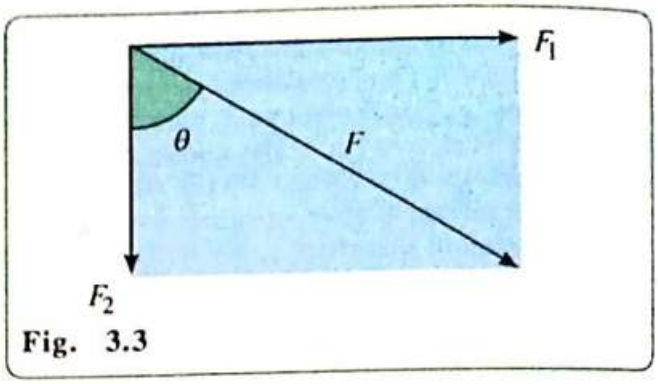


Fig. 3.3

$$\text{Net force} = \sqrt{F_1^2 + F_2^2}$$

$$= \sqrt{(2\times10^{11})^2 + (3.6\times10^{11})^2}$$

$$= \underline{4.12\times10^{11}\ \text{N}}$$

$$\tan\theta = \frac{F_1}{F_2} = \frac{2\times10^{11}}{3.6\times10^{11}} = 0.5555$$

$$\theta = \underline{32.28^\circ}$$

Net force is 4.12×10^{11} N in a direction 32.28° to the line joining the +10 C and −16 C charges.

3.2 Electric Field, Electric Field Intensity and Electric Potential

In section 3.1 we found that there is an electric force between two point charges. We here ask the question, How can one electric charge exert a force on a second charge even though they are not touching? We can obtain the answer from the concept of electric field. According to Michael Faraday, an electric field extends outwards from every charge and spreads out through space with the charge as centre. Any charge placed near the first charge experiences a force because of the effect of the electric field of the first charge. The electric field of the first charge interacts directly with the second charge to produce a force.

We can define **electric field** as a region where an electric force is experienced by a charged body.

An electric field is investigated by measuring the force on a small positive *test charge*. A test charge is a charge so small that the force it exerts does not

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significantly alter the field it is testing. With the aid of the test charge, we can map out an electric field using a number of lines called *electric lines of force*. Thus an electric field is represented by electric lines of force.

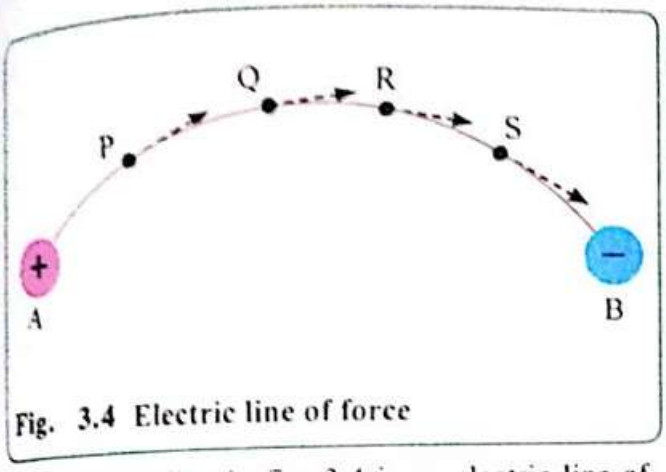


Fig. 3.4 Electric line of force

If the curved line in fig. 3.4 is an electric line of force, then the tangents to this line at points P, Q, R, S represent the direction a small positive test charge would go if placed at these points. The arrows indicate the direction of motion of the test charge.

Electric lines of force are imaginary lines drawn in such a way that the direction at any point (i.e. the direction of the tangent) is the same as the direction of the field at that point, OR, The line of force is the path which an isolated small positive charge would follow if placed in the field.

The student should review chapter 7 of Book 1 for the nature and properties of electric lines of force and for the patterns of different electric fields.

The strength of an electric field is defined in terms of the force on a positive test charge. The force itself depends on the charge of the body and on the field strength or Intensity.

The **Strength of an Electric Field** (or **Electric Field Intensity**) E, at any point is defined as the force per unit positive charge (q) at that point.

$$E = \frac{F}{q} \qquad 3.7$$

If F is measured in Newtons and q in Coulombs, then the field Intensity E is in Newtons per Coulomb (NC^{-1}). Since E is a vector, the electric field is an example of a *vector field*. Electric fields or field intensity are added by the usual vector methods. Thus the electric intensity at a point due to several charges is the vector sum of the intensities due to the individual charges.

From equation 3.7, we also see that

$$F = qE \qquad 3.8$$

Electric Force = Electric Charge × Electric Field Intensity

We can obtain the equation for the Field Intensity E due to a point charge q at a distance r from that charge making use of equations 3.6 and 3.7. We assume the point charge is situated in vacuum.

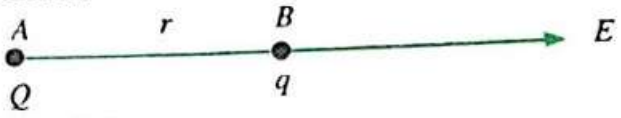


Fig. 3.5

The force between point charge Q and test charge q situated at B is

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}$$

Field Intensity at B is

$$E = \frac{F}{q} = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2 q} = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Hence we can write

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad 3.9$$

This is the field intensity due to a charge Q at a distance r from the charge.

Example 3.3

Calculate the electric field intensity in vacuum at a distance of 10 cm from a charge of 5×10^{-4} C. (Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ Nm^2C^{-2}$)

Solution

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{9\times10^9 \times 5\times10^{-4}}{(0.1)^2}$$

$$= \underline{4.5\times10^8\ NC^{-1}}$$

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Example 3.4

a. Explain what is meant by:
   (i) Electric field intensity
   (ii) Electric lines of force.

b. Two similar but opposite point charges $-q$ and $+q$ each of magnitude 5.0×10^{-8} C are separated by a distance of 8.0 cm in vacuum as shown in the diagram below

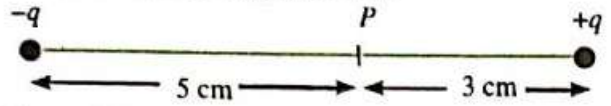


Fig. 3.6

Calculate the magnitude and direction of the resultant electric field intensity E at the point P. Draw lines of force due to this system of charges. (Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{Nm}^2\text{C}^{-2}$)

(SSCE)

Solution

For question (a) see text

b. At P the direction of the field vector is towards the left since the positive charge at P is attracted towards the $-q$ charge.

$$\therefore\quad E_- = \frac{-q}{4\pi\varepsilon_0 r^2} = \frac{5\times10^{-8}\times9\times10^9}{(0.05)^2}$$

$$= \underline{1.8\times10^5\ \text{NC}^{-1}}$$

The field vector due to $+q$ charge is also directed towards the left since the +ve charge at P is repelled by +.

$$\therefore\quad E_+ = \frac{+q}{4\pi\varepsilon_0 r^2} = \frac{5\times10^{-8}\times9\times10^9}{(0.03)^2}$$

$$= \underline{5\times10^5\ \text{NC}^{-1}}$$

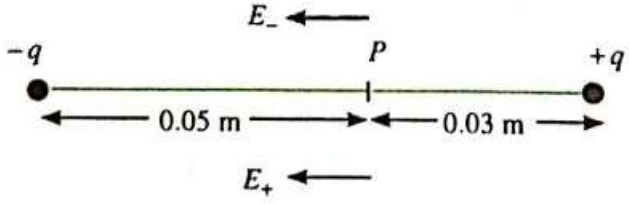


Fig. 3.7

Resultant electric field intensity, $E = E_- + E_+$

$$E = (1.8\times10^5 + 5.0\times10^5)\ \text{NC}^{-1}$$

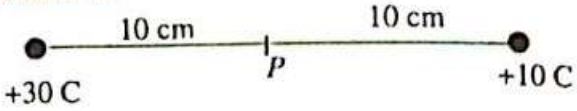
$$= \underline{6.8\times10^5\ \text{NC}^{-1}}$$

The field is directed towards the left or towards $-q$ charge.

Example 3.5

Two point charges +30 C and +11 C are separated by a distance of 20 cm. Compute the electric intensity and the force on a $+5\times10^{-6}$C charge placed midway between the charges.

Solution



$$E = \frac{q}{4\pi\varepsilon_0 r}$$

A positive charge at P will be repelled towards the right by the + 30 C charge, and repelled towards the left by the +10 C charge.

Hence the resultant field intensity E at P is given by

$$E = E_{30} - E_{10}$$

$$= 9\times10^9\left[\frac{30}{(0.1)^2} - \frac{10}{(0.1)^2}\right]\text{NC}^{-1}$$

toward the right

$$= \frac{9\times10^9}{(0.1)^2}(30-10) = \frac{9\times10^9\times20}{10^{-2}}$$

$$= \underline{1.8\times10^{13}\ \text{NC}^{-1}}\ \text{toward the right}$$

$$F = qE = 5\times10^{-6}\times1.8\times10^{13}\ \text{N}$$

$$= \underline{9.0\times10^7\ \text{N}}$$

3.3 Electric Potential

In a gravitational field, forces of attraction act between masses. Therefore in order to separate two masses, work must be done against the field.

Similarly, in an electrostatic field where forces of attraction or repulsion act between charges, work must be done against this field in order to move charges against this field.

The *gravitational potential* at any point in a gravitational field is the work done per unit mass in bringing mass to that point from an arbitrarily chosen zero level, the ground or sea level of the earth. Similarly in an electrostatic or electric field the *electric potential* at a point is defined as the work done per unit positive charge in bringing charge to that point from an arbitrarily chosen zero of potential. If work is done against the field, the potential is positive. If work is done by the field the potential is negative. Points of positive potential are said to be at a higher potential than those of negative potential. Thus in an electric cell,

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the positive terminal is at a higher potential than the negative terminal. The arbitrarily chosen zero of potential is taken at infinity, but in practical terms, the earth is taken to be at zero potential. Thus whenever a conducting body is connected to the earth it is said to be 'grounded' or 'earthed' or to have a zero potential.

Electric potential (V) at a point is defined as the work done in bringing unit positive charge from infinity to that point against the electrical forces of the field.

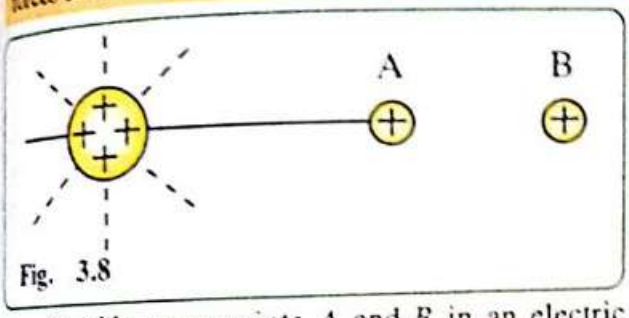


Fig. 3.8

Consider two points A and B in an electric field. If more work is done in bringing unit positive charge from infinity to A than to B we say that A is at a higher potential than B. There is then a *potential difference* (P.d.) between A and B. This difference in potential is equal to the extra work done in moving the positive charge from B to A against the electric field.

Potential difference (V_{AB}) between two points A and B is the work done in taking unit positive charge from one point to the other in the electric field.

Both the potential and the potential difference are scalar quantities having the dimension of work/charge. The unit is the volt.

$$1 \text{ volt} = 1 \frac{\text{Joule}}{\text{Coulomb}}$$

The potential difference between two points is one Volt if the work done in taking one Coulomb of positive charge from one point to the other is one Joule.

$$W = qV \qquad 3.10$$

or (Joules) = (Coulomb × Volts)

$$V = \frac{W}{q} \qquad 3.11$$

Also it can be shown that electric intensity (E) is related to the electric potential difference between two points through the equation.

$$E = \frac{V}{d} \qquad 3.12$$

where d is the distance between the two points in an electric field. Hence from 3.12 we have that

$$V = Ed \qquad 3.13$$

$$\text{Volt} = \text{NC}^{-1}\text{m}$$

Also from equation 3.12 we see that the unit of E can also be in volt per metre. But from equation (3.9) we have

$$E = \frac{q}{4\pi\varepsilon_0 r^2} \text{ or } \frac{q}{4\pi\varepsilon_0 d^2}$$

Putting this in equation 3.13 we have that

$$V = \frac{q}{4\pi\varepsilon_0 d^2} \times d$$

or $$V = \frac{q}{4\pi\varepsilon_0 d} \qquad 3.14$$

where V is the potential at a point due to a charge q at a distance d from the charge.

Example 3.6

Two parallel plates are charged to a voltage of 40 V. If they are separated by a distance of 10.0 cm, calculate the electric intensity between them.

Solution

$$E = \frac{V}{d}$$

$$= \frac{40 \text{ (volt)}}{0.1 \text{ m}} = \underline{400 \text{ Vm}^{-1}}$$

Example 3.7

Calculate the potential at a point 40 cm from a charge of 3×10^{-6} C. (Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2C^{-2})

Solution

$$V = \frac{q}{4\pi\varepsilon_0 d}$$

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$$= \frac{9\times10^{9}\times3\times10^{-6}}{0.4}$$

$$= \underline{6.75\times10^{4} \text{ volts}}$$

Example 3.8

How much work is required to carry a charge of 3×10^{-5} C from a point 50 cm from a charge of 2×10^{-4} C to a point 20 cm from it.

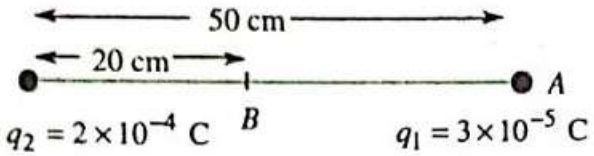


Fig. 3.9

Solution

$W = q_1 V_{AB}$

$V_{AB} = \dfrac{q}{4\pi\varepsilon_0 r}$

V_{AB} = potential difference between A and B

Potential at A due to a charge of 2×10^{-4} C

$$= \frac{2\times10^{-4}}{4\pi\varepsilon_0}\times\frac{1}{0.5}.$$

Potential at B due to charge of 2×10^{-4} C

$$= \frac{2\times10^{-4}}{4\pi\varepsilon_0}\times\frac{1}{0.2}.$$

Potential difference V_{AB} is given by

$$V_{AB} = \frac{2\times10^{-4}}{4\pi\varepsilon_0}\left(\frac{1}{0.2}-\frac{1}{0.5}\right)$$

But $\dfrac{1}{4\pi\varepsilon_0} = 9\times10^{9}$

Hence $V_{AB} = 9\times10^{9}\times2\times10^{-4}\times3$

$= 54\times10^{5}$ Volts

Hence $W = qV_{AB}$

$= 3\times10^{-5}\times54\times10^{5}$ Joules

$= \underline{162 \text{ Joules}}$

3.4 Capacitors and Capacitance

A capacitor is a device for storing electric charges or for storing electricity. It consists essentially of two conductors (metal plates) carrying opposite charges. The metal plates are separated a small distance, d, by an insulator

The insulating material placed between the plates of a capacitor is called a *dielectric substance*. The capacitance increases with the area between the plates, decreases with increasing distance, and increases with dielectric constant.

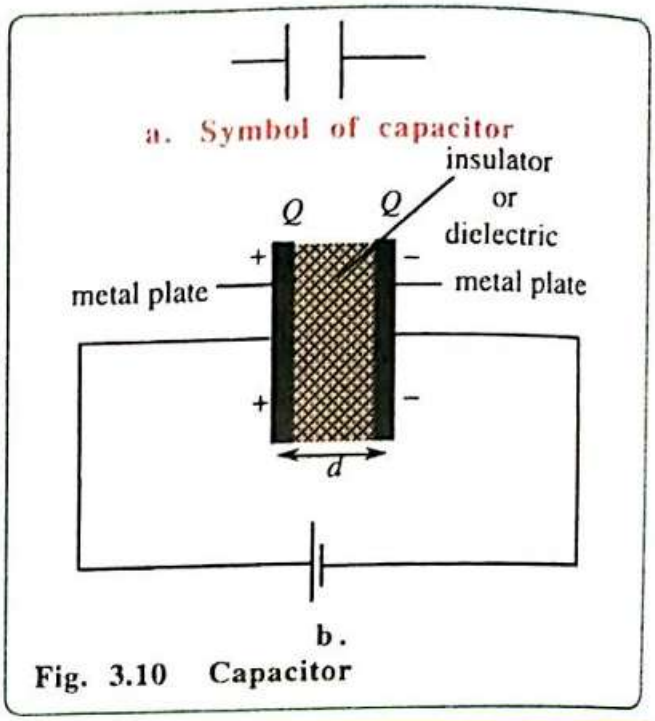


Fig. 3.10 Capacitor

The **Capacitance** (C) of a capacitor is defined as the ratio of the charge Q on either plates or conductors to the potential difference V between them.

$$C = \frac{Q}{V} \qquad 3.15$$

The unit of the Capacitance is the *Farad*. The Capacitance of a capacitor is one *Farad* if one Coulomb of charge is transferred from one conductor to the other per unit volt of potential difference between the conductors. Thus the units of equation 3.15 are

$$\text{Farad} = \frac{\text{Coulomb}}{\text{Volt}}$$

1 Microfarad (μF) = 10^{-6} Farad (F)
1 Picofarad (pF) = 10^{-12} Farad

Quantitatively, the capacitance of a capacitor is a measure of its ability to store up electricity. The capacitance, C is a constant for a given capacitor. The value of the Capacitance depends on the size, shape, distance separating the two plates, and also on the nature of the material that separates them and the common area of plates.

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Example 3.9

A capacitor of charge, 5×10^{-8} Coulomb has a potential of 100 Volts. What is the capacitance of the capacitor?

Solution

$$C=\frac{Q}{V}$$
$$=\frac{5\times10^{-8}}{100}$$
$$=5\times10^{-10} \text{ Farad}$$
$$=\underline{5\times10^{-4}\ \mu F}$$

Arrangement of Capacitors – Series and Parallel arrangements

Figure 3.11 shows two different arrangements of capacitors in a circuit. In (a) the capacitors are arranged in series, in (b) they are arranged in parallel.

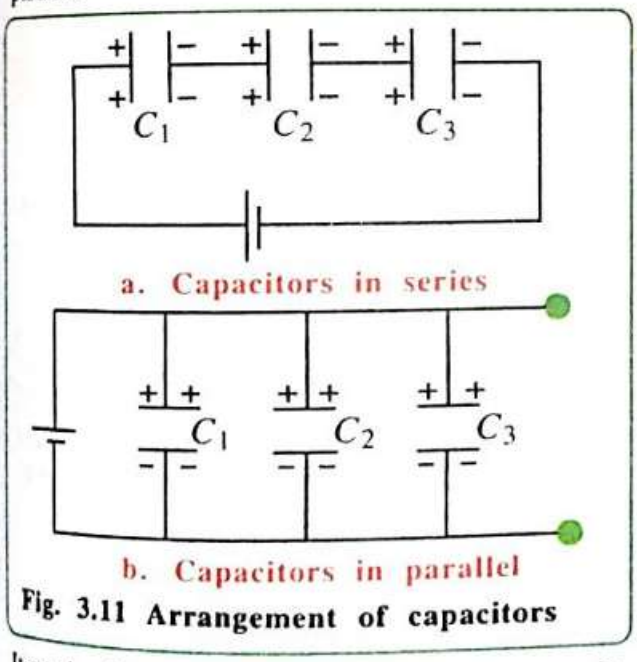


Fig. 3.11 Arrangement of capacitors

It can be shown that the equivalent capacitance, C, for the capacitors in series is given by

$$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3} \qquad 3.16$$

Capacitors in series

For capacitors in parallel, it can be shown that the equivalent capacitance C is given by

$$C=C_1+C_2+C_3 \qquad 3.17$$

Capacitors in parallel

The student should note the difference between the combination of capacitances and the combination of resistances.

The resultant or equivalent capacitance of any number of capacitors connected in parallel is the sum of their individual capacitances. The reciprocal of the equivalent capacitance of any number of capacitors connected in series is the sum of the reciprocals of the individual capacitances

Example 3.10

Two capacitors of capacitances 5 μF and 3 μF are connected in series and the resulting combination is connected across 600 volts. What is (i) the equivalent capacitance of the combination (ii) the total charge on the combination?

Solution



Fig. 3.12

(i) $\frac{1}{C}=\frac{1}{5}+\frac{1}{3}=\frac{8}{15}$

$$C=\frac{15}{8}\mu F=1.875\ \mu F=1.875\times10^{-6}\ F$$

(ii) $q=CV$

$$=(1.875\times10^{-6}\times600) \text{ Coulombs}$$
$$=\underline{1.125\times10^{-3} \text{ Coulomb}}$$

Example 3.11

Two capacitors of capacitances 2 μF and 3 μF are connected in parallel. What is the equivalent capacitance?

Solution

$$C=C_1+C_2=(2+3)\ \mu F$$
$$=\underline{5\ \mu F}$$

Example 3.12

Find the equivalent capacitance for a combination of capacitors shown in the fig. 3.13.

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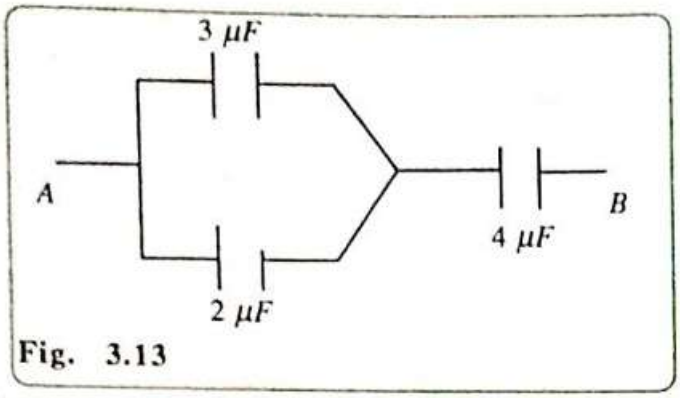


Fig. 3.13

Solution

For the capacitors in parallel, the equivalent capacitance C is

$$C = 3 + 2 = 5\ \mu F$$

The 5 μF combination is now in series with the 4 μF capacitance.

$\therefore$ the net capacitance is given by

$$\frac{1}{C} = \frac{1}{5} + \frac{1}{4}$$

$$= \frac{9}{20}\ \mu F$$

$$C = \frac{20}{9}\ \mu F = \underline{2.22\ \mu F}$$

3.5 Energy Stored in a Capacitor

A capacitor is charged so that it can store electrical energy. While the capacitor is being charged, its charge builds up from an initial value of zero to a final value, q, and the potential difference between its plates builds up from zero to a final value V.

During the charging of the capacitor, the average value of the potential difference is given by $\frac{0+V}{2} = \frac{1}{2}V$. The work done in charging the capacitor through an average potential difference of $\frac{1}{2}V$ is

$$W = \frac{1}{2}V \times q$$

$$= \frac{1}{2}qV$$

But $C = \frac{q}{V}$ or $V = \frac{q}{C}$

$$\therefore \quad W = \frac{1}{2}qV = \frac{1}{2}q \times \frac{q}{C}$$

$$= \frac{1}{2}\frac{q^2}{C}$$

Also $q = CV$, hence

$$W = \frac{1}{2}qV = \frac{1}{2}CV \cdot V$$

$$= \frac{1}{2}CV^2$$

This work done in charging a capacitor is the energy stored in the capacitor. Hence

$$W = \frac{1}{2}qV = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}CV^2 \qquad 3.19$$

W is in joules if q is in coulombs, V in volts and C in farads.

Example 3.13

A capacitor of capacitance 5 μF is subjected to a potential difference of 1000 volts across its terminals. Calculate the energy stored in the capacitor.

Solution

$$W = \frac{1}{2}CV^2$$

$$= \frac{1}{2} \times 5 \times 10^{-6} \times 10^6$$

$$= \underline{2.5 \text{ Joules}}$$

Application of Capacitors

Capacitors are used in

(i) radio circuits for tuning
(ii) Ignition system of motor vehicles
(iii) the elimination of sparks when a circuit containing inductance is suddenly opened e.g. in induction coils.

Summary

1 *Electric field* is an area around a charge, q_1, in which another charge q_2 can experience a force.

2 The force, F, between two charges q_1 and q_2 is given by

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

where r is the distance between the two charges and $\frac{1}{4\pi\varepsilon_0}$ is a constant $= 9.0 \times 10^9$ Nm^2C^{-2}. F acts along the line joining the two charges.

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3 *Electric line of force* is an imaginary line whose direction at any point in the electric field is the same as the direction a small positive charge would follow if placed at that point in the electric field.

4 *Electric field intensity* E is the force per unit positive charge. It is a vector.

$$E = \frac{F}{q} = \frac{Q}{4\pi\varepsilon_0 r^2}$$

5 *Electric Potential* (V) is the work done in bringing unit positive charge from infinity to the point.

6 *Potential difference* (P.d) between two points A and B is the work done in bringing unit positive charge from B to A where A is at a higher potential than B.

7 A *capacitor* is a device for storing electric charge. The capacitance of a capacitor is the charge per unit P.d. between the plates of a capacitor ($C = \frac{Q}{V}$). The capacitance of a parallel plate capacitor is dependent on (i) the area of the plates (ii) the distance of separation between the plates and (iii) the dielectric between the plates.

8 When capacitors are connected in series, the equivalent capacitance is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$

When connected in parallel, the equivalent capacitance is

$$C = C_1 + C_2 + C_3 + \cdots + C_n$$

9 Energy stored in a capacitor is given by

$$W = \frac{1}{2}qV = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}CV^2$$

Exercise 3

In the following problems assume

$$\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ \text{Nm}^2\text{C}^{-2}$$

1

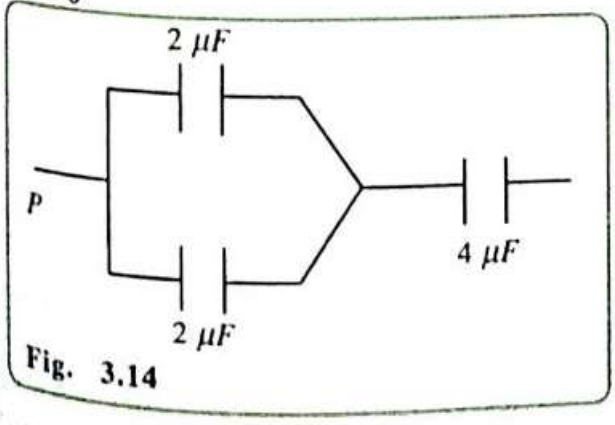


Fig. 3.14

The net capacitance in the circuit above is

A 8.0 μF B 6.0 μF C 4.0 μF D 2.0 μF

(JAMB)

2

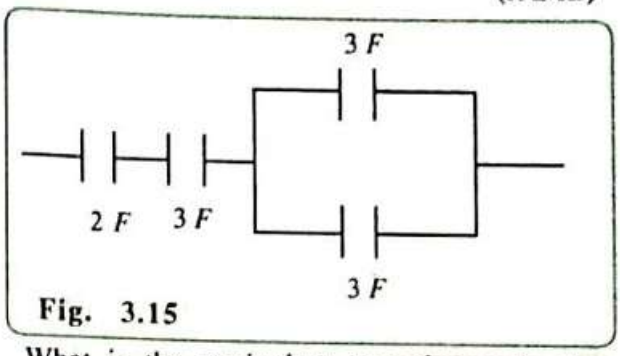


Fig. 3.15

What is the equivalent capacitance in the circuit above?

3 What is the electric potential energy between two protons of charge q and at a distance of r apart? (Permittivity of free space = ε_0)

A $4\pi\varepsilon_0 q^2 r$ B $\frac{qr}{4\pi\varepsilon_0}$

C $\frac{q}{4\pi\varepsilon_0 r}$ D $\frac{q^2}{4\pi\varepsilon_0 r}$

E $\frac{q^2}{4\pi\varepsilon_0 r^2}$

(SSCE)

4 A work of 30 Joules is done in transferring 5 milliCoulombs of charge from a point B to a point A in an electric field. The potential difference between B and A is

A 1.7×10^{-4} V B 3.4×10^{-4} V

C 1.5×10^{-1} V D 6.0×10^{3} V

E 1.2×10^{4} V

(SSCE)

5 A charge of 1.0×10^{-5} Coulombs experiences a force of 40 N at a certain point in space. What is the electrical field intensity?

A 8.00×10^{6} NC^{-1}

B 4.0×10^{6} NC^{-1}

C 4.00×10^{-4} NC^{-1}

D 2.0×10^{-4} NC^{-1}

E 2.5×10^{-7} NC^{-1}

(SSCE)

6 State the law of force between two charges, q_1 and q_2 separated by a distance r. Write down the units of F, q, r and the constant of proportionality.

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7 Find the force of attraction between two equal but opposite charges, of 2×10^{-6} C if the distance between them is 25 cm.

8 Explain the terms

(i) *Electric field* (ii) *Electric Intensity.*

Write down the expression for the electric intensity at a distance r metre from a point charge Q Coulomb.

Write down the expression for the force felt by a charge q Coulomb in a field of intensity E Newtons per Coulomb.

9 Calculate the electric field intensity at a point 15 cm from a charge of 10 μC.

10 A small object carrying a charge of -5×10^{-9} C experiences a downward force of 10 Newton when placed at a certain point in an electric field.

a. What is the electric intensity at that point.

b. What would be the magnitude and direction of the force acting on an electron placed at that point? (Charge on an electron = 1.6×10^{-19} C)

11 Define *electric potential.* Determine the electric potential due to a charge of 12×10^{-9} C at a point 6 cm from the charge.

12 *What is a capacitor?* Explain the term *capacitance* of a capacitor.

Four capacitors of capacitances C_1, C_2, C_3 and C_4 are connected in series. Write down the expression for the equivalent capacitance C. If they are connected in parallel, what is the equivalent capacitance?

13 A capacitor of capacitance 3.0 μF is subjected to a 2000 V potential difference across its terminals. Calculate the energy stored in the capacitor.

14 Derive the expressions for the energy stored by a capacitor of charge Q, potential difference between the plates of V and capacitance C. Name two applications of capacitors. On what factors does the capacitance of a parallel plate capacitor depend?

15 (a) (i) What is meant by ***dielectric substance***?

(ii) List the factors which determine the capacitance of a parallel plate capacitor and state the effect each of them has on the capacitance.

16 (a) Explain the statement the **capacitance of a capacitor is 5 μf**.

(b) (ii) State the factors upon which the capacitance of a parallel plate capacitor depend.

(iii) State how the capacitance depends on each of these factors stated in (b)(i).

(c) A series arrangement of three capacitors of value 8 μf, 12 μf and 24 μf is connected in series with a 90-V battery.

(i) Draw an open circuit diagram for this arrangement.

(ii) Calculate the effective capacitance in the circuit.

(iii) On close circuit, calculate the charge on each capacitor when fully charged.

(iv) Determine the pd across the 8 μf capacitor.

(SSCE)

17 (a) Explain

(i) Electric potential.

(ii) Electric potential energy

(iii) State the S.I unit of each term in (a) above,

(b) An isolated electrically-charged sphere of radius r, and charge Q is supported on an insulator in air of permittivity ε_o. Write down

(i) an expression for the electric field intensity on the surface of the sphere.

(ii) an expression for the electric potential at the surface of the sphere.

(iii) a relationship between the electric field intensity and the electric potential at the surface of the sphere.

(c) The plates of a parallel plate capacitor, 5.0 x 10^{-3}m apart are maintained at a potential difference of 5.0 x 10^{4}V. Calculate the magnitude of the

(i) Electric field intensity between the plates.

(ii) Force on the electron.

(iii) Acceleration of the electron. (electronic charge = 1.6 x 10^{-19}C, mass of electron - 9.1 x 10^{-31}kg).

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4 ELECTRIC FIELD 2: CURRENT ELECTRICITY

This second section of our discussion of electric field, is a revision and extension of current electricity treated in Book 1 chapter 7. The student should thoroughly review that chapter before embarking on the present chapter.

At the end of this chapter, the student should be able to:

1 Identify all the component parts of a simple cell and an accumulator.
2 Solve problems involving series and parallel connections of resistance and cells.
3 Convert a galvanometer to an ammeter and to a voltmeter.
4 State the factors on which the resistance of a uniform wire depends.
5 State and demonstrate the condition for a balanced wheatstone bridge circuit and deduce the condition for balanced metre bridge circuit.
6 Explain the basic principle of the potentiometer circuit.

4.1 Production of Continuous Charges

A continuous flow of charges is produced from primary and secondary cells.

a. Primary Cells

Primary cells are those from which current is produced as a result of non-reversible chemical changes taking place between the various components of the cell

A *simple cell* is an example of a primary cell. When all the zinc of such a cell is used up through the chemical action with sulphuric acid, by which the current is produced, the cell cannot be restored or recharged to its original condition by passing a charging current through the cell in the reverse order. In order to cause primary cells to produce current once again, they must be resupplied with fresh active materials (zinc, copper, sulphuric acid).

The disadvantage of the primary cells is that the chemicals in them are gradually used up when the cells are in use.

b. Secondary Cells (or Accumulators)

The second class of cells are called *secondary cells* or *accumulators*.

Secondary cells or accumulators are cells whose chemical actions can be reversed by driving a current through them in a direction opposite to the current they supply. Such cells can therefore be recharged and used for a long time.

The main advantage of accumulators is that they have very low internal resistance (unlike primary cells). They can therefore provide large currents for a long time with very little change in their *terminal potential difference*.

They are used in motor cars to provide the energy for the spark igniting the petrol in the engine, and also in telephone exchanges to provide current in telephone cables when required.

Examples of Primary Cells

There are three types of primary cells - the simple cell, the Daniel cell and the Leclanché cell.

The Daniel Cell

It consists of (i) a copper vessel filled with a saturated copper sulphate solution. (ii) a porous pot containing dilute sulphuric acid (H_2SO_4) which is the electrolyte for this cell. (iii) An amalgamated zinc rod immersed in the acid. The zinc rod acts as the negative terminal of the cells while the copper vessel acts as the positive terminal. The copper sulphate solution acts as the depolarizer for the Daniel cell.

The initial e.m.f. produced by the Daniel cell is about 1.1 volts and the cell provides only a small current for some time.

The Daniel cell is no longer in use.

The Leclanché Cell

a. The Wet Type

The internal structure of this cell consists of (i) the carbon rod contained in a porous pot. This carbon rod acts as the positive terminal of the cell. (ii) A porous pot containing a mixture of manganese dioxide and powdered carbon, parked around the carbon rod. (iii) A glass or porcelain jar filled with a solution of ammonium chloride. (iv) A zinc rod immersed in the ammonium chloride solution.

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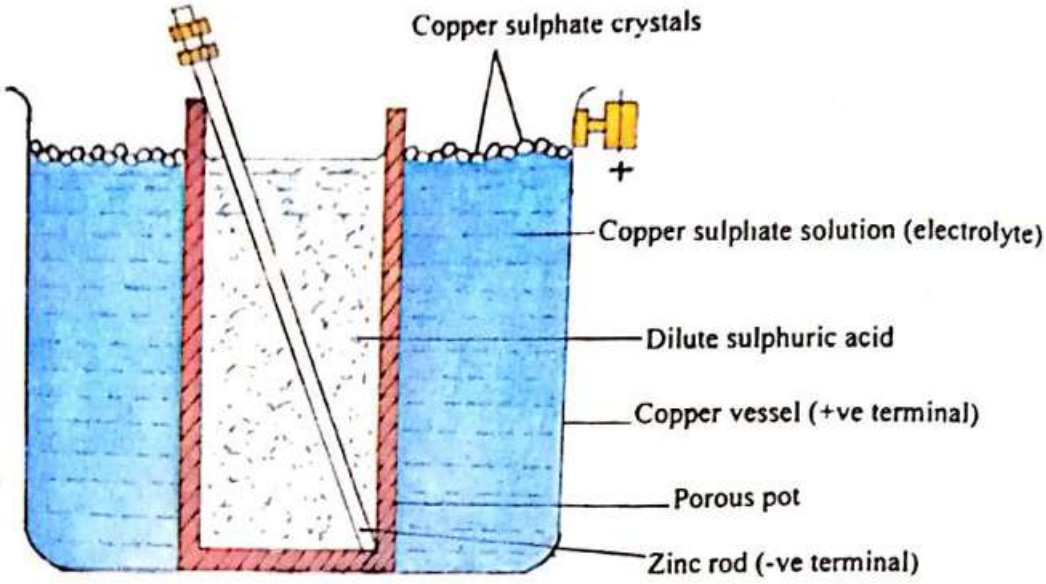


Fig. 4.1 Daniel cell

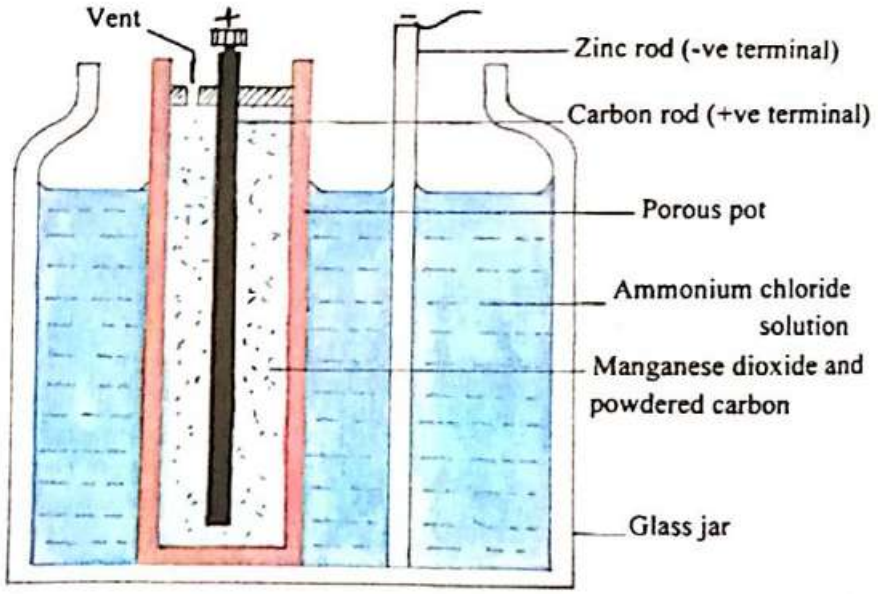


Fig. 4.2 Leclanché cell

In this cell, the zinc rod is the negative terminal, the carbon rod is the positive terminal, the ammonium chloride is the electrolyte and the manganese dioxide is the depolarizing agent. Depolarization however takes place more slowly than the rate at which hydrogen is liberated by the action of zinc and ammonium chloride solution. Therefore when the cell is in continuous use, some depolarization takes place and the e.m.f. of the cell drops. Another defect of the wet leclanché cell is that it is cumbersome to carry about without spilling the liquid.

The Leclanché cell is therefore used only where intermittent current is required, e.g. in electric bells. The advantage of the cell is that its e.m.f. is relatively high, about 1.5 volts, and its chemicals are cheap.

b. Leclanché Cell - the Dry Type

In the dry type of leclanché cell,

1 the electrolyte is ammonium chloride in the form of a paste or jelly and is mixed with starch and flour.
2 The positive terminal is a carbon rod.
3 The negative terminal is the zinc container.
4 The depolarizer of manganese dioxide mixed with powdered carbon is contained in a muslin bag round the carbon rod.
5 A cardboard disc is used at the bottom to prevent the carbon pole from touching the negative zinc container to prevent the short circuiting of the cell.

The e.m.f. of a dry leclanche cell is about 1.5 volts and the chemical reaction taking place within this cell is the same as that of the wet type. The dry cell like the wet leclanché cell cannot maintain

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current for a long time because it polarizes if used continuously owing to the hydrogen liberated at the carbon rod. Hydrogen is produced at a faster rate than can be depolarized by the manganese dioxide.

Dry cells are used in torches, bicycle lamps, electric bells and transistor radios.

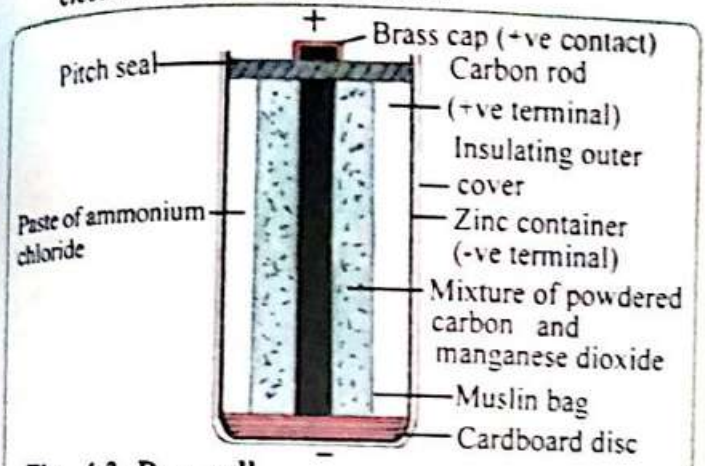


Fig. 4.3 Dry cell

Secondary Cells

There are two types -

a. the *lead-acid accumulator* and
b. the alkaline or Nickel - Iron (NIFE) accumulator.

1 Lead - Acid Accumulator

The simple form of the cell consists of (i) a positive pole of lead peroxide (chocolate or dark brown in colour) (ii) a negative pole of lead plate, (grey in colour) and (iii) an electrolyte of dilute sulphuric acid.

In modern commercial accumulators, both the positive and negative plates are made of grids of lead-antimony alloy, the holes and grooves of which are filled with either the lead or the lead peroxide. The positive plates containing the lead peroxide and the negative plates containing lead are assembled alternatively in groups and are separated by insulators. All the positive plates are connected together to form one positive terminal of the accumulator, and the negative plates are also connected together to form the negative terminal. The whole framework is encased in a plastic container filled with sulphuric acid.

When fully charged, the e.m.f. of the cell is about 2.2 V and the relative density of the dilute sulphuric acid is about 1.25. The cell is considered completely discharged when the relative density of the acid falls to 1.15.

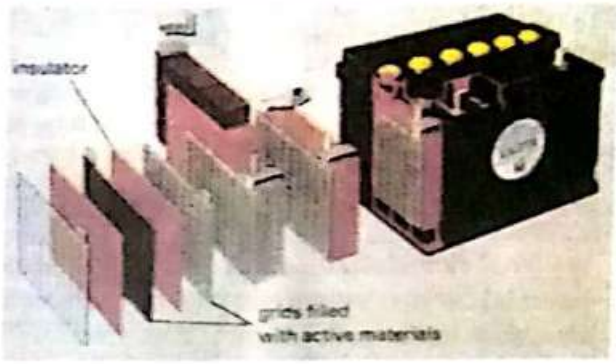
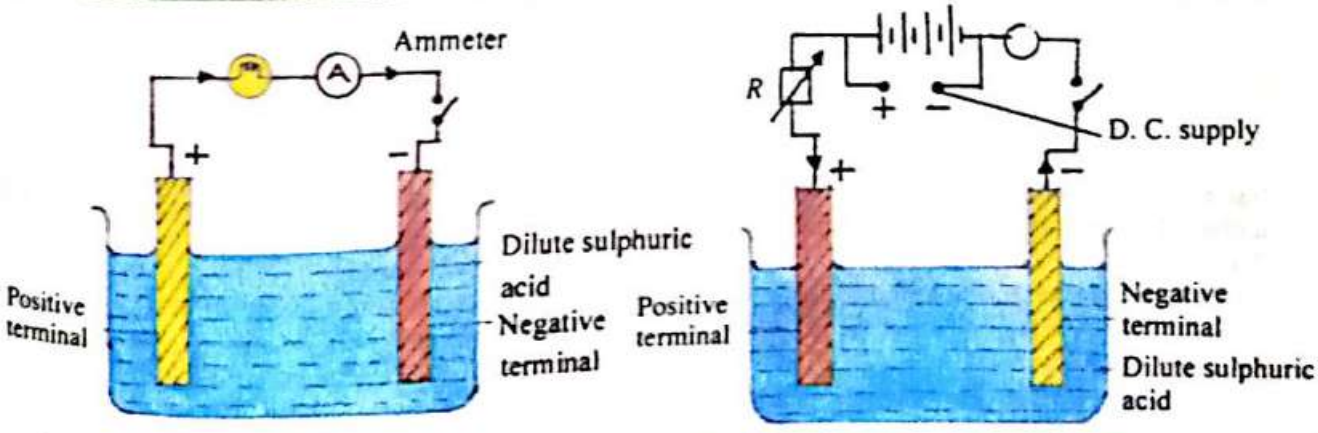


Fig. 4.4

Using or Discharging the Accumulator

In fig 4.5a the accumulator is used to supply current to an external resistance e.g. an electric bulb.

After some time of such use, the current will cease to flow and will be indicated by both the ammeter and the bulb. The accumulator is said to be *discharged*. A little red sulphate is formed at



a. Discharging an accumulator

b. Charging an accumulator

Fig. 4.5

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both of the plates and the relative density of the acid and the e.m.f. of the cell decrease below the original levels.

2 Nickel-Iron (NIFE) Accumulator

The poles of this accumulator are positive nickel plates and negative iron plate. The electrolyte is a solution of potassium hydroxide. The e.m.f. when fully charged is about 1.5 volts and falls to 1.3 volts after some time.

The NIFE accumulator has a longer life span than the lead acid accumulator.

Charging and Recharging an Accumulator

To charge an accumulator, we pass direct current (d.c.) through it in the opposite direction to that which it supplies (fig 4.5b). The positive terminal of the d.c. is connected to the positive terminal of the accumulator and the negative terminal of the d.c is connected to the negative terminal of the accumulator.

The rheostat R in the charging circuit is essential as the sulphuric acid has a very low resistance such as 0.02 ohm. The ammeter A is also included in the circuit to ensure that only the charging current recommended by the manufacturer is used.

Charging the accumulator is considered complete when the relative density of the acid attains the recommended value of 1.25 and the e.m.f. of the accumulator is about 2.2 volts.

4.2 Arrangement of Cells

Series Connection

A simple cell has an e.m.f. of about 0.89 V. When a p.d greater than this is required, a series arrangement of cells is used. In such an arrangement, the positive terminal of each cell is connected to the negative terminal of the next cell as shown in fig 4.6a.

The total e.m.f. of the cells arranged in series is equal to the sum of the individual e.m.f. of each cell. $E = E_1 + E_2 + E_3$. The internal resistance, r, of the series arrangement is the sum of the internal resistance of each cell. $r = r_1 + r_2 + r_3$.

Such an arrangement increases the current flowing in the circuit since the cells assist each other.

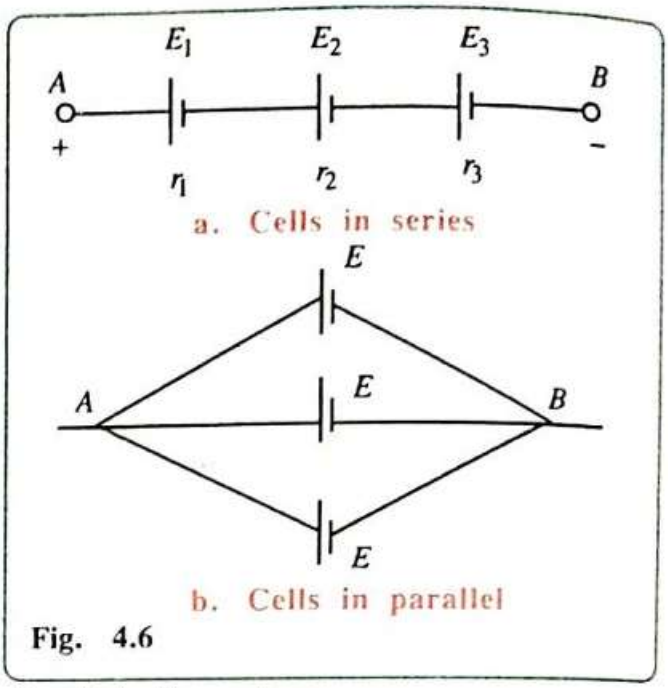


Fig. 4.6

Parallel Connection

In a parallel arrangement of cells, all the positive terminals are joined to one point, A , and all the negative terminals are joined to another point B.

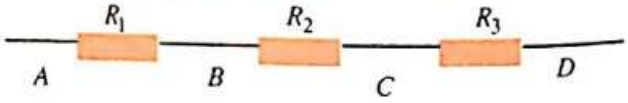
If each cell has an e.m.f., E, the net e.m.f. of the parallel arrangement is also E, that of a single cell.

Since all the positive terminals of the cells are joined to the same point, they must be at the same potential. Similarly for all the negative terminals. Hence the p.d. between AB must be the same as the p.d. across any of the cells.

Arrangement of Resistors

Resistors in Series

When resistors are connected end to end as shown in fig. 4.7 they are said to be in series.



$R = R_1 + R_2 + R_3$

Fig. 4.7

In such an arrangement, the current passing through each resistor is the same, but the p.d. across each resistor is different.

The combined or equivalent resistance (R) of a number of resistors in series is equal to the sum of their individual resistances (fig. 4.7).

$$R = R_1 + R_2 + R_3 + \cdots + R_n \qquad 4.1$$

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We can prove the above equation using Ohm's law.

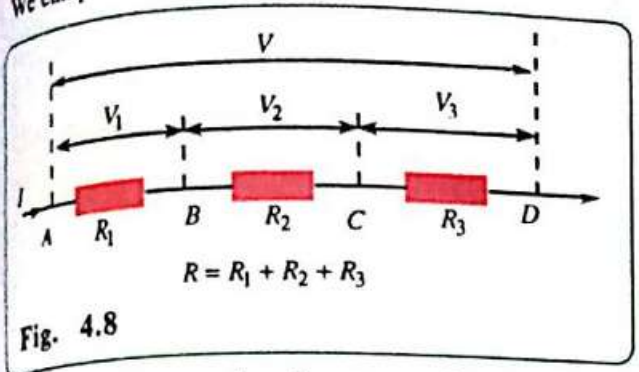


Fig. 4.8

Let V be the total p.d. across the combined resistors, and R the combined resistance.

Using Ohms law we have

$$V = IR$$

where I is the current flowing through each resistor. The total p.d. (V) across the combined resistances is equal to the sum of the individual p.d.'s across each resistor, hence

$$V = V_1 + V_2 + V_3 + \cdots$$

$$IR = IR_1 + IR_2 + IR_3 + \cdots$$

Eliminating I we have

$$R = R_1 + R_2 + R_3 + \cdots$$

If there are n resistors in series the same arguements can be used to show that

$$R = R_1 + R_2 + R_3 + \cdots + R_n \qquad 4.2$$

Example 4.1

Five resistances of value 0.1 ohms, 1.7 ohms, 2.0 ohms, 1.0 ohms and 0.9 ohms are connected in series. Calculate the value of the combined resistances.

Solution

$$R = R_1 + R_2 + R_3 + R_4 + R_5$$

$$= (0.1 + 1.7 + 2.0 + 1.0 + 0.9) \text{ ohms}$$

$$= \underline{5.7 \text{ ohms}}$$

Resistors in Parallel

When resistors are arranged side by side such that their corresponding ends join together at two common junctions P and Q as shown in fig. 4.9, the set up is known as a parallel arrangement of resistors. In such an arrangement, the p.d across the resistors is the same but the current through each resistor is different. The combined resistance R is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \qquad 4.3a$$

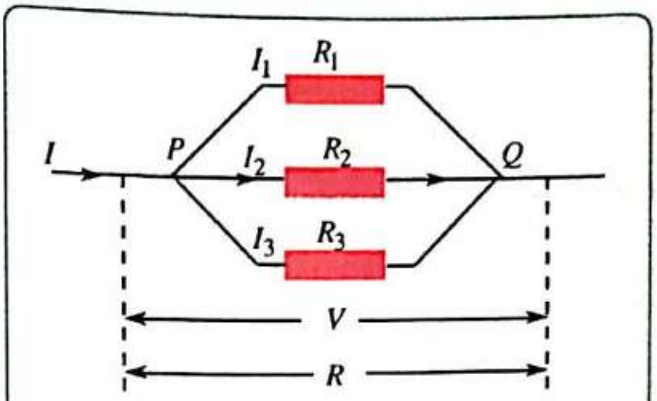


Fig. 4.9 Resistors in parallel

We can prove this as follows:

By ohms law

$$I = \frac{V}{R}$$

where I is the main current flowing, V is the total p.d. across the parallel resistors and R is the combined resistance. I divides into three branches I_1, I_2, I_3 as shown in fig. 4.9. The total current I is given by

$$I = I_1 + I_2 + I_3$$

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

Eliminating V from the above we have

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

If there are n resistors in parallel we can show similarly that

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n} \qquad 4.3b$$

Example 4.2

Four resistors of resistances 1 ohm, 5 ohms, 3 ohms and 4 ohms are connected in parallel. Calculate the value of the combined resistance.

$$\frac{1}{R} = \frac{1}{1} + \frac{1}{5} + \frac{1}{3} + \frac{1}{4} \text{ ohm}^{-1}$$

$$= 1.00 + 0.20 + 0.33 + 0.25 = 1.78 \text{ ohm}^{-1}$$

$$R = \frac{1}{1.78} = \underline{0.56 \text{ ohm}}.$$

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Internal Resistance of a Cell

The internal resistance, r, of a cell is the opposition to current flow offered by the chemicals between the poles of the cell.

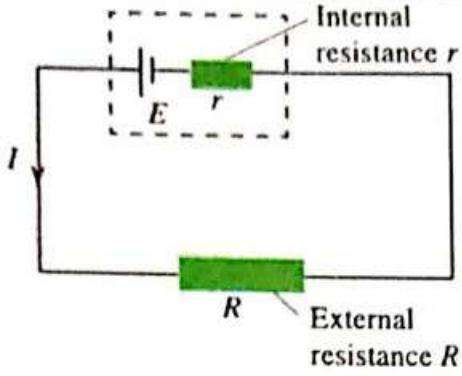


Fig. 4.10 Internal resistance of cell

This internal resistance determines the maximum current that can be supplied by a cell of given e.m.f. (E). We recall that:

The e.m.f. of a cell is the p.d. across its terminals when it is in an open circuit, i.e. not supplying current to an external circuit.

By Ohm's law, such a current is given by

$$I_{max} = \frac{E}{r} \qquad 4.4$$

Thus the more the internal resistance (r) the less current the cell is capable of supplying. Hence cells of low internal resistance can supply a greater current to an external resistance than cells of high internal resistance.

If in fig. 4.10, the cell of e.m.f. E, supplies a current I (A) to an external resistance R, then the p.d. across R, known as *Terminal p.d.* is given by

$$V = E - v \qquad 4.5a.$$

or $$E = V + v \qquad 4.5b.$$

where v is the p.d. across the internal resistance, r.

Using Ohm's law in equation 4.5b, we have

$$E = IR + Ir \qquad 4.6$$
$$= I(R + r)$$

or $$I = \frac{E}{R + r} \qquad 4.7$$

The term Ir in equation 4.6 is called the *lost voltage*, and is equal to v, the potential drop across the internal resistance.

From Equation 4.5 we see that the terminal p.d., V, across the external resistance R, is less than the e.m.f., E, of the cell, by the *lost voltage* (v)

$$E - V = v$$

or $$V = E - v = E - Ir \qquad 4.8$$

Terminal p.d **is defined as the p.d between the terminals of a cell when it is delivering current to an external circuit.**

Example 4.3

A battery made up of 2 cells joined in series supplies current to an external resistance of 5 Ω. If the e.m.f. and internal resistance of each cell is 0.6 V and 3 Ω respectively, calculate the current flowing in the external resistor, the terminal p.d, and the lost voltage.

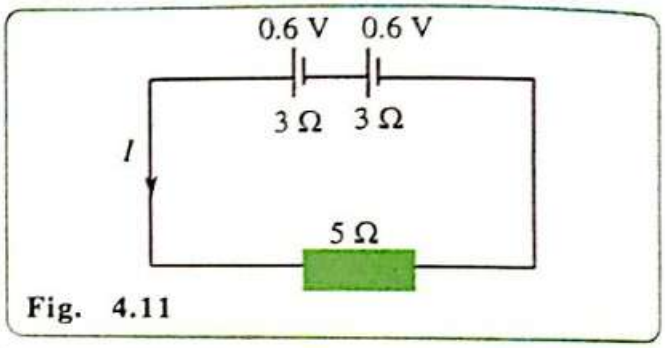


Fig. 4.11

Solution

Total internal resistance of the cells = 3 Ω + 3 Ω
= 6 Ω

Total resistance R, of the circuit is given by

$R = 6 + 5 = 11$ Ω

Total E.m.f. = 0.6 + 0.6 = 1.2 V

Current I is given by

$$I = \frac{E}{R} = \frac{1.2}{11} \text{ A}$$
$$= \underline{0.11 \text{ A}}$$

Terminal p.d is given by

$V = IR = 0.11 \times 5 = \underline{0.55 \text{ volts}}$

Lost voltage, $v = 1.2 - 0.55$
$= \underline{0.66 \text{ volts}}$

Example 4.4

A cell can supply currents of 0.80 A and 0.40 A through a 2 ohm and a 5 ohm resistor respectively. Calculate the internal resistance of the cell. (SSCE)

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Solution

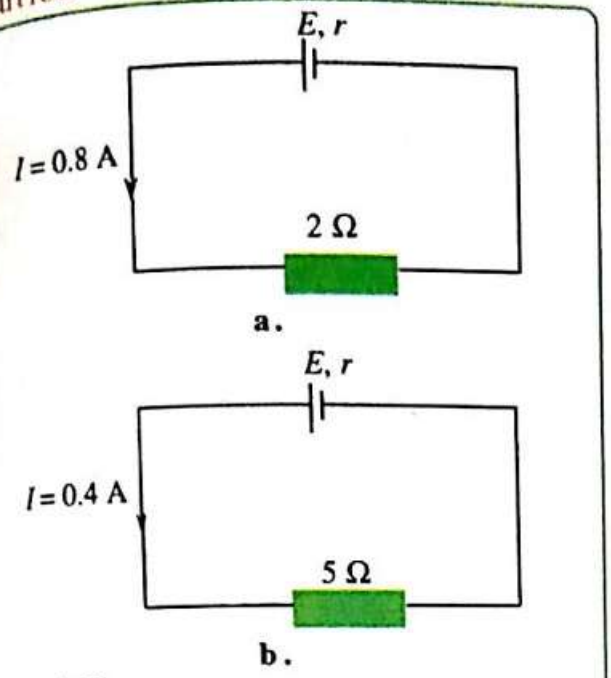


Fig. 4.12

Let r be the internal resistance of the cell.

From $I = \frac{E}{R+r}$

we have

From (a) $0.80 = \frac{E}{2+r}$ (i)

From (b) $0.40 = \frac{E}{5+r}$ (ii)

From $\frac{\text{Eq. (i)}}{\text{Eq. (ii)}}$ we have

$\frac{0.8}{0.4} = \frac{5+r}{2+r}$

$r = \underline{1\ \Omega}$

Example 4.5

A battery of e.m.f. 12 volts and internal resistance $r = 1.5$ ohms is connected to an external resistance of 3 ohms. Calculate

a. the current flowing in the circuit.
b. the potential drop across the 3 ohm resistance and in the internal resistance.
c. the terminal voltage of the battery.
d. the reading of a voltmeter connected across the battery terminal in open circuit.

Solution

a. $I = \frac{E}{R+r} = \frac{12}{3+1.5}$

$= \underline{2.67\ A}$

b. (i) $V_R = IR = 2.67 \times 3$

$= \underline{8.01\ \text{volts}}$

(ii) $V_r = Ir = 2.67 \times 1.5$

$= \underline{4.00\ \text{volts}}$

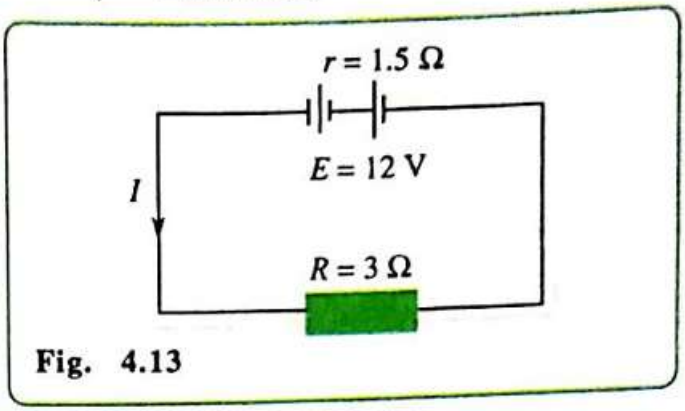


Fig. 4.13

c. Terminal p.d., $V = E - Ir$

$= 12 - 4$

$= \underline{8\ \text{volts}}$

This is the same as the potential drop across the external resistance.

d. The voltmeter connected across the battery terminals in open circuit will read the E.m.f of the battery i.e. 12 volts.

Standard Resistors and Rheostats

We define *resistance* as opposition to the flow of current, and a resistor as a component which is specially designed to provide a known amount of resistance in a circuit.

Standard resistors are resistors which have fixed resistance values e.g. 1 Ω, 2 Ω, 5 Ω, 10 Ω. They are made by cutting up measured lengths of resistance materials such as constantan and manganin and mounting them on a wooden block and providing them with terminals for connection. A resistor whose resistance can be varied is known as *variable resistor* whose symbol is as shown in fig 4.14b.

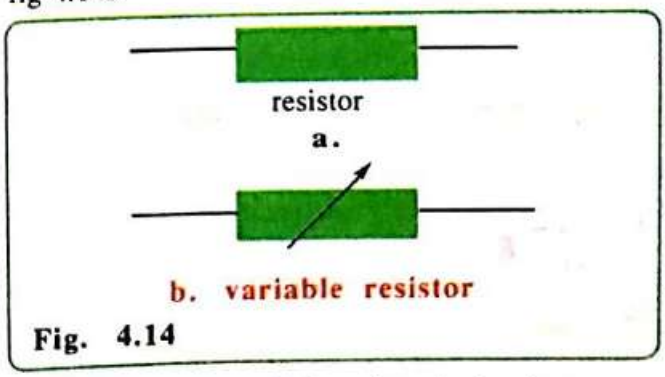


Fig. 4.14

A special type of variable resistor is the *rheostat*.

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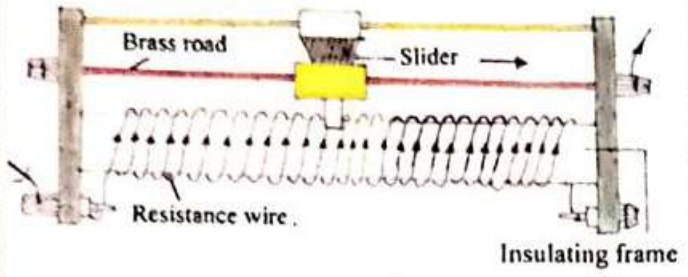


Fig. 4.15 The Rheostat

It consists of a coil of resistance wire wound round an insulating frame, and a slider which makes contact at various points on the wire. The current enters the rheostat at one end of the coil, passes through a part of the resistance wire (determined by the position of the movable slider), and goes through the slider and brass rod to the other end. By moving the slider, we vary the resistance of the rheostat.

The rheostat can also be used as a *potential divider* to produce a varying potential difference.

A way of providing standard resistors into a circuit is through what is called a resistance box. A resistance box contains several standard resistors. The working of the plug type resistance box is as shown in fig 4.16. When the plug is in, current passes across the top and bypasses the resistance R_1 below.

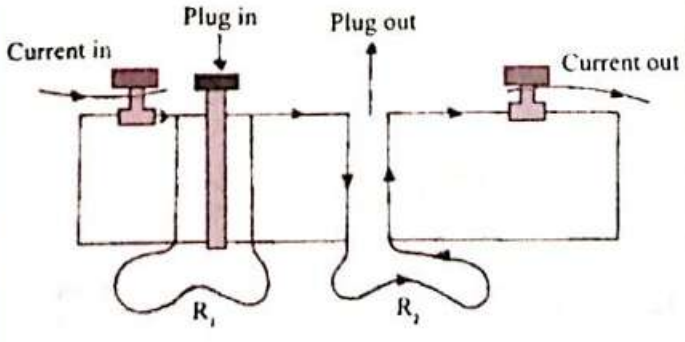


Fig. 4.16 The working of the resistance box

When the plug is out, the current passes through the coil of resistance, R_2 below, whose known value is marked by the plug. R_2 is now the value of the resistance in the circuit. For example, if the top of the resistance box is as shown in fig. 4.17, where shaded circles indicate that plugs are inserted and unshaded circles indicate that the plugs have been removed, the total resistance in circuit is 14 ohms in fig 4.17a but 9 ohms in fig. 4.17b.

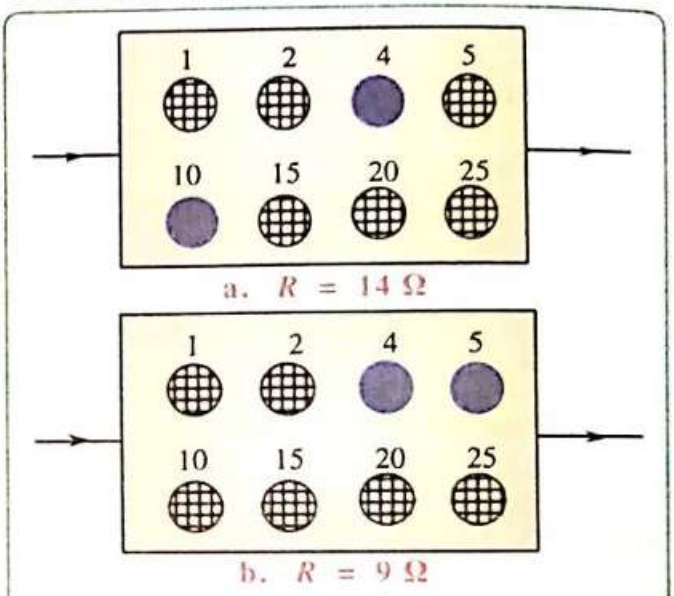


Fig. 4.17 Resistance box

4.3 Resistivity and Conductivity

We recall from Book 1, chapter 7, that four factors affect the resistance of a conducting wire. These are the length, the area, the temperature and the type or substance of the material.

The resistance of a wire or material conductor maintained at a constant temperature is related to its length (l) and its cross-sectional area (A) by the expression

$$R = \rho \frac{l}{A} \qquad 4.9$$

where ρ is a constant of proportionality known as the *Resistivity* of
the material.
From eqn. 4.9, we have that

$$\rho = \frac{RA}{l} \qquad 4.10$$

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Thus $\rho = R$ when $l = 1$ and $A = 1$. Hence we define Resistivity as follows:

Resistivity is the resistance of unit length of material of unit cross-sectional area.

When R is measured in ohms, A in m and l in m, the unit of ρ is in ohm-metre (Ωm).

We recall our definition of resistance as the ability of a material to oppose the flow of current through it. The greater the resistivity of a wire, the poorer it is as an electrical conductor. Because of this, the term, conductivity is used to specify the current-carrying ability of a material. The greater the conductivity the more easily can current flow through the material. Thus materials of high conductivity also have low resistivity. Conductivity, σ, is the reciprocal of the resistivity, ρ.

$$\sigma = \frac{1}{\rho} \qquad 4.11$$

Electrical conductivity is a measure of the extent to which a material will allow current to flow easily through it when a p.d is applied at a specified temperature. It is the reciprocal of the resistivity.

From Equation 4.11, we see that the unit of conductivity is $(\Omega m)^{-1}$.

Example 4.6

The resistance of a wire of length 100 cm and diameter 0.3 mm is found to be 3.0 ohms. Calculate (i) the resistivity, (ii) the conductivity of the material of the wire.

Solution

$$\rho = \frac{RA}{l}$$

$R = 3\ \Omega$, $l = 100$ cm $= 1.0$ m, $r = \left(\frac{0.3}{2} \times 10^{-3}\right)$ m

$$\text{Area} = \pi r^2 = \pi\left(\frac{0.3}{2} \times 10^{-3}\right)^2 \text{ m}^2$$

$$= \pi\left(1.5 \times 10^{-4}\right)^2 \text{ m}^2$$

$$\rho = 3 \times \frac{22}{7} \times \frac{(1.5 \times 10^{-4})^2}{1}\ \Omega m$$

$$= 21.21 \times 10^{-8}\ \Omega m$$

$$= \underline{2.12 \times 10^{-7}\ \Omega m}$$

$$\text{Conductivity, } \sigma = \frac{1}{\rho}$$

$$= \frac{1}{2.12 \times 10^{-7}}\ (\Omega m)^{-1}$$

$$= \underline{4.7 \times 10^{6}\ (\Omega m)^{-1}}$$

4.4 Galvanometer Conversions

Conversion of a Galvanometer to an Ammeter. Shunt

An ammeter is used for measuring currents. A galvanometer is used for detecting and measuring very small currents. The features and working principle of a moving coil galvanometer will be discussed in chapter 7.

We can convert the galvanometer into an ammeter by connecting a suitable resistor in parallel with the galvanometer. A resistor used for this purpose is known as a *shunt*. The shunt is a low resistance wire and is used to divert a large part of the current being measured but to allow only a small current to pass through the galvanometer.

Example 4.7

A galvanometer of resistance 5 ohms gives a full scale deflection when a current of 50 mA flows through it. How will you convert it to an ammeter capable of measuring 2 A?

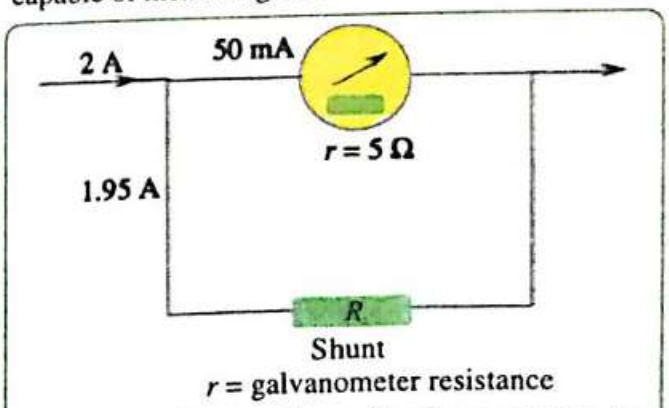


Fig. 4.18 Conversion of galvanometer to ammeter

To convert a galvanometer to an ammeter we connect a shunt of resistance $R\ \Omega$ in parallel with the galvanometer to carry off any current in excess of 50 mA or 0.05 A.

Current flowing through the shunt = (2 – 0.05) A = 1.95 A

p.d across shunt = $(1.95 \times R)$ volts

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p.d. across galvanometer resistance = 0.05 × 5 volts.

Since p.d across shunt = p.d across the galvanometer resistance, we have:

$$1.95 \times R = 0.05 \times 5$$

$$R = \frac{0.05 \times 5}{1.95}$$

$$= \underline{0.128\ \Omega}$$

Thus a small resistance of value 0.128 Ω will convert the galvanometer to an ammeter reading up to 2 A.

Conversion of a Galvanometer to a Voltmeter. The Multiplier

A galvanometer used for measuring very small currents can be converted to a voltmeter by connecting a high resistance or *multiplier* in *series* with the galvanometer.

Suppose we wish to convert a galvanometer whose full scale deflection is 10 mA to a voltmeter capable of measuring voltages up to 20 V. Let the resistance of the galvanometer be 5 ohm.

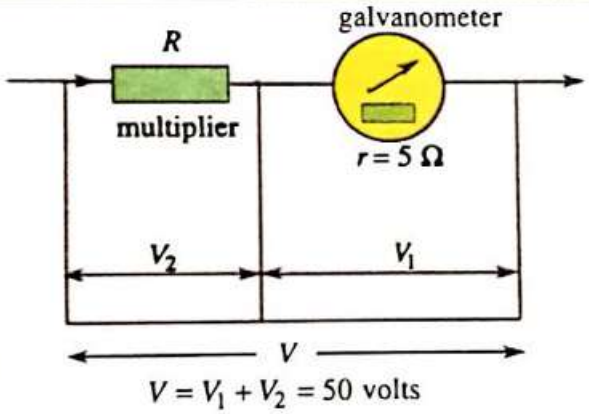


Fig. 4.19 Conversion of galvanometer to voltmeter

The maximum current that can pass through the galvanometer is 10 mA or 0.01 A.

p.d across the galvanometer at full scale deflection is given by

$$V_1 = 0.01 \times 5 = 0.05 \text{ Volt}$$

Since the total p.d across the multiplier and galvanometer at full scale deflection is 20 Volts, the p.d across the multiplier alone must be given by

$$V_2 = 20 - 0.05 = 19.95 \text{ Volts}$$

Current passing through the multiplier is 0.01 A. Hence if R is the resistance of the multiplier, we have that

$$V_2 = 0.01 \times R = 19.95 \text{ V}$$

$$\therefore \quad R = \frac{19.95}{0.01}$$

$$= \underline{1995 \text{ Ohm}}$$

Thus a high resistance is needed.

4.5 The Principle of a Potentiometer

The *Potentiometer* is a device to precisely measure potential differences. It is an apparatus commonly used to measure or to compare the e.m.f.s of cells. It can also be used to compare or measure resistances.

It consists of a metallic wire AB of uniform cross-sectional area, about 100 cm long, mounted on a board between thick brass or copper strips of negligible resistance. A metre-rule is attached below the uniform resistance wire AB as shown in fig. 4.20. An accumulator and a rheostat connected across AB maintain a constant current flowing through the wire and a steady potential difference between the ends of the wire.

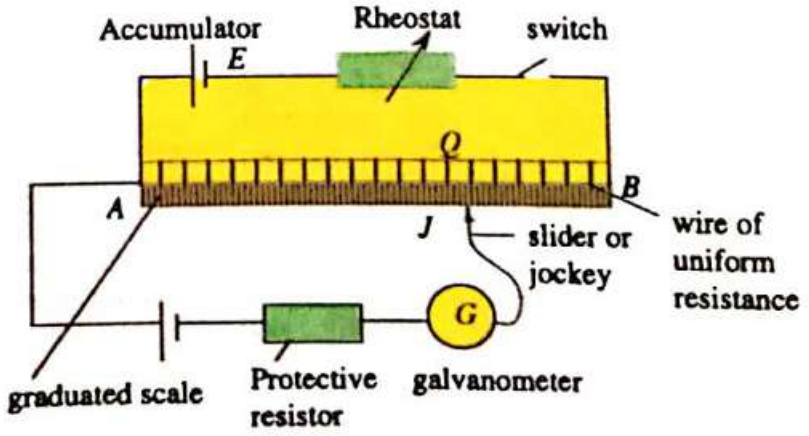


Fig. 4.20 The Potentiometer

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A centre-zero sensitive galvanometer, G, is attached to the sliding contact J and to the point A through a protective resistor as shown in the diagram. The sliding contact or Jockey is a metal with a knife edge which can slide along the wire and make contact with any point.

Working Principle

When a steady current (I) flows along the uniform wire, there is a steady potential drop throughout its length. Thus each unit length of the wire will have a constant p.d. across it. Therefore p.d. across 1 cm length of wire = IR, where R is the resistance per unit length of wire. We recall that the resistance of a uniform wire is directly proportional to its length. Therefore with constant current I along the wire, the p.d. between A and any point Q on the wire is directly proportional to the length AQ of the wire.

$\therefore \quad V_{AQ} \propto AQ$

or $V_l \propto l$, where l is the length of AQ and V_l is the p.d across the length l.

Use of the Potentiometer to Compare or Measure E.M.F.s

A cell, C_1 whose e.m.f., E_1 is required, is inserted in the galvanometer circuit of the potentiometer as shown in fig. 4.21. We join the positive pole of C_1 to the point A, the same terminal to which the positive pole of the Accumulator is joined.

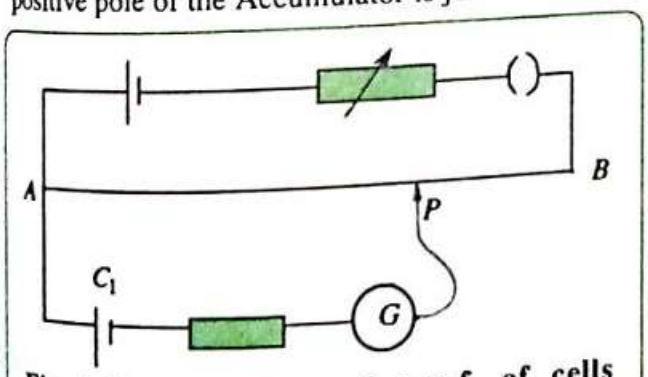


Fig. 4.21 Comparison of e.m.f. of cells

We join the negative pole of C_1 to the jockey through the sensitive centre-zero galvanometer.

By tapping gently with J on the wire we find the point of contact P when no deflection is indicated by the galvanometer. This is said to be the 'null' or 'balance' point. At this point the p.d. (V_1) across AP is just equal to the e.m.f. E_1 of C_1, i.e. $V_1 = E_1$. But p.d. between A and P is proportional to l_1, the length of AP.

$\therefore \quad E_1 \propto l_1$

We replace the cell C_1 with another cell, C_2 whose e.m.f. (E_2) is to be compared with that of C_1. The experiment is then repeated and the new balance point obtained say at N. If the length $AN = l_2$ then we can write that

$$E_2 \propto l_2$$

Hence $$\frac{E_1}{E_2} = \frac{AP}{AN} = \frac{l_1}{l_2}$$

Thus the ratio of the e.m.f.s of the two cells can be calculated from the ratio of the two lengths, l_1 and l_2.

If we want to obtain the actual e.m.f. of a cell, e.g., that of C_1 we use a *standard cell* whose e.m.f. is known as the second cell C_2. Suppose the e.m.f. of the standard cell is 1.02 V, then from the relation

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

we have that

$$\frac{E_1}{1.02} = \frac{l_1}{l_2}$$

$$\therefore \quad E_1 = \left(\frac{l_1}{l_2} \times 1.02\right) \text{ volts}$$

It is important to note that unlike the voltmeter method for measuring p.d., this potentiometer method gives the actual value of the e.m.f. of the cell as the cell is not driving any current through the circuit when the measurement is taken.

Advantage of Potentiometer over the Voltmeter Method for Measuring e.m.f.

1. The potentiometer measures e.m.f. when the cell is giving no current (i.e. in open circuit). Therefore it gives the correct measurement of e.m.f. of the cell since no correction need be made for the lost volt.
2. Apart from the fact that the voltmeter does not give the e.m.f. to a high accuracy because it takes current from the cell, it also uses a pointer which could cause reading error.
3. The potentiometer measures the e.m.f. to a higher degree of precision than the voltmeter which may have a zero error.

Example 4.8

The current in a potentiometer is adjusted with a rheostat so that the galvanometer reads zero when a

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standard cell of e.m.f. 1.02 volts connected to the slider intercepts 40.0 cm of the potentiometer wire. When a cell, C_1 of unknown e.m.f. is substituted for the standard cell, the balance point is at 49.0 cm. Calculate the e.m.f. of the cell.

Solution

$$\frac{\text{E.m.f. of standard cell}}{\text{E.m.f. of cell, } C_1} = \frac{40.0}{49.0}$$

$$\text{E.m.f. of } C_1 = \frac{49.0}{40.0} \times 1.02$$

$$= \underline{1.25 \text{ volts}}$$

The Wheatstone bridge

The wheatstone bridge circuit consists of four resistances arranged as shown in fig. 4.22. G is a sensitive galvanometer connected across the junctions B and D of the circuit. E is a dry cell of e.m.f. 1.5 volts which drives a current through the various arms of the bridge. Keys, k_1 and k_2 are connected to the arms of the cell and galvanometer.

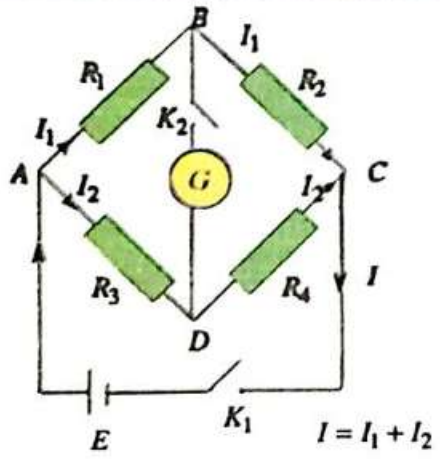


Fig. 4.22 Wheatstone bridge

Working Principle

Let us suppose that R_1 is an unknown resistance and R_2, R_3, R_4 are resistance boxes. For random values of the resistances a current will flow through the circuit when the keys are closed, and the galvanometer will show a deflection. By varying one or more of the resistances, a point is reached where the galvanometer shows no deflection, i.e. no current flows along BD when the keys are closed. At this balance point, B and D are at the same potential (i.e. there is no p.d. between B and D). At this balance point, the current (I_1) which flows through R_1 should also flow through R_2. Likewise a current I_2 flows through R_3 and R_4.

Since B and D are at the same potential,

p.d. across AB = p.d. across AD,

p.d. across BC = p.d. across CD,

$$\therefore \quad I_1R_1 = I_2R_3 \quad \text{(i)}$$

$$I_1R_2 = I_2R_4 \quad \text{(ii)}$$

Dividing (i) by (ii) we have

$$\frac{I_1R_1}{I_1R_2} = \frac{I_2R_3}{I_2R_4}$$

or

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \qquad 4.12$$

The Metre bridge

The metre bridge is the practical arrangement of the wheatstone bridge. The metre bridge thus utilizes the principle of the wheatstone bridge to measure an unknown resistance.

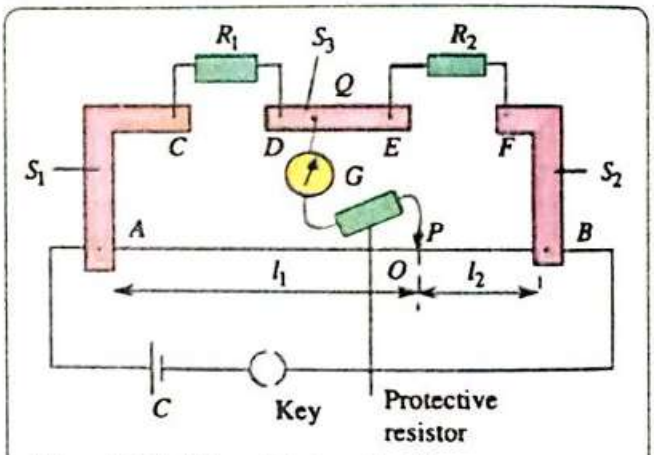


Fig. 4.23 The Meter bridge

It consists of a uniform wire AB 100 cm long, the ends of which are soldered to two copper or brass strips, S_1 and S_2. The wire is stretched over a graduated scale. A third strip, S_3 is placed parallel to the wire. The unknown resistances R_1 and a standard resistor R_2 are connected as shown across the gaps CD, EF. With the key closed the jockey attached to the centre-zero galvanometer is tapped along the wire until the galvanometer shows no deflection.

By applying the wheatstone bridge relation (equation 4.12) to the balanced metre bridge, we have

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

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where R_3 and R_4 are the resistances of the wire segment AP and PB of lengths l_1 and l_2 respectively. If K is the resistance per unit length of the wire, then we have

$$\frac{R_1}{l_1 K} = \frac{R_2}{l_2 K}; \text{ hence } \frac{R_1}{R_2} = \frac{l_1}{l_2}$$

or $$R_1 = \frac{l_1}{l_2} R_2$$

Since l_1 and l_2 can be read off from the graduated scale attached to the wire, R_1, the unknown resistance can be calculated if the standard resistance R_2 is known.

Precautions

In the use of the metre bridge for finding an unknown resistance the following precautions should be taken to get an accurate result:

1 It is necessary to ensure that all electrical connections are tight.
2 The jockey should not be scrapped along the wire but tapped lightly on the wire.
3 The readings of l_1, l_2 should be repeated and the average of each found.
4 Ensure that the switch is always open when readings are not being taken.
5 The value of the known resistance R_2 should be chosen so that the balance point falls somewhere between the 35 cm and 70 cm mark.
6 R_1 and R_2 should be interchanged to obtain a second pair of the values of l_1 and l_2.
7 The battery key should always be closed before the jockey is tapped on the wire.

Example 4.9

The metre bridge (fig. 4.24) consists of an unknown resistance X and a 20-ohm resistor in the two arms. The balance point is 40 cm from P as shown in the figure. What is the value of X?

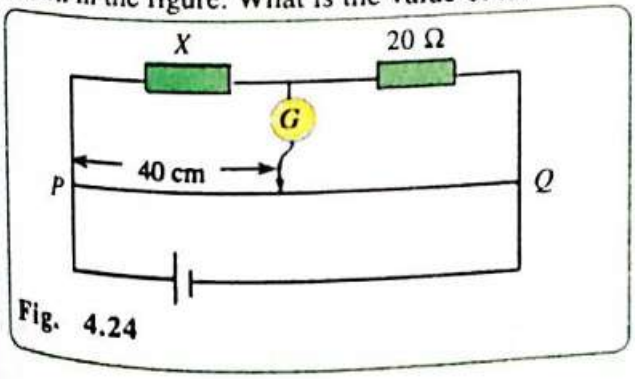


Fig. 4.24

A 60 ohms B 40 ohms C 30 ohms
D 13.33 ohms E 8 ohms
(SC/GCE)



Solution

Let the resistance per unit length of wire be k Ωcm^{-1}. Using wheatstone bridge principle we have

$$\frac{X}{40k} = \frac{20}{(100-40)k}$$

$$\frac{X}{40k} = \frac{20}{60k}$$

$$\frac{X}{40} = \frac{20}{60}$$

$$X = \frac{40 \times 20}{60}\ \Omega$$

$$= \underline{13.33\ \Omega}$$

4.6 Measurement of Resistance by Ammeter-voltmeter Method

The resistance of a wire can also be measured using ohm's law, as we now show with the following activity.

Activity 4.1

Set up the apparatus as shown in fig. 4.25.

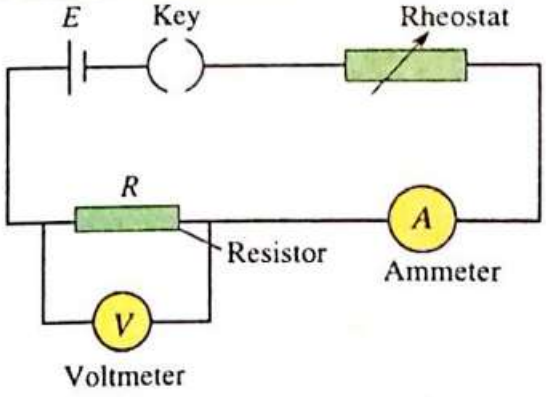


Fig. 4.25 Measurement of resistance by Ohm's law

R is a resistor whose resistance is required. V is a voltmeter connected in parallel with the resistor to measure the p.d. across the resistor. A is an ammeter connected in series with the resistor, and E is a cell or battery.

With the key plugged in, adjust the rheostat until an appreciable current is shown by the ammeter. Record the readings of the Ammeter (I)

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and voltmeter (V). Repeat the procedure five more times by adjusting the rheostat to obtain different readings of the ammeter and the voltmeter. Record your readings as shown in table 4.1.

Table 4.1

V (volt)	I (A)

Plot a graph of V against I and obtain the slope of the graph.

From Ohm's law, $\frac{V}{I} = R$.

Hence the slope of graph of V against I gives the resistance of the wire.

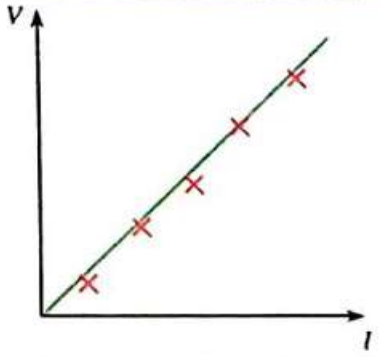


Fig. 4.26 Graph of V against I, Ohm's law

In this method of measuring the resistance of a wire, some current is always diverted through the voltmeter to make it work. So the p.d. across R is reduced. This therefore makes the value of R less accurate than that obtained through the metre bridge method which uses a 'null method' whereby no current flows in the galvanometer when measurements are made.

Summary

1 A continuous flow of charge can be provided by primary cells and secondary cells or accumulators.

2 The chemical actions that take place in primary cells are irreversible, but those of accumulators can be reversed by driving a current through them in a direction opposite to that of the current they supply. Accumulators can thus be recharged.

3 See table below.

4 For resistances in series

$$R = R_1 + R_2 + R_3 + \cdots + R_n$$

For resistances in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

5 The internal resistance of a cell is the opposition to the flow of current offered by the chemicals between the poles of the cell.

6. a. For cells in series, total e.m.f

$$E = E_1 + E_2 + E_3 + \cdots + E_n$$

total internal resistance

$$r = r_1 + r_2 + r_3 + \cdots + r_n.$$

b. For cells in parallel with similar cells, total e.m.f = e.m.f. of one cell, but

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \cdots + \frac{1}{r_n}$$

7 Resistivity $\rho = \frac{RA}{l}$ ohm-meter

Conductivity, $\sigma = \frac{1}{\text{Resistivity}}$.

8 To convert galvanometer to an ammeter connect a suitable (small) resistance (shunt) in parallel with the galvanometer; to convert a galvanometer to a voltmeter, connect a

3

Cell	+ve pole	−ve pole	electrolyte	depolarizer
Leclanché (wet)	Carbon rod	zinc rod	Ammonium chloride solution	manganese dioxide
Leclanché (dry)	Carbon rod	zinc container	Ammonium chloride	manganese dioxide + powdered carbon
Daniel	copper container	zinc rod	dilute sulphuric acid	copper sulphate solution
Accumulator	Lead peroxide	Lead	dilute sulphuric acid	

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large (large) resistance (multiplier) in series with the galvanometer.

9 The principle on which the potentiometer works is that the p.d. between any two points on a wire of uniform cross-sectional area is directly proportional to the length of the wire between the two points.

The potentiometer is used to compare or measure e.m.f.s and resistances.

10 Resistances can be measured by either the ammeter-voltmeter method or by the metre bridge principle. Metre bridge method is more accurate because it uses the 'null method' when no current flows in the galvanometer.

Activity 4.2 To find the internal resistance of a cell

Set up the circuit as shown in fig. 4.27.

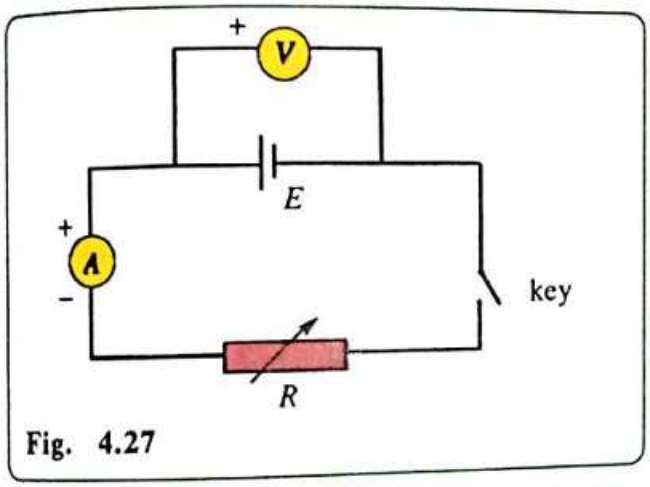


Fig. 4.27

E is a dry battery of e.m.f. of about 1.5 volt. A is an ammeter that can read 0 – 1A; V is a voltmeter that can read 0 – 2 V. R is a rheostat, and K is a key. V is connected in parallel with the battery and A is connected in series with the battery and the rheostat.

Find the e.m.f. of the battery by recording the reading of the voltmeter when the key is open.

With key closed adjust the rheostat until the ammeter shows an appreciable value. Record the readings of I and V of the ammeter and the voltmeter.

Adjust the rheostat so that more current flows in the circuit. Record the new readings of ammeter and voltmeter. Repeat the experiment four more times and tabulate your readings of I and V. Plot a graph of V against I and find the slope of the graph. What precautions will you take to obtain an accurate result?

From the relation $V = E - Ir$, a graph of V against I should be a straight line graph with a negative slope equal to r. When no current flows in the circuit (i.e. when in open circuit) $I = 0$ and $V = E$. Therefore the intercept on the voltage axis gives the e.m.f. of the battery.

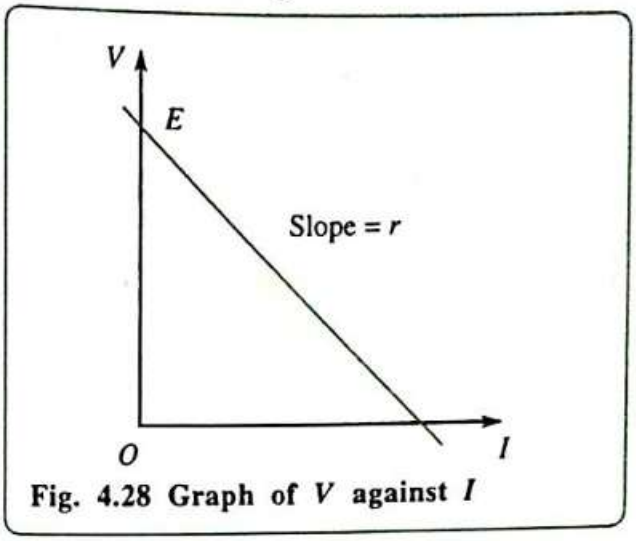


Fig. 4.28 Graph of V against I

Exercise 4

1 Calculate the value of R when G shows no deflection in the circuit illustrated below.

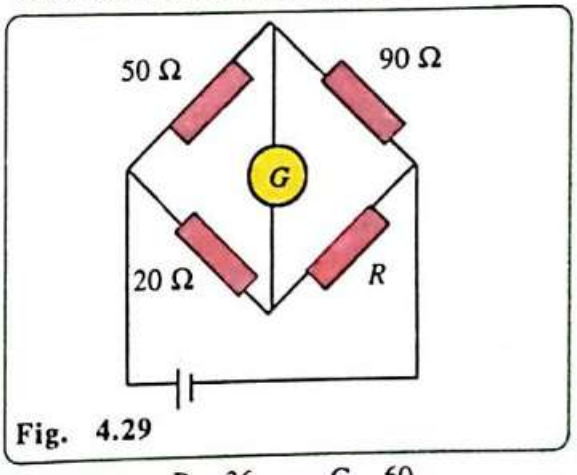


Fig. 4.29

A 11 B 36 C 60
D 120 E 225

(SSCE)

2

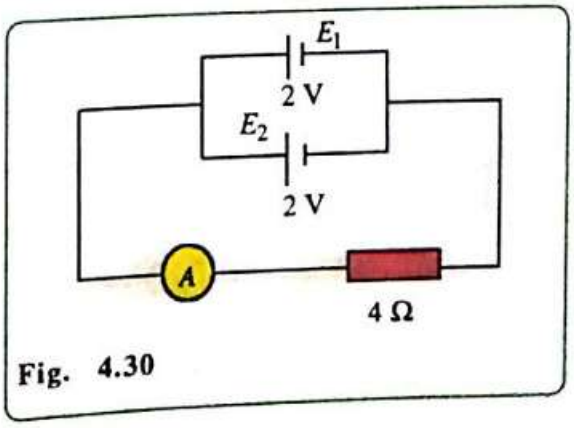


Fig. 4.30

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The internal resistance of each of the cells E_1 and E_2 shown in the figure above is 2 Ω. Calculate the total current in the circuit.

A 0.80 A B 0.50 A
C 0.40 A D 0.004 A

(JAMB)

3 A cell of e.m.f. 1.5 V and internal resistance 2.5 ohm is connected in series with an ammeter of resistance 0.5 ohm and a resistor of resistance 7.0 ohm. Calculate the current in the circuit.

A 0.15 A B 0.20 A C 0.60 A
D 3.00 A E 6.67 A

(SSCE)

4 A cell of internal resistance 2 ohms supplies current to a 6-ohm resistor. The efficiency of the cell is

A 12.0% B 25.0%
C 33.3% D 75.0%

(JAMB)

5 When a resistance R is across a cell, the voltage across the terminals of the cell is reduced to two-thirds of its nominal value. The internal resistance of the cell is

A $\frac{1}{3}$R B $\frac{1}{2}$R C $\frac{2}{3}$R D R

(JAMB)

6 Name the two advantages which a lead-acid accumulator has over a primary cell.

7 Draw and label (a) a Leclanché cell (b) Lead-acid accumulator.

8 Draw a circuit diagram for the recharging of an accumulator.

9 Name the poles and the electrolytes for the following: (i) Dry Leclanche cell (ii) lead-acid accumulator, (iii) NIFE accumulator.

10 What is the value of the e.m.f. of (i) Dry Leclanche cell (ii) a lead-acid accumulator (iii) a simple cell.

11 Explain the terms *Electromotive force*, *Terminal p.d.*

12 a. Explain the terms e.m.f. and internal resistance of a cell.
b. Describe an experiment to illustrate Ohm's law.
c. A voltmeter is connected in parallel with a variable resistance R which is in series with an ammeter and a cell. The cell is of e.m.f. E and internal resistance r. For one value R_1 of the variable resistance, the ammeter reads 0.3 A and the voltmeter reads 0.9 V. For another value R_2 of the variable resistance, the ammeter reads 0.25 A and the voltmeter reads 1.0 V. Neglecting the resistances of the meters, find the values of R_1, R_2, E and r.

(SC/GCE)

13

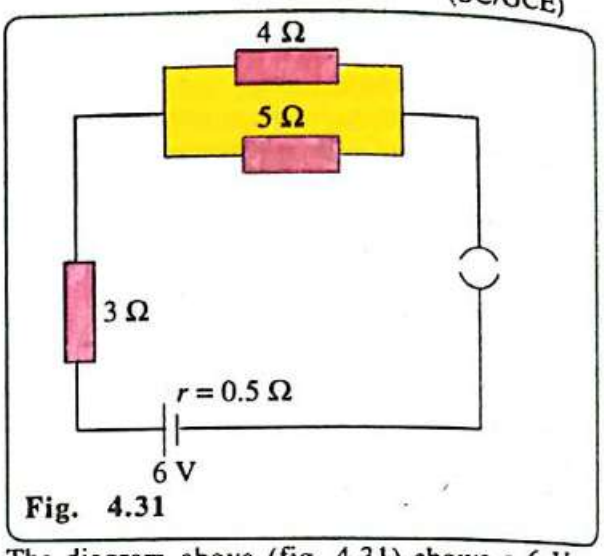


Fig. 4.31

The diagram above (fig. 4.31) shows a 6 V battery of internal resistance 0.5 ohm connected to three resistors of values 3 ohm, 4 ohm, and 5 ohm as shown. Calculate the current in each resistor, the p.d. across the 3 ohm resistor, and across the parallel resistors.

14 Calculate the length of a constantin wire of diameter 0.6 mm and resistivity 1.1×10^{-6} Ωm required to construct a standard resistor of resistance 35 Ω.

(SC/GCE)

15 A battery of three cells in series, each of e.m.f. 2 V and internal resistance 0.5 Ω is connected to a 2 Ω resistor in series with a parallel combination of two 3 Ω resistors. Draw the circuit diagram and calculate:
a. the effective external resistance
b. the current in the circuit
c. the lost volts in the battery
d. the current in one of the 3 Ω resistors.

(SC/GCE)

16 Explain why the e.m.f. of a cell is greater than the p.d. across the cell when it is passing a current through an external resistance. A cell of e.m.f. E and internal resistance r was connected in series with two external resistors A and B, of 4 Ω and 2.5 Ω respectively. When a high resistance voltmeter was connected across A, the reading was found to be 2 V. When another resistor of 4 Ω was connected parallel to A and the voltmeter connected across A and B, the voltmeter reads 3 V. Draw circuit diagrams of

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the two arrangements. Calculate the e.m.f. and the internal resistance of the cell.

(SC/GCE)

17 a. With the aid of a circuit diagram, describe how a simple slider wire potentiometer is used to measure the e.m.f. of a cell. State any two precautions that are necessary to obtain accurate results in this experiment. State one advantage of the potentiometer over ordinary voltmeter.

b. A potentiometer wire of length 1.0 m and resistance 2.0 Ω is connected in series with a variable resistance and a 2.0 V accumulator of negligible resistance. What value must be given to the variable resistance so that the fall of electric potential along the potentiometer wire may be one millivolt per centimeter?

(SC/GCE)

18 A galvanometer of internal resistance 100 Ω gives a full scale deflection for a current of 10 mA. Calculate the values of the resistance necessary to convert the galvanometer to:

(i) a voltmeter reading up to 5 V

(ii) an ammeter reading up to 10 A

(SC/GCE)

19 A galvanometer of resistance 50 Ω which gives a full-scale deflection for 1 mA, is to be adapted to measure currents of up to 5 A.

(i) Calculate the resistance of the resistor required.

(ii) If this resistor is made of a material of cross-sectional area 4×10^{-4} cm^2 and resistivity 2×10^{-6} Ωcm, calculate its length.

(SC/GCE)

20 a. Define electric resistance and name four factors which can determine the resistance of a metal wire.

b. Derive an expression for the resistance R of two resistors of resistances R_1 and R_2 connected

(i) in series

(ii) in parallel

c. A potentiometer circuit consists of a battery of e.m.f. 5 V and internal resistance 1.0 Ω connected in series with a 3.0 Ω resistor and a potentiometer wire AB of length 1.0 m and resistance 2.0 Ω. Calculate:

(i) The total resistance of the circuit

(ii) The current flowing in the circuit

(iii) The lost volt from the internal resistance of battery across the battery terminals

(iv) The p.d. across the wire AB

(v) The e.m.f. of a dry cell which can be balanced across 60 cm of the wire AB.

Assume the wire has a uniform cross-sectional area.

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5 ELECTRIC FIELD 3: CONDUCTION THROUGH LIQUIDS AND GASES

This third section on electric field will deal with the effect of passing electric current through liquids and gases.

At the end of this chapter the student should be able to:

1 Explain the conditions under which liquids and gases conduct electricity.
2 Explain the behaviour of charges or charge carriers in liquids and gases in the electric field.
3 Describe an application each of electrical conduction through liquids and gases.

5.1 Conduction Through Liquids

Electrolysis

Electrolysis is the chemical change in a liquid due to the flow of electric current through the liquid.

Michael Faraday studied this process extensively and laid the foundation for the theory of electrolysis. Some liquids are good conductors while others are poor conductors of electricity. Good conductors are known as *electrolytes*, poor conductors are *non-electrolytes*.

Liquids such as solutions of acids, bases and salts are generally good conductors. Liquids such as benzene and paraffin or kerosene are poor conductors. Organic compounds are generally poor conductors. While pure water is also a poor conductor, water containing some dissolved salts conducts moderately.

A *Voltameter* is a device for studying the flow of current through a liquid. It consists of the following parts:

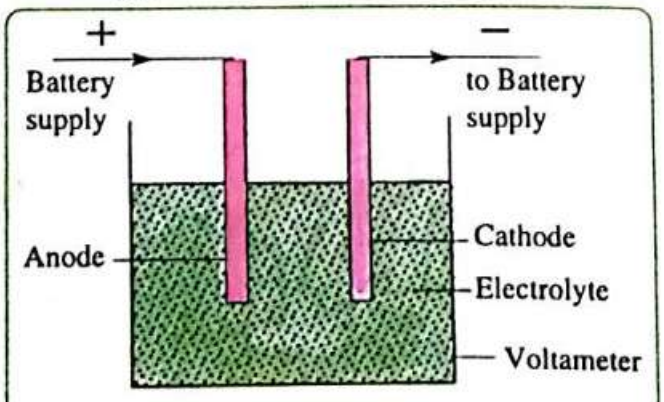


Fig. 5.1 A Voltameter

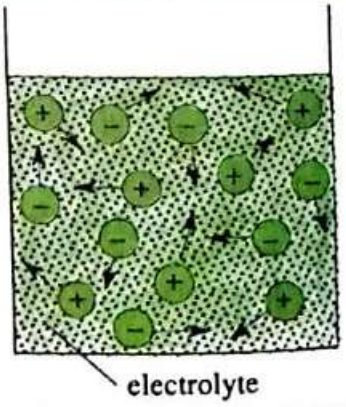
1 *The Electrolyte* is the liquid or molten substance which conducts a current and is decomposed by it.
2 Electrodes are materials in the form of a rod or plate through which current enters or leaves the electrolyte. There are two electrodes: (a) the positive electrodes through which current enters the electrolyte is called the *Anode*. (b) The negative electrode through which liquid leaves the electrolyte is called the *Cathode*.

5.2 Dynamics of charged particles (ions) in electrolytes

The Ionic theory

In an electrolyte there are positively and negatively charged particles called *ions*. The molecules that constitute the electrolyte are split in solution into these ions through the process known as *electrolytic dissociation*. The dissociation of an electrolyte occurs irrespective of whether or not an electric field is applied to the electrolyte.

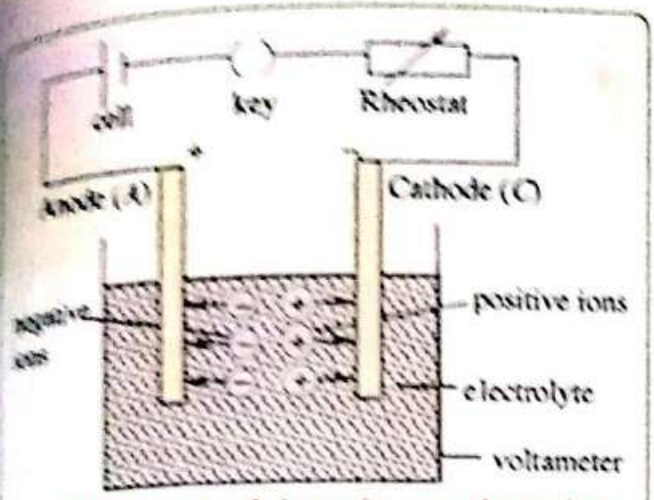
Ions in an electrolyte execute random movements until a battery is connected to the electrodes of a voltameter. As soon as a p.d. is set up across the electrodes the positive ions drift to the cathode, which is at a negative potential while the negative ions drift to the anode which is at a positive potential.



a. Random movement of ions in an electrolyte

Fig. 5.2

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b. Movement of ions in an electrolyte under a p.d.

Fig. 5.2 Dynamics of ions in an electrolyte

This directional movement of ions is the electric current flowing through the electrolyte. Such movement ceases as soon as the battery is disconnected, and the ions move randomly once again.

Thus electrolytic solutions are able to conduct electricity because the electrolytes in solution can dissociate into ions. Non electrolytes in solution do not easily dissociate into ions.

5.3 Examples of Electrolysis

Electrolysis of acidulated water

Use is made of the Hoffman voltameter. It consists of two identical graduated tubes and a funnel tube through which the electrolyte is introduced.

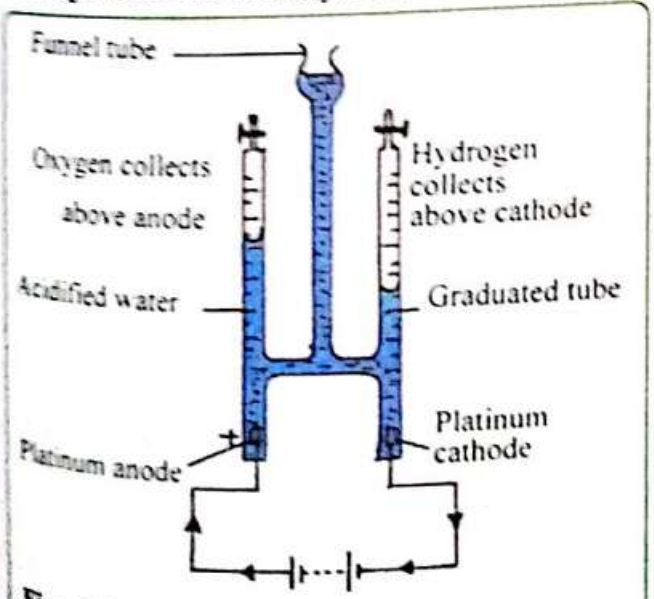


Fig. 5.3 Electrolysis of water

The electrolyte is water to which a few drops of sulphuric acid have been added to make the water more conducting. The electrodes consist of a pair of platinum plates at the bottom of the tubes.

With the taps closed, current is passed through the electrolyte. A steady stream of bubbles soon forms around each electrode. The gas formed at each electrode collects in the upper parts of the tubes above the electrode. It is observed that oxygen forms in the tube above the Anode and hydrogen above the Cathode. It is also observed that the ratio of the volume of oxygen to the hydrogen formed is 1:2.

In this process, water dissociates into hydrogen ions(H^+) and hydroxyl ions (OH^-)

$$H_2O \longrightarrow H^+ + OH^-$$

The sulphuric acid dissociates into Hydrogen ions and sulphate ions

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

When current is passed through the platinum electrodes in acidulated water, H^+ ions drift to the cathode and are discharged there:

$$2H^+ + 2e^- \longrightarrow H_2$$

OH^- ions drift to the anode and are discharged more easily than SO_4^{2-} ions, resulting in the formation of oxygen and water

$$4OH^- \longrightarrow O_2 + 2H_2O + 4e^-$$

Electrolysis of Copper Sulphate Solution with Copper electrodes

The set up of the apparatus is as shown in fig. 5.4.

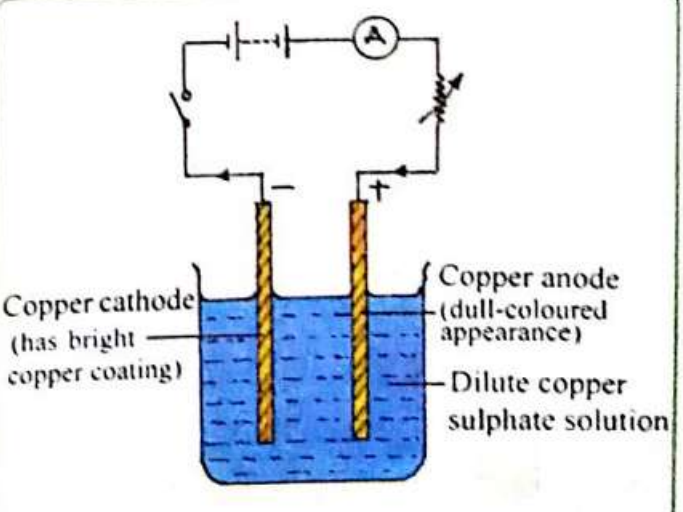


Fig. 5.4 Electrolysis of copper sulphate solution

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Two copper electrodes are immersed in a solution of copper sulphate. Current is passed into the solution through the electrodes. After about 30 minutes, it will be observed that a fresh bright coating of copper has formed over the immersed end of the cathode, while the lower end of the anode shows a dull coloured appearance. If the electrodes are weighed before and after the current is passed, it will also be observed that the cathode has gained some mass while the anode has lost the same mass. Also the concentration of the copper sulphate solution remains the same no matter how long the current flowed, and is therefore not affected by the passage of electricity through it.

The copper sulphate in solution dissociates into copper ions and sulphate ions. The water breaks down into hydrogen ions and hydroxyl ions.

$$CuSO_4 \qquad Cu^{2+} + SO_4^{2-}$$

$$H_2O \longrightarrow H^+ + OH^-$$

When electricity is passed, both the copper and Hydrogen ions drift to the cathode where the copper is discharged in preference to the hydrogen. Hence copper is deposited at the cathode.

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

The SO_4^{2-} and OH^- drift to the anode but here the discharge of copper ions into solution takes precedence over the discharge of SO_4^{2-} and OH^- Hence we have

$$Cu \longrightarrow Cu^{2+} + 2e^-$$

The overall result of the processes described is the flow of copper from the anode to the cathode and a flow of electrons round the external circuit. There is no net gain or loss of copper from the solution. Hence the density of the copper sulphate solution is unchanged throughout the electrolysis.

Electrolysis of Copper Sulphate solution using platinum electrodes

What is observed in this process is that the platinum cathode is covered with a bright-red layer of pure copper. The copper sulphate solution loses its blue colour.

During electrolysis, Cu^{2+} and H^+ ions drift to the cathode where the copper is discharged in preference to hydrogen. At the anode, (OH^-) ions are discharged and combine in pair to give water and oxygen

$$4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$$

The net result is the deposit of copper on the cathode, the production of oxygen around the anode and the reduction in the concentration of the copper sulphate solution.

5.4 Application of electrolysis

Electrolysis finds several applications in industry, for example, in the electroplating of metals, the purification of impure metals and the electrolytic production of metals from their compounds.

1 Electroplating

This is the process of coating a metallic object with another metal in order to improve its appearance or to safeguard it against corrosion. Copper, silver, chromium, nickel, or gold can be used to electroplate other metallic objects. For example, bumpers of cars or the handlebars of bicycles are often chromium-plated to give them a bright appearance and protect them from rusting. Cutlery such as spoons are often coated with silver.

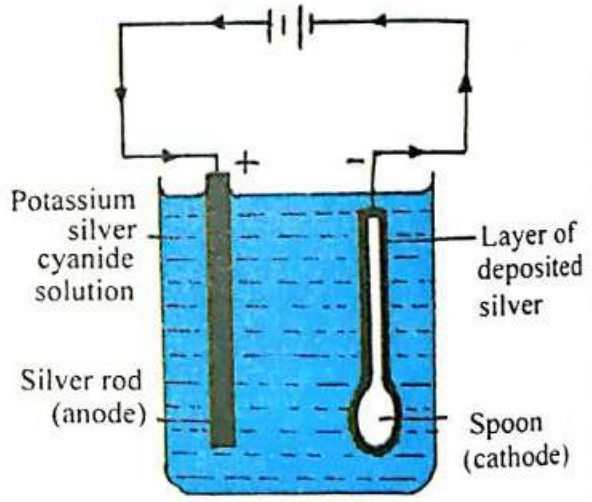


Fig. 5.5 Electroplating

In general the material to be electroplated is used as the cathode of a voltameter which has an anode made of the electroplating material. The electrolyte used is a solution of the salt of the metal to be deposited. For example the silver-plating of spoons is carried out by making them the cathode of a voltameter containing a silver nitrate solution and passing current through the solution. The anode is made of pure silver. The silver nitrate in solution dissociates into Ag^+ and NO_3^- ions. During the passage of electricity, Ag^+ ions drift to the cathode where they are deposited on the spoon cathode. The spoon therefore gets covered

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with the silver. The NO_3^- remains in solution, combining with silver from the anode to form more silver nitrate, thus maintaining the concentration of the silver nitrate solution at the original level.

Calibration of an Ammeter

The apparatus is set up as shown in fig 5.6. The ammeter to be calibrated is used in the set up.

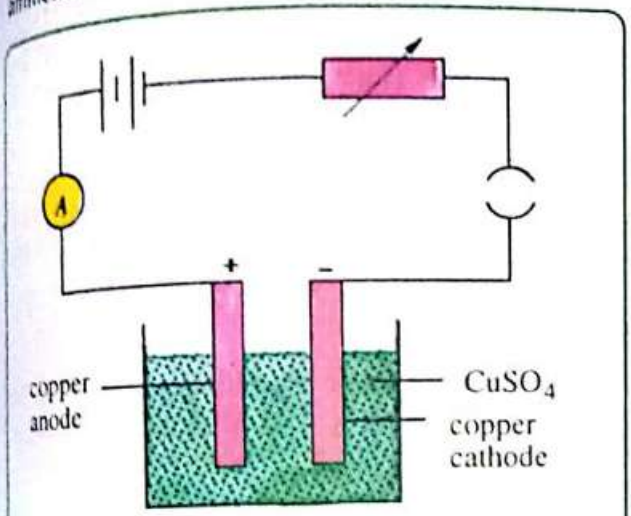


Fig. 5.6 Calibration of an ammeter

The copper cathode is weighed before the current is switched on. By the adjustment of the rheostat a suitable current of about 1.5 Amp is allowed to flow through the copper Voltameter for about 30 minutes. The current is maintained at a steady value throughout the 30 minutes of the experiment by adjusting the rheostat from time to time.

At the end of the 30 minutes, the cathode is washed, dried and reweighed to obtain the mass of copper deposited.

Let the mass of copper be M and the electrochemical equivalent of copper (e.c.e.) be Z.

Then from $M = ZIt$, we have that the true current is given by

$$I = \left(\frac{M}{Zt}\right) \text{ Amp.}$$

Since M, Z and t are known, I can be calculated. This value of I is then compared to the value recorded by the ammeter. The difference gives the correction to be applied to any reading as recorded by the ammeter.

5.5 Faraday's Laws of Electrolysis

From the result of his experiments, Faraday discovered two laws of electrolysis.

Faraday's 1st law states that the mass, M, of a substance liberated in electrolysis is directly proportional to the quantity of electricity Q, which has passed through the electrolyte.

Faraday's 2nd law states that the masses of different substance deposited or liberated by the same quantity of electricity are directly proportional to the chemical equivalents of the substances.

In symbols, the 1st law states that

$$M \propto Q$$

where Q is the quantity of electricity in coulombs, but since $Q = It$, we can say that

$$M \propto It$$

where I is the current (in amperes) and t is the time (in seconds) during which the current flows.
Hence

$$M = ZIt \qquad 5.1$$

Z is a constant of proportionality known as the electrochemical equivalent (e.c.e.) of the substance. Hence when a current of 1 amp flows through the electrolyte in one second, the mass of the substance deposited is equal to the electrochemical equivalent.

The **electrochemical equivalent** (e.c.e.) of a substance is the mass of that substance deposited by one Coulomb of electricity (or by 1 amp in 1 sec.) in a process of electrolysis. (Unit = g/Coulomb).

Experimental Verification of Faraday's 1st Law of Electrolysis

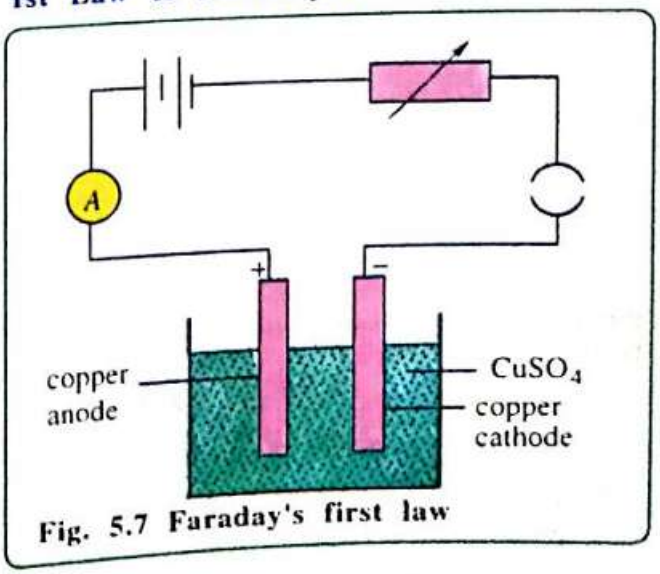


Fig. 5.7 Faraday's first law

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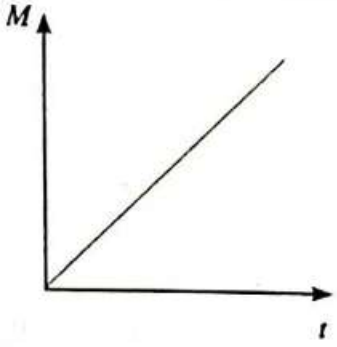
The apparatus is set up as shown in Fig. 5.7. The copper cathode is washed, dried and weighed, and the weight is recorded. By adjusting the rheostat, 1 amp of current is made to flow for about 20 minutes, after which the cathode is rinsed with distilled water, dried and reweighed to obtain the weight of copper deposited during the period. The experiment is repeated about four more times, allowing 20 minutes for each process. The mass deposited after each 20 minutes is measured and recorded as shown in Table 5.1a.

Table 5.1a

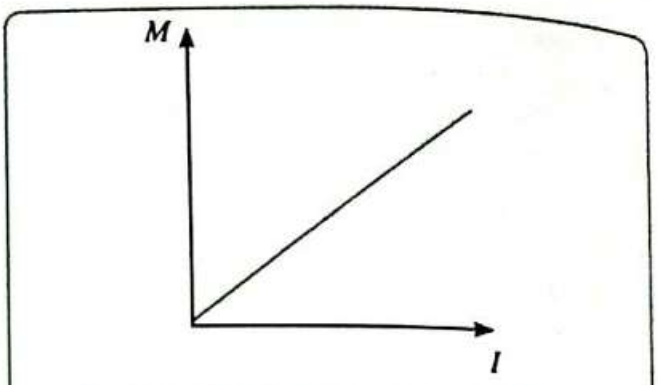
Time /min	Current A	Mass of cathode	Mass of copper deposited
0	0	M_0	0
20	1.0	M_1	$M_1 - M_0$
40	1.0	M_2	$M_2 - M_0$
60	1.0	M_3	$M_3 - M_0$
80	1.0	M_4	$M_4 - M_0$
100	1.0	M_5	$M_5 - M_0$

A graph of total mass deposited against total time the current flowed is plotted and a straight line graph passing through the origin is obtained. This shows that $M \propto t$.

In a second set of experiment, different values of current (e.g. 0.5, 1, 1.5, 2.0, 2.5 Amps) are each passed for about 20 minutes through the electrolyte, and the mass of copper deposited for such current is obtained. A graph of mass against current is plotted and a straight line graph passing through the origin is again obtained, thus showing that M is proportional to I.



a. Graph of mass versus time



b. Graph of mass versus current

Fig. 5.8

From the two sets of experiments we showed that M is proportional to I (constant t) and proportional to t (constant I); i.e. $M \propto It$.

Example 5.1

Find the mass of copper deposited on the cathode of a copper voltameter if a current of 0.53 A is passed through it for 30 minutes (e.c.e. of copper = 3.3×10^{-4} gC^{-1}).

Solution

By Faraday's 1st law of electrolysis we have that

$$M = ZIt$$
$$= 3.3\times10^{-4}\times0.53\times(30\times60)$$
$$= [3.3\times10^{-4}\ (gC^{-1})\times0.53(A)\times(30\times60)\ \text{sec}]g$$
$$= \underline{0.315\ g}$$

Example 5.2

Calculate the time in minutes required to electroplate an article of area 300 cm^2 with a layer of copper 0.06 cm thick if a constant current of 2 A is maintained. Assume that the density of copper is 8.8 gcm^{-3} and that one coulomb liberates 0.00033 g of copper.

Solution

Total vol. of copper deposited
= area × thickness
= $(300 \times 0.06)\ cm^3 = 18\ cm^3$
Mass of copper = vol. × density
$= 18 \times 8.8 = 158.4$ g.
From $M = ZIt$

$$t = \frac{M}{ZI} = \frac{158.4}{0.00033\times2}\ \text{secs}$$
$$= 2.4\times10^5\ \text{secs}$$
$$= \underline{4\times10^3\ \text{minutes}}$$

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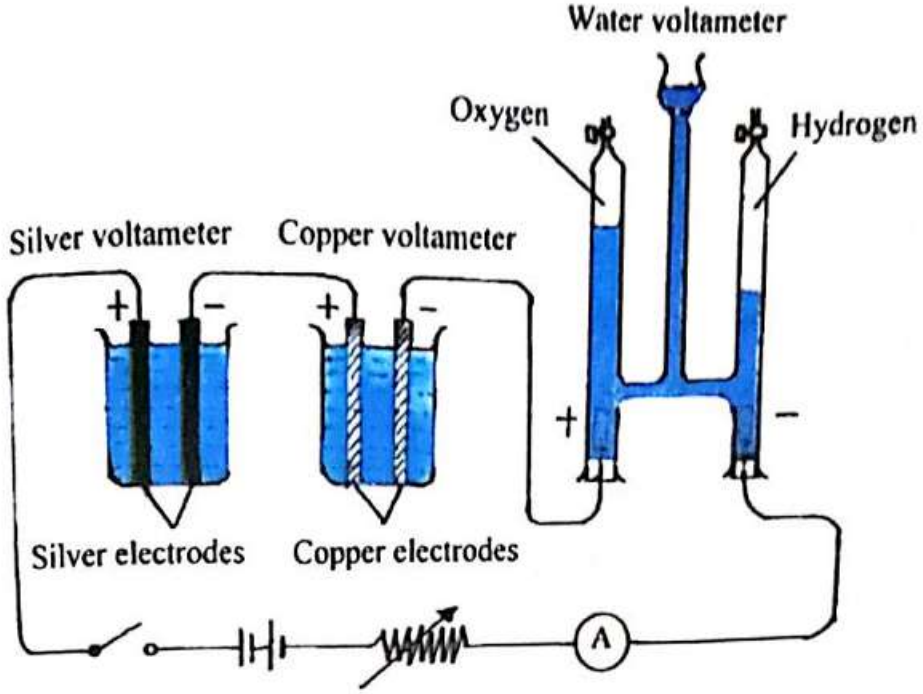


Fig. 5.9 Faraday's second law of electrolysis

Experimental verification of Faraday's 2nd Law

A copper voltameter, a silver voltameter and a water voltameter are connected in series as shown in fig. 5.9. The copper and silver cathodes are weighed and the same quantity of electricity is passed through the three voltameters for about 30 minutes. The copper and silver cathodes are then rinsed with distilled water, dried and reweighed to obtain the masses of copper and silver deposited. The volumes of oxygen and hydrogen are read and converted into masses using their densities.

If the masses are compared they will be found to be in the following ratio

Copper : Silver : Hydrogen : Oxygen = 32 : 108 : 1 : 8. This is the same as the ratio of their chemical equivalents.

Example 5.3

A copper and a silver voltameter are connected in series, and at the end of a period of time, 5.0 g of copper was deposited, calculate the mass of silver deposited at the same time. Chemical equivalent of copper = 31.5. Chemical equivalent of silver = 108.

Solution

$$\frac{\text{Mass of copper deposited}}{\text{Mass of silver deposited}} = \frac{\text{chemical equivalent of copper}}{\text{chemical equivalent of silver}}$$

$$\frac{5.0}{\text{mass of silver}} = \frac{31.5}{108}$$

$$\therefore \quad \text{mass of silver} = \frac{108 \times 5}{31.5}$$

$$= \underline{17.14\ \text{g}}$$

5.6 Conduction of Electricity Through Gases

Discharge through Gases

The conduction of electricity through gases is studied using a *discharge tube*. This consists of a long glass tube with metal electrodes sealed to each end of the tube (fig. 5.10).

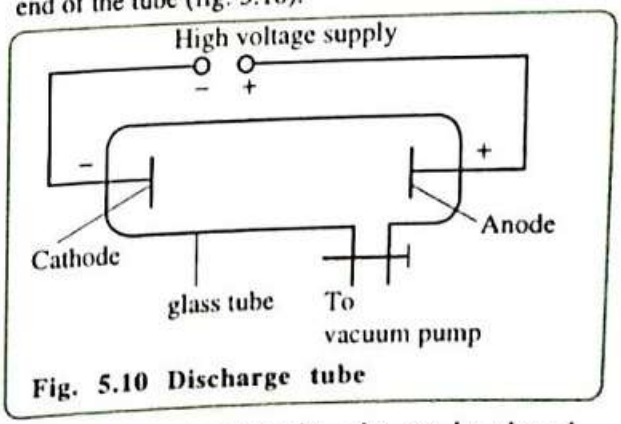


Fig. 5.10 Discharge tube

The pressure of gas in the tube can be altered through the vacuum pump connected to a side-arm of the tube.

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Conditions under which gases conduct electricity

Experiments with the discharge tube show that gases conduct electricity under low pressure and high potential difference (> 100 V). Under these conditions, a gas in a discharge tube will glow because its atoms are given energy by the flow of electricity through the tube.

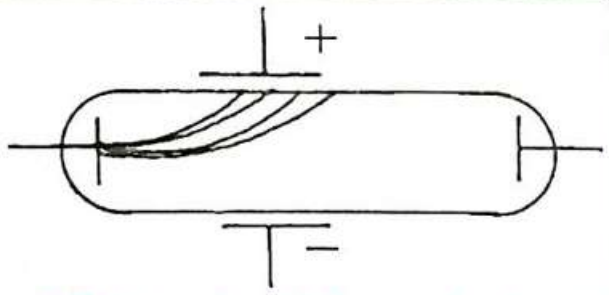
At very low pressure (~0.01 mm.Hg) and high voltage (1000 V) the gas in the tube breaks into ions. The positive ions move towards the cathode, the negative ions and free electrons move towards the anode. The positive ions knock off electrons from the metal plate of the cathode. These electrons produced at the cathode are called *cathode rays*.

Nature and Properties of Cathode Rays

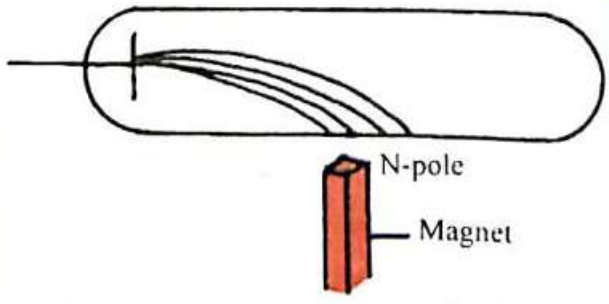
1 They consist of streams of fast moving particles of negative electricity called electrons. Because they are negatively charged, they are attracted towards the anode.

2 They cause glass and other materials (e.g. zinc sulphide) to fluoresce or glow with a greenish colour.

3 They travel in straight lines.



a. Deflection of cathode rays by electric field



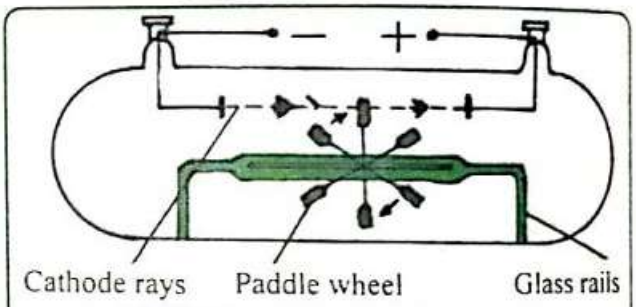
b. Deflection of cathode rays by magnetic field

Fig. 5.11

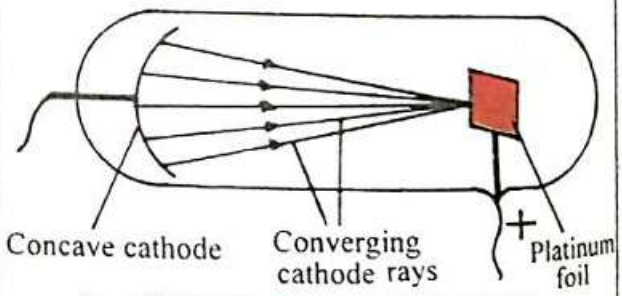
4 They are deflected by electric and magnetic fields. If passed between positive and negative electrostatic plates, they are deflected towards the positive plate showing they are negatively charged. This is confirmed by placing the N-pole of a powerful magnet near the tube as shown in fig 5.11b. The rays are deflected downward as shown. If the polarity of the magnet is reversed, the path of the cathode rays is bent up. The bending of the cathode rays in the presence of a magnetic field shows that they are charged, and the direction of bending shows they are negatively charged.

5 They can ionize a gas.

6 They will turn a light paddle wheel in the tube showing that they have momentum and therefore mass and energy. This will suggest that they are streams of particles rather than electromagnetic waves.



a. Cathode rays possess energy



b. Cathode rays produce heat

Fig. 5.12

7 They can produce intense heat on objects which stop them, confirming that they are highly energetic particles.

8 They can affect photographic plates.

9 They can produce X-rays from high density metals when they are suddenly stopped by such metals.

10 They are highly penetrating and can penetrate through metals such as aluminium plate, steel plate, and gold foil.

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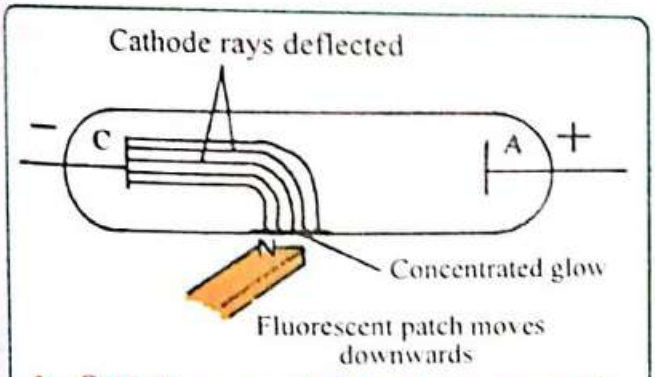
From the nature and properties of cathode rays stated above, we conclude that they are highly energetic electrons.

Activity 5.1

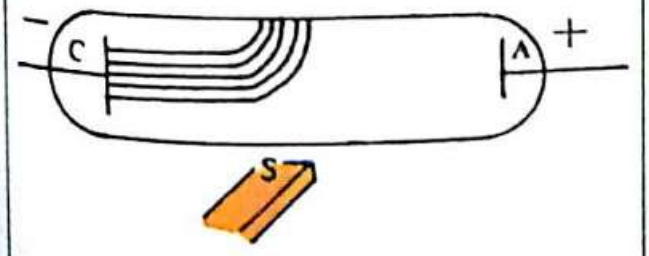
To show that cathode rays are deflected by a magnetic field

Connect the cathode and anode of a discharge tube to the secondary terminals of an induction coil using copper wires. Reduce the pressure in the tube by pumping out air until the pressure is so low that streams of cathode rays move towards the anode. The production of cathode rays will be noticed by the glow or fluorescence that appears in the tube.

Place the N–pole of a strong bar magnet above or below the tube and at right angle to the tube and note the effect on the electron-stream. Move the N-pole up and down and observe the movement of the glow. Repeat with the S–pole and note the effect.



a. Cathode rays deflected by magnetic field



b.

Fig. 5.13

Applications of gas discharge: Fluorescent lamps and Neon Signs

One application of cathode rays is in fluorescent tubes used commercially for lighting and display signs. The tubes contain mercury vapour or sodium vapour, which at low pressure fluoresce or glow at the passage of cathode rays. The cathode rays (or electrons) produced in the tube cause the emission of ultra-violet rays when they bombard the vapourised atoms of mercury. As the interior of the fluorescent lamps are coated with phosphor, this gives off visible light when the ultra-violet rays strike it. Fluorescent tubes are also used as display signs. The most used gas is the neon gas which gives bright orange-red colour.

Fluorescent lamps are more efficient than filament lamps.

Neon signs are produced when an evacuated discharge tube is filled with neon gas, and put under very low pressure of about 5mm of Hg. The gas discharged between the electrodes has a characteristic colour. The colour produced by discharge tubes depends on the nature of the gas in the tube and the nature of the coating inside the tube.

5.7 The Hot Cathode, Thermionic Emission

In gas discharge tubes, cathode rays (electrons) are produced from a cold cathode. Another way of producing free electrons is from a hot cathode.

Whenever a metal is heated to a sufficiently high temperature, electrons are emitted from the surface of the metal in a process known as *thermionic emission.*

Thermionic emission is the emission of electrons from the surface of a hot metal.

In an experiment first performed by Thomas Edison in 1883, it is observed that if the electrode *A* was sealed into the evacuated bulb of a carbon filament lamp, whose filament is heated by connecting it to a battery, a current flows through the electrode if it was made positive with respect to the filament (fig. 5.14). More current flows when the temperature of the filament is raised.

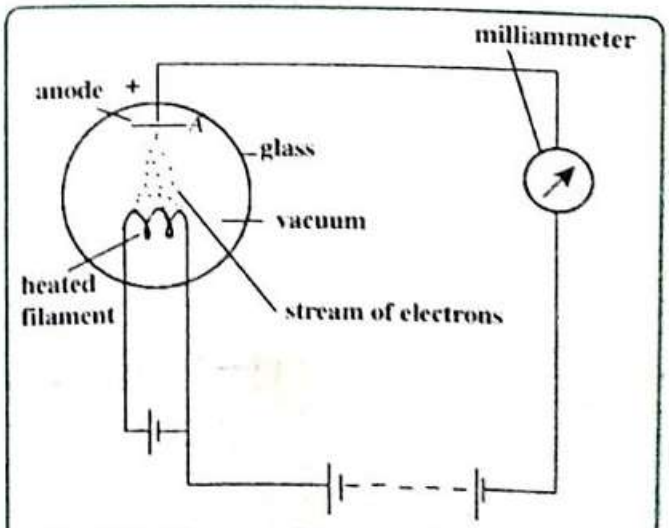


Fig. 5.14 Thermionic emission from hot filament

When the electrode is made negative with respect to the hot filament, no current is detected by the milliammeter no matter how hot the filament.

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These observations are explained as follows: When the tungsten filament is heated to a high temperature, the extra energy thus given to its free electrons at the surface of the metal enables them to break through the surface of the metal and exist, outside it as an 'electron cloud' or 'space charge'. This is the process of thermionic emission, and the emitting filament is the hot cathode. When plate *A* is positive with respect to the hot cathode, it attracts the free electrons of the electron cloud and current flows as indicated by the ammeter but when negative it repels the electrons and no current flows.

The Diode Valve

The diode valve is a simple application of the principle of thermionic emission. It consists of an anode, *A*, usually in the form of a cylinder, a hot filament, or heater made of thin tungsten wire, and a cathode surrounding the filament. All these component parts are enclosed in a highly evacuated glass bulb fig 5.15.

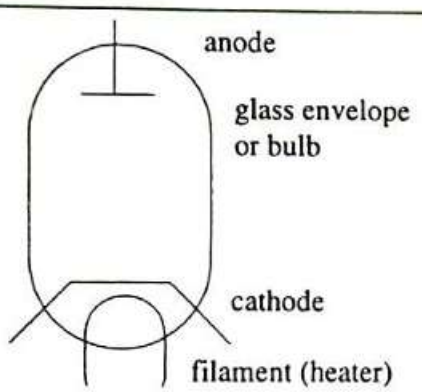


Fig. 5.15 Diode valve

Action of a diode. Diode valve characteristics

The action of a diode can be studied using a circuit as shown in fig. 5.16.

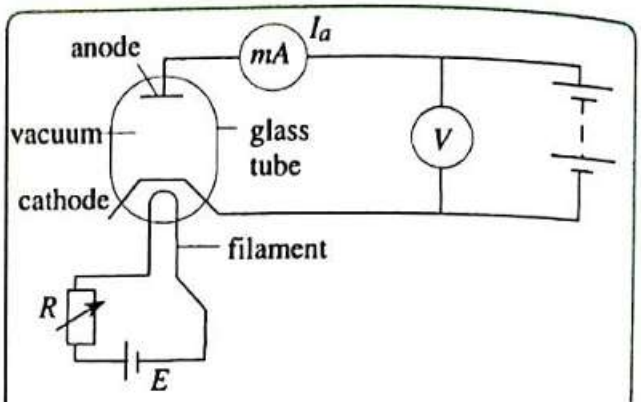


Fig. 5.16 Diode valve characteristics circuit

The filament supplies free electrons when heated by the current from the battery *E*. When the anode is made positive in potential with respect to the cathode, electrons flow towards the anode and constitute the *Anode Current* I_a, which is registered by the milliameter. No current flows (i.e. infinite resistance) when the anode is made negative in potential relative to the cathode. Thus the action of the diode allows current to flow only in one direction. The anode potential V_a relative to the cathode is recorded by the voltmeter (V). Fig 5.17 shows the variation of I_a with respect to V_a.

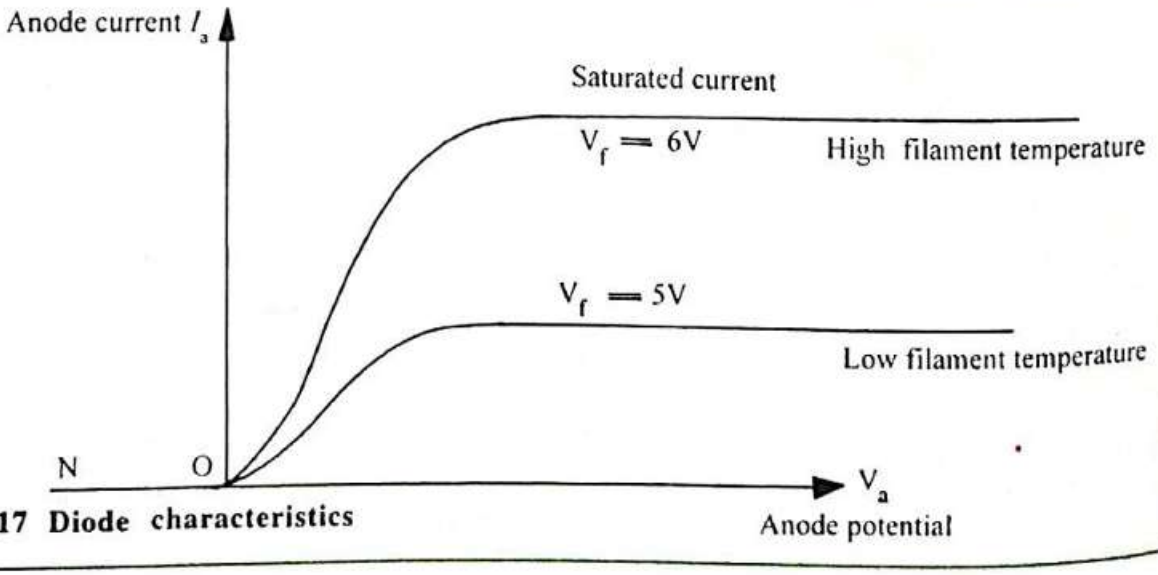


Fig 5.17 Diode characteristics

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As V_a is increased from zero, it is observed that I_a increases until it reaches a maximum value where it remains constant no matter the value of anode potential. This maximum constant value is known as the *saturation current*.

We can repeat the experiment by adjusting the heater rheostat, R, so that heater current is altered. The graph in each case shows the same form, but the more the heater current, the higher the saturation current obtained. The portion ON of the graph shows that no current flows when V_a is negative.

Diode characteristic curve shows that the diode valve does not obey Ohm's law. It is therefore called a *non-ohmic conductor*.

Because the action of diode allows current to flow only in one direction, the valve can be used as a *rectifier* to produce a d.c. voltage from an a.c. supply.

Similarities and differences between thermionic emission and liquid vaporization.

Similarities

1 Heat is involved in both processes
2 Both occur at the surface of the material.
3 In both cases, rate of phenomenon increases with temperature
4 Particles are released in both and both can reach a saturation point whereby as much particles are leaving the surface of the material as are returning to it.

Differences

1 Vaporization takes place from liquid surfaces. Thermionic emission takes place from metal surfaces.
2 Water molecules are released in vaporization while electrons are emitted in thermionic emission.
3 Vaporization occurs at a definite temperature for a given substance (the boiling point). Thermionic emission may not necessarily occur at a fixed temperature.

Application of Hot-Cathode (thermionic) emission

Cathode-ray Oscilloscope

An application of thermionic emission is in cathode-ray oscilloscope, an instrument used for the investigation of currents and voltages in electronic circuits.

The cathode ray oscilloscope is a vacuum tube containing an *electron gun* at one end, and a fluorescent screen at the other end. Between these are two pairs of deflector plates near the middle of the tube.

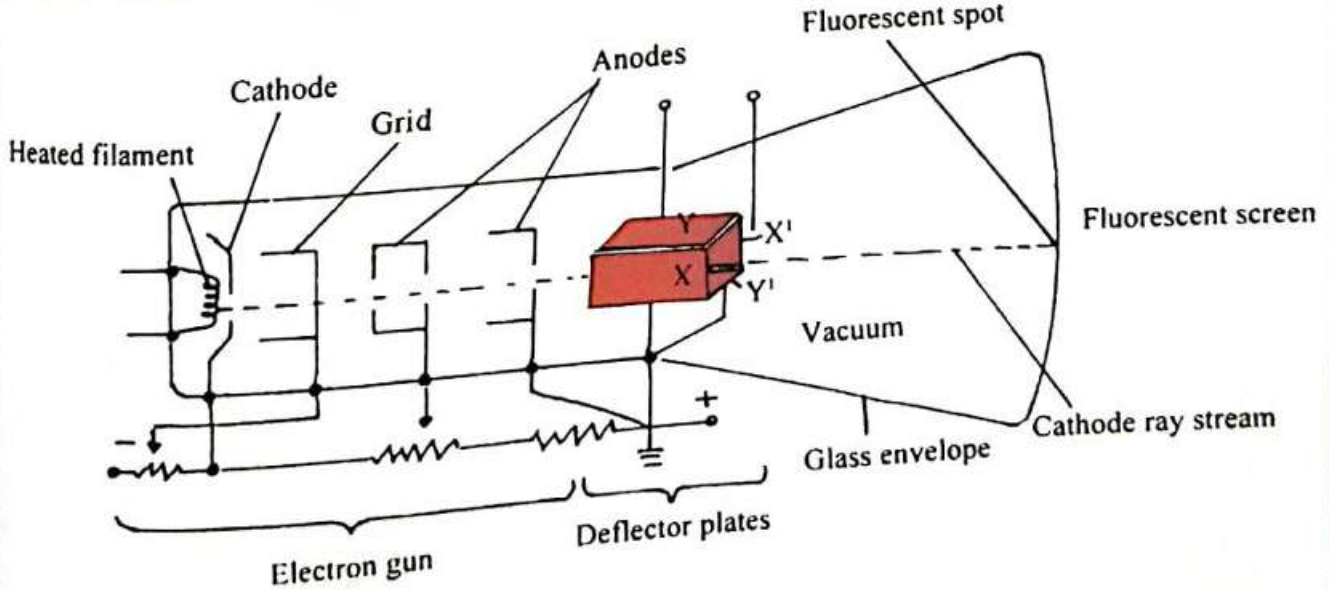


Fig. 5.18 Cathode-ray oscilloscope

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The '*electron gun*' consists of (i) the heated filament to supply electrons by thermionic emission (ii) the cathode (iii) the anode, made more positive with respect to the cathode so as to attract the electrons emitted from the cathode. The anode also acts as a focusing lens to accelerate and focus the electrons on to a spot in the fluorescent screen.

The deflector plates XX' and YY' are arranged in pairs to control the path of the electron beam. The YY' pair deflects the beam vertically, the XX' pair deflects it horizontally. Such deflections occur when an alternating potential difference is applied across the plates.

The cathode ray oscilloscope is used for studying all types of wave forms especially the alternating current waveform, and to measure frequencies and amplitudes of voltages of electronic devices.

Summary

1 *Electrolysis* is the chemical change in a liquid due to the flow of electric current through the liquid.
2 A *voltameter* is a device for studying the flow of current through a liquid.
3 An electrolyte is a liquid which conducts electricity. It contains charged particles called ions which carry the current in the electrolyte.
4 In the electrolysis of acidulated water, oxygen is liberated at the anode and hydrogen in the cathode in the ratio 1:2.
5 In the electrolysis of copper sulphate solution using copper electrodes, copper is deposited on the cathode and an equal amount of copper goes into the solution from the anode. If platinum electrodes are used, copper is deposited on the cathode. Density of copper sulphate solution is reduced.
6 In electroplating, the material to be electroplated is used as the cathode of a voltameter whose anode is made of the electroplating material, the electrolyte used is a solution of the salt of the metal to be deposited.
7 *Faraday's first law of electrolysis*: The mass, M, of any substance deposited on an electrode during electrolysis is proportional to the quantity of electricity Q which is passed through the electrolyte
$M = ZQ = ZIt$
8 *Faraday's second law of electrolysis*: The masses of different substances deposited or liberated by the same quantity of electricity are proportional to their chemical equivalents.
9 The electrochemical equivalent Z of an element is the mass of the element liberated or deposited in electrolysis by one coulomb of electricity or by one ampere of current flowing for one second.
10 Cathode rays are produced in gas discharge tubes under low pressure and high voltages.
11 Cathode rays consist of streams of high-velocity electrons. They are deflected by magnetic and electric fields. They travel in a straight line and cause certain materials (e.g. glass) to fluoresce. They have mass, momentum, and energy.
12 Thermionic emission is the emission of electrons from the surface of a hot metal.
13 A diode valve consists of an anode and a cathode in an evacuated glass bulb. Electrons are released from a hot filament and attracted towards the anode when the anode is at a positive potential with respect to the cathode. The diode valve can be used as a rectifier to convert a.c. voltages to d.c. voltages.
14 An application of thermionic emission is the cathode ray oscilloscope.

Exercise 5

1 Why will one Coulomb of electricity not liberate the same masses of different substances in an electrolytic process?
Because
A the mass of a substance liberated is related to the quantity of electricity which passes.
B the mass of the substance liberated is proportional to the concentration of the electrolytes.
C the electrochemical equivalents of different substances are different.
D the mass of substance liberated is inversely proportional to the chemical equivalent.
E One Coulomb of electricity takes different time to flow through different electrolytes.
(SC/GCE)

2 In an electrolysis experiment, a cathode of mass 5 g is found to weigh 5.01 g after a current of 5 A flows for 50 seconds. What is the electrochemical equivalent of the deposited substance?
A $0.00004\ gC^{-1}$ B $0.00002\ gC^{-1}$
C $0.02500\ gC^{-1}$ D $0.05000\ gC^{-1}$

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E $0.00001\ gC^{-1}$

(JAMB)

3 The electrochemical equivalent of platinum is $5\times10^{-7}\ kgC^{-1}$. To plate-out 1. 0 kg of platinum, a current of 100 A must be passed through an appropriate vessel for

A 5.6 hours B 56 hours

C 1.4×10^{4} hours

D 2.0×10^{4} hours.

(JAMB)

4 The electrochemical equivalent of a metal is $0.126\times10^{-6}\ kgC^{-1}$. The mass of the metal that a current of 5 A will deposit from a suitable bath in 1 hour is

A 0.0378×10^{-3} kg

B 0.227×10^{-3} kg

C 0.378×10^{-3} kg

D 0.595×10^{-3} kg

E 2.268×10^{-3} kg

(JAMB)

5 When a metal is heated to a high temperature, electrons are emitted from its surface. This process is known as

A. photoelectric emission
B artificial radioactivity
C field emission
D secondary emission
E thermionic emission

(SSCE)

6 Which of the following is not true of a discharge tube?

A The pressure of gas must be very low
B The glass tube must have two electrodes at the end
C The gas in the tube must be carbon dioxide
D Air is gradually pumped out of the tube
E The coating on the walls of the tube at a certain stage glows.

(SSCE)

7 Explain the meaning of the following terms: *electrolysis*, *voltameter*, *electrode*, *electrolyte*, *anode*, *cathode*.

8 Which of the following are electrolytes: mercury, alcohol, petrol solution of sodium chloride, dilute sulphuric acid, molten copper.

9 What do you understand by the statement: the electrochemical equivalent of copper is $0.00033\ gC^{-1}$?

In an electrolysis experiment, a current causes a deposit of 1.05 g of copper on the cathode of the voltameter in 3 minutes. If the electrochemical equivalent of copper is $0.00033\ gC^{-1}$ what is the value of the current?

10 State Faraday's laws of electrolysis. Briefly describe an experiment to illustrate the first law.

How long will it take for a 1.5 A current to coat evenly a layer of copper 0. 1 mm thick on both sides of a 12 cm × 12 cm copper plate (e.c.e. of copper = $0.000331\ gC^{-1}$; density of copper 9 gcm^{-3}).

11 Describe how you would use the electrolysis of copper sulphate solution to calibrate an ammeter. Draw a circuit diagram of a suitable arrangement to illustrate the method you have described. If the ammeter records a steady current of 1.1 A for 30 minutes while the copper deposited is 0.75 g calculate the error in the ammeter reading (e.c.e. of copper = $0.00033\ gC^{-1}$).

12 Name two industrial applications of electrolysis.

Calculate the time taken to deposit 1.5 6 g of nickel using a current of 2.34 A in a nickel-plating process. (Assume that 1.20 g of nickel are deposited per ampere-hour.

(SC/GCE)

13 The cathode plate of a copper voltameter gains 1.65 g of copper in 20 minutes. Determine the average current. (e.c.e. of copper = 0.00033 gC^{-1}).

14 Describe briefly how electrons can be liberated from (a) a cold cathode (b) a hot cathode.

15 a. What is thermionic emission?
b. State two similarities and two differences between thermionic emission and vaporization in liquids.

(SC/GCE)

16 What are carriers of electric current in (i) metals (ii) electrolytes?

17 What are cathode rays, and what is their nature? List four properties of cathode rays.

18 Draw a diagram of a diode valve and label its parts. Explain why current can pass through it in only one direction. Draw a circuit diagram of the set up you will use to study diode characteristics. Draw a graph of the variation of anode current with anode potential.

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